F79SU Survival Models 2025–2026

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Adapted from F79SU Lecture Notes by I.D. Currie and A.S. Macdonald

Module Information

- Lecturers:
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- Contributes to exemption from IFoA Subject CS2
- Programming language used: R
- Assessment:
  - Project 1 (Weeks 3-5), 15%,
  - Project 2 (Weeks 9-11), 25%,
  - Final Exam (Weeks 13+), 60%.
- Course website: Please check Canvas for all official information.

Schedules (Malaysia)

- Lectures:
  - Mondays 3-4pm, WG-LR4
  - Thursdays 9-11am, WG-LR6
- Tutorials:
  - TG1: Mondays 1-3pm, W3-SR11
  - TG2: Tuesdays 11am-1pm, W1-LS8
  - TG3: Wednesdays 12-2pm, W1-LS8
- Consultation Hour:
  - Thursdays 11am-12pm, E1 MACS Office

Course Content

1. Introduction to Survival Models
2. Estimating the Lifetime Distribution
3. Markov Models
4. Binomial and Poisson Models of Mortality
5. Methods of Graduation
6. Graduation and Statistical Tests
7. Mortality Projection
8. Exposed to Risk

Note: These slides are merely supplementary
to the Malaysia lectures; they are not standalone!
For complete information, please refer to
the official lecture notes on Canvas.

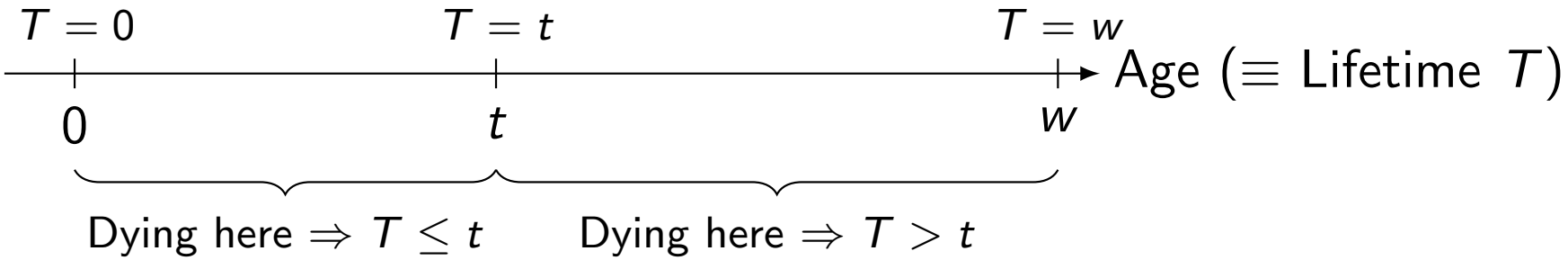
Chapter 1

Introduction to Survival Models

1. Notation and Basic Formulae:
   - Lifetime – $T, F(t), S(t)$,
   - Future Lifetime – $T_x,\ F_x(t) \equiv {}_tq_x,\ S_x(t) \equiv {}_tp_x$,
   - Force of mortality – μ_{x+t}
   - Relationships between ${}_tp_x, {}_tq_x$ and μ_{x+t}

Notation and Basic Formulae

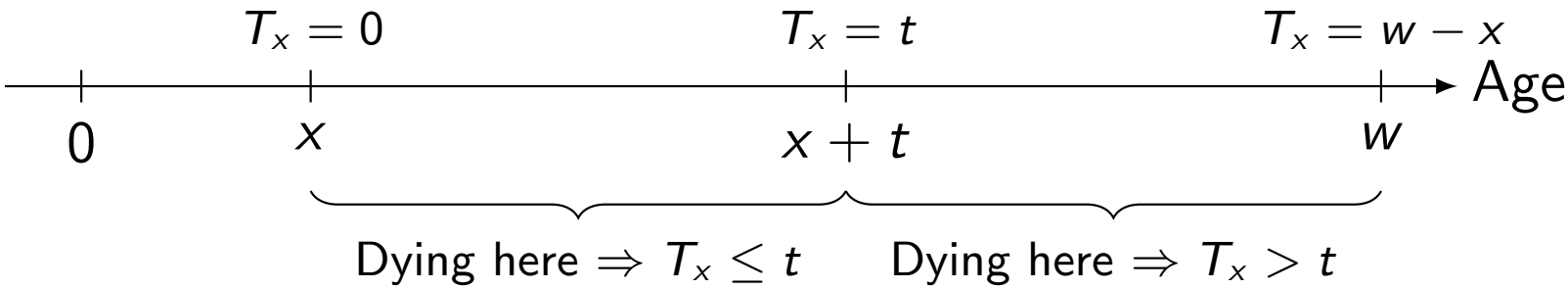
Lifetime (of a newborn)



- Random variable: T = Future lifetime of a newborn $\in [0, w]$,
- (Cumulative) Distribution Function $F(t) = \mathbb{P}(T \leq t)$,
- Survival Function $S(t) = \mathbb{P}(T > t) = 1 - F(t)$.

Notation and Basic Formulae

Future Lifetime (of a life aged x)



- Random variable: T_x = Future lifetime of a life aged x,

$$T_x = (T - x \mid T > x) \in [0, w - x]$$

- (Cumulative) Distribution Function:

$$F_x(t) = \mathbb{P}(T_x \leq t) = \mathbb{P}(T \leq x + t \mid T > x) \equiv {}_tq_x$$

- Survival Function:

$$S_x(t) = \mathbb{P}(T_x > t) = \mathbb{P}(T > x + t \mid T > x) \equiv {}_tp_x$$

Notation and Basic Formulae

Naturally, we can use Bayes' rule $\mathbb{P}(A|B) = \mathbb{P}(A \cap B)/\mathbb{P}(B)$ to expand $F_x(t)$ and $S_x(t)$, giving the following relations:

$$\begin{aligned} F_x(t) &= \mathbb{P}(T \leq x + t \mid T > x) \\ &= \frac{\mathbb{P}(x < T \leq x + t)}{\mathbb{P}(T > x)} \\ &= \frac{\mathbb{P}(T \leq x + t) - \mathbb{P}(T \leq x)}{\mathbb{P}(T > x)} \\ &= \frac{F(x + t) - F(x)}{S(x)} \end{aligned}$$

$$\begin{aligned} S_x(t) &= \mathbb{P}(T > x + t \mid T > x) \\ &= \frac{\mathbb{P}(T > x + t)}{\mathbb{P}(T > x)} \\ &= \frac{S(x + t)}{S(x)} \end{aligned}$$

Notation and Basic Formulae

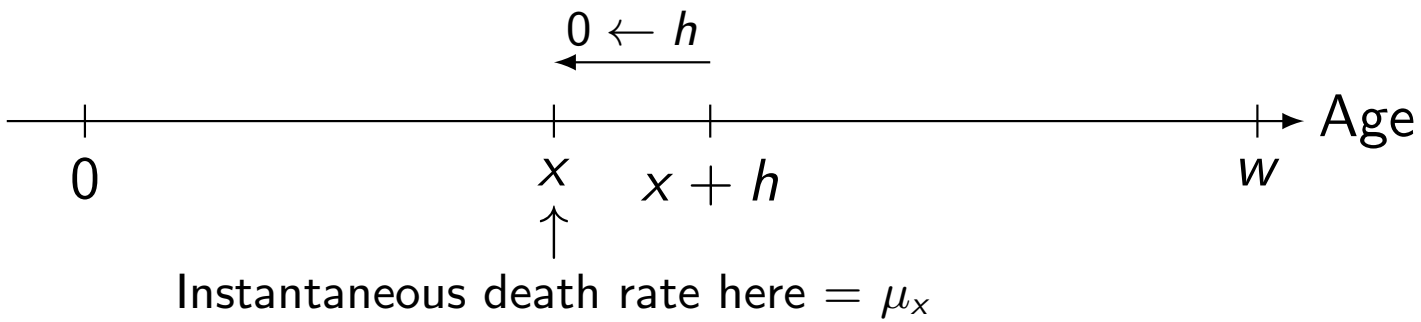
Force of Mortality

- Also known as hazard rate, or instantaneous death rate.
- For age $x \in [0, w]$,

$$\mu_x := \lim_{h \to 0^+} \frac{\mathbb{P}(T \leq x + h \mid T > x)}{h}$$

$$= \lim_{h \to 0^+} \frac{\mathbb{P}(T_x \leq h)}{h} \equiv \lim_{h \to 0^+} \frac{{}_hq_x}{h} = \frac{d}{dt} {}_tq_x \Big|_{t=0}$$

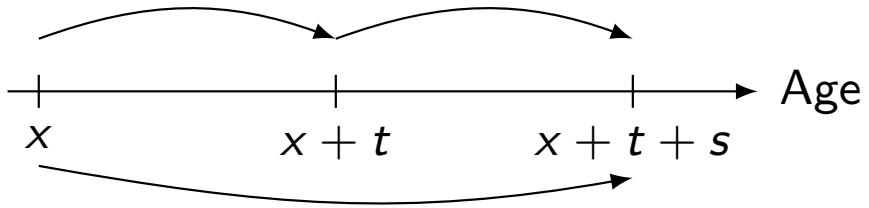
- Particularly, ${}_hq_x \approx h \cdot \mu_x$.



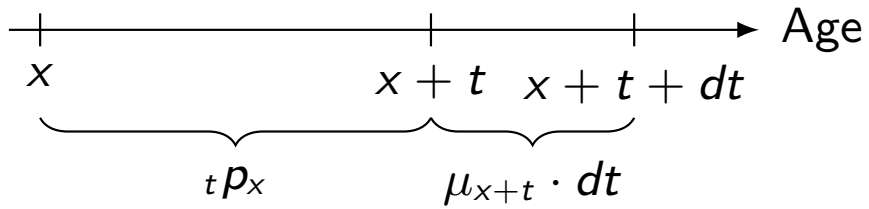
Notation and Basic Formulae

Further relationships

- $_{s+t}p_x = {}_tp_x \cdot {}_sp_{x+t}$



- $\frac{d}{dt}{}_tq_x = f_x(t) = {}_tp_x \cdot \mu_{x+t}$



- ${}_tq_x = \int_0^t {}_sp_x \cdot \mu_{x+s}\, ds$

$${}_tp_x = \exp\left\{ -\int_0^t \mu_{x+s}\, ds \right\}$$

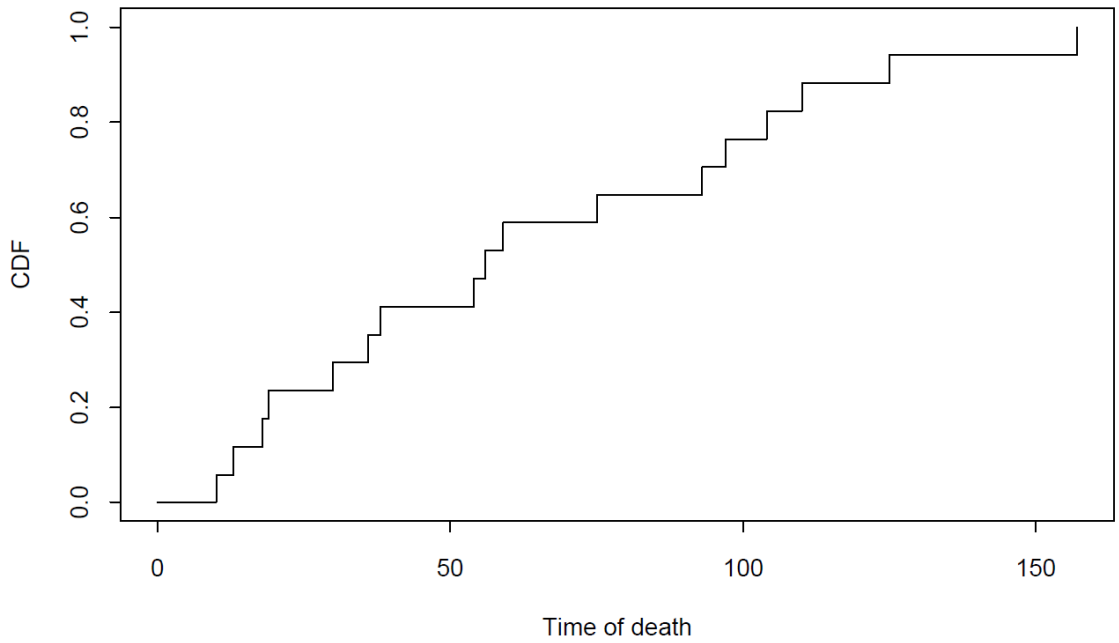
Chapter 2

Estimating the Lifetime Distribution

1. Study Types and Censoring
2. The Kaplan-Meier estimate of the survivor function
3. The Cox Regression Model

Study Types and Censoring

- Objective: want to estimate T, T_0 and T_x.
- In other words, we want to estimate $F(t), F_x(t)$, $S(t), S_x(t)$ etc.
  A typical estimate $\hat{F}(t)$ may look like this:



- How can we do this?
- What issues arise?

Study Types: Cohort Studies

- Cohort studies follow a large number of new-born lives until their death.
- Pros:
  - Simple estimations
  - No mixing of mortalities
- Cons:
  - Extremely long time
  - High proportion of subjects leave study $\Rightarrow$ Right censoring

Study Types: Cohort Studies

- Cross-sectional studies follow a large number of lives by observing a *section* of their lives.

  E.g. dividing an investigation into single years of age $x \to x + 1$ and following each "mini-cohort" for 3-4 years.

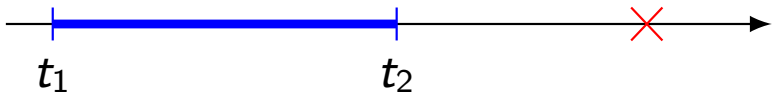
- Pros:
  - Short time of study
  - Smaller proportion of subjects leave study

- Cons:
  - Mixing of mortalities
  - Most lives do not die $\Rightarrow$ Right censoring

Censoring

Censoring occurs when a waiting time T is not known exactly due to certain mechanisms in a study.

- Right censoring: observations in progress cut short



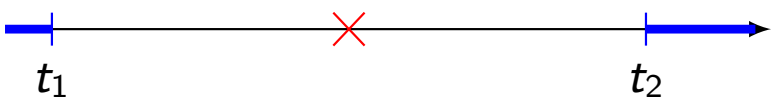
$(T > t_2 - t_1)$

- Left censoring: Time of entry into state of interest is unknown



$(T > t_2 - t_1)$

- Interval censoring: event of interest occurs between 2 times



$(t_1 \leq T \leq T_2)$

Censoring

The following are *ways* in which right, left or interval censoring can arise.

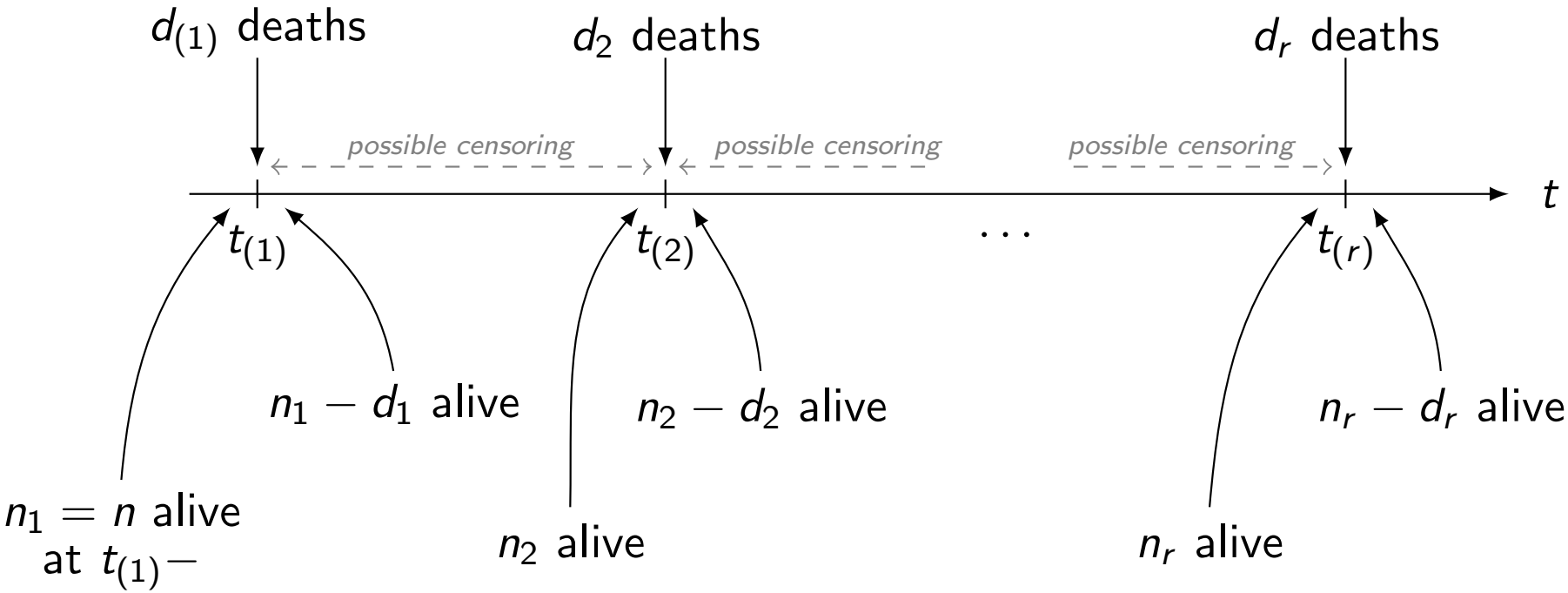
- Random censoring: Censoring time random variable $C_i < T_i$
- Non-informative censoring: $C_i \perp\!\!\!\perp T_i$
- Type I censoring: Known $C_i = c$
- Type II censoring: C_i depends on fixed number of failures/deaths.

The Kaplan-Meier Estimate

Motivation

The Kaplan-Meier Estimate: Objective

- Objective: Want to estimate $S_x(t)$ (and thus $F_x(t)$) from n possibly right-censored survival times $t_1, t_2, ..., t_n$.
- A visualization:



The Kaplan-Meier Estimate: Formulation

- MLE of probability of dying in $(t_{(j)} - \delta, t_{(j)}]$ is therefore d_j/n_j,

  $\Longrightarrow$ MLE of probability of surviving $(t_{(j)} - \delta, t_{(j)}]$ is $(n_j - d_j)/n_j$.

  $\Longrightarrow$ MLE of Probability of surviving $(t_{(j)} - \delta, t_{(j+1)} - \delta)$ is $(n_j - d_j)/n_j$ (assuming censored observations do not count as deaths).

  $\stackrel{\delta \to 0}{\Longrightarrow}$ MLE of Probability of surviving $[t_{(j)}, t_{(j+1)})$ is $(n_j - d_j)/n_j$.

The Kaplan-Meier Estimate: Results

- Assuming independent deaths, MLE of probability of survival until $t \in [t_{(k)}, t_{(k+1)})$ is a product of the probability of survival in the current interval and all preceding intervals:

$$\hat{S}(t) = \prod_{j=1}^{k} \left(\frac{n_j - d_j}{n_j} \right), \ t \in [t_{(k)}, t_{(k+1)}),$$

  with $\hat{S}(0) = 1$. This is known as the Kaplan-Meier estimate for the survivor function.

- Last observation is a death $\Rightarrow$ $n_r = d_r$ $\Rightarrow$ $\hat{S}(t) = 0 \ \forall t \geq t_{(r)}$.
- Last observation is censored at t^* $\Rightarrow$ $\hat{S}(t)$ undefined $\forall t > t^*$.

The Kaplan-Meier Estimate: Results

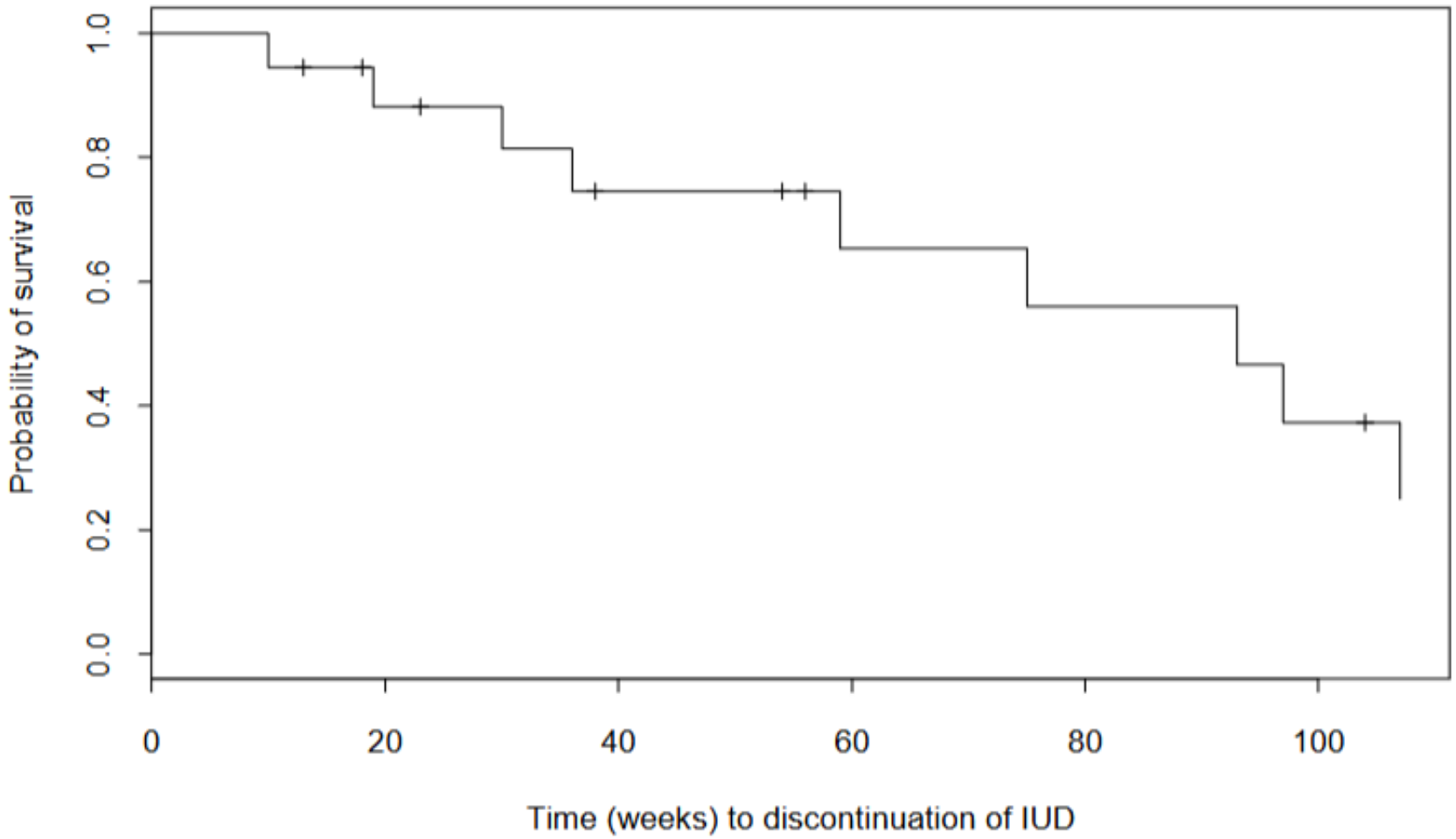
- Compute the standard error of the Kaplan-Meier estimate using Greenwood's formula:

$$\text{Var}\left(\hat{S}(t)\right) = \left(\hat{S}(t)\right)^2 \sum_{j=1}^{k} \frac{d_j}{n_j(n_j - d_j)},$$

for $t \in [t_{(k)}, t_{(k+1)})$.

The Kaplan-Meier Estimate

- The Kaplan-Meier Estimate $\hat{S}(t)$ is a *step function*. An example:



The Cox Regression Model

Motivation

The Cox Regression Model: Introduction

- Also known as Cox proportional hazards model.
- It models the hazard function, i.e. force of mortality of a life as a function of certain covariates.
- Based on the Cox regression model, we write:

$$\lambda(t; \underline{z}_i) = \lambda_0(t) \exp\left\{\beta_1 z_{i1} + \beta_2 z_{i2} + \cdots + \beta_p z_{ip}\right\}$$
$$\equiv \lambda_0(t) \exp\{\underline{\beta}' \underline{z}_i\}$$

where $\lambda_0(t)$ is the baseline hazard, and $\underline{z}_i = (z_{i1}, z_{i2}, ..., z_{ip})'$ are the covariates of the i-th life.

The Cox Regression Model: Introduction

$$\lambda(t; \underline{z}_i) = \lambda_0(t) \exp\left\{\beta_1 z_{i1} + \beta_2 z_{i2} + \cdots + \beta_p z_{ip}\right\}$$

- What does this form imply about the effects of covariates?
- Why isn't there an intercept coefficient β_0?
- How do we use this model?

The Cox Regression Model: Objectives

- Objective: Estimate $\underline{\beta} = (\beta_1, \beta_2, \cdots, \beta_p)'$, given n possibly right-censored survival times and their respective covariates.

- Dataset: An example of a dataset well-suited for a Cox regression model:

Male	2	5+	9	11+	14
Female	4	7+	10+	12+	15

Here, $p = 1$ only, and so $\lambda(t; z_i) = \lambda_0(t) \exp\{\beta z_i\}$.

The Cox Regression Model: Formulation

- Before going further, let us first define some useful notations: For $j = 1, ..., r$,

$$t_{(j)} = \text{Time of the } j\text{-th death},$$

$$R(t_{(j)}) = \text{The risk set at time } t_{(j)}$$

$$\equiv \text{The set of lives alive just before time } t_{(j)}$$

The Cox Regression Model: Formulation

- First consider the 1st death at $t_{(1)}$ – can we find a likelihood for $\underline{\beta}$?

  In other words, what is the probability that a life with covariates $\underline{z}_{(1)}$ dies at $t_{(1)}$, given that we know that there's a death at $t_{(1)}$, written in terms of $\lambda(t; \underline{z}_{(1)})$?

- How about:

$$\mathbb{P}\Big(\text{Life with } \underline{z}_{(1)} \text{ dies at } t_{(1)} \;\Big|\; \text{One death from } R(t_{(1)}) \text{ at } t_{(1)}\Big)$$

- There's an issue here - the probability that one dies *at* a certain time is 0. Let's consider small intervals instead.

- Additionally, due to the condition of "one death from $R(t_{(1)})$ at $t_{(1)}$", we also assume no ties in death times.

The Cox Regression Model: Formulation

$$\mathbb{P}\Big(\text{Life with } \underline{z}_{(1)} \text{ dies in } \big[t_{(1)}, t_{(1)}+\delta\big) \,\Big|\, \text{One death from } R(t_{(1)}) \text{ in } \big[t_{(1)}, t_{(1)}+\delta\big)\Big)$$

$$= \frac{\mathbb{P}(\text{Life with } \underline{z}_{(1)} \text{ dies in } \big[t_{(1)}, t_{(1)}+\delta\big))}{\mathbb{P}(\text{One death from } R(t_{(1)}) \text{ in } \big[t_{(1)}, t_{(1)}+\delta\big))}$$

$$\approx \frac{\delta \cdot \lambda(t_{(1)}; \underline{z}_{(1)})}{\sum_{\ell \in R(t_{(1)})} \delta \cdot \lambda(t_{(1)}; \underline{z}_{\ell})}$$

$$= \frac{\delta \cdot \lambda_0(t_{(1)}) \cdot \exp\{\underline{\beta}' \underline{z}_{(1)}\}}{\sum_{\ell \in R(t_{(1)})} \delta \cdot \lambda_0(t_{(1)}) \cdot \exp\{\underline{\beta}' \underline{z}_{\ell}\}} = \frac{\exp\{\underline{\beta}' \underline{z}_{(1)}\}}{\sum_{\ell \in R(t_{(1)})} \exp\{\underline{\beta}' \underline{z}_{\ell}\}}$$

...and this is only for the first death!

The Cox Regression Model: Results

- Applying the same logic to all deaths at $t_{(1)}, \ldots, t_{(r)}$ with no ties, we have the likelihood function:

$$L(\underline{\beta}) = \prod_{j=1}^{r} \frac{\exp\{\underline{\beta}' \underline{z}_{(j)}\}}{\sum_{\ell \in R(t_{(j)})} \exp\{\underline{\beta}' \underline{z}_{\ell}\}}$$

  and the MLE $\hat{\underline{\beta}}$ can be found by maximizing $L(\underline{\beta})$ or $\ell(\beta)$.

- Notice that the derivation of $L(\underline{\beta})$ uses only the *order* of death, and not the actual *times* of death – hence $L(\underline{\beta})$ is only a partial likelihood, not a true likelihood.

The Cox Regression Model: Results

- The partial likelihood behaves very much like an ordinary likelihood function. If β_0 is the true unknown value of β, then

$$\mathbb{E}(\hat{\beta}) \longrightarrow \beta_0 \ ,$$

$$\text{Var}(\hat{\beta}) \longrightarrow -\frac{1}{\mathbb{E}\left(\frac{\partial^2 \ell}{\partial \beta^2}\right)\Big|_{\beta=\beta_0}} \approx -\frac{1}{\frac{\partial^2 \ell}{\partial \beta^2}\Big|_{\beta=\hat{\beta}}} = \frac{1}{I(\hat{\beta})} \ ,$$

asymptotically, i.e. as $n \to \infty$.

The Cox Regression Model: Hypothesis Testing

- We've obtained an estimate for $\hat{\beta}$, but is it statistically significant?
- Often, we test for $H_0 : \beta = 0$ vs. $H_1 : \beta \neq 0$, since this tells us whether the relevant covariate has any effect on the hazard function.
  - If we do not reject H_0, then we do not have evidence that the mortality of the two groups differ.
  - If we reject H_0, then there is evidence that the mortality of the two groups differ.
- More generally, we can also test for $H_0 : \beta = \beta_0$ vs. $H_1 : \beta \neq \beta_0$, where β_0 is some hypothesized value.

The Cox Regression Model: Hypothesis Testing

Test	Test Statistic	Asymptotic Distribution
Wald Test	$z^2 = \frac{(\hat{\beta} - \beta_0)^2}{\frac{1}{I(\hat{\beta})}}$	χ^2_1
Likelihood Ratio Test	$-2 \log \Lambda = -2 \left[\ell(\beta_0) - \ell(\hat{\beta}) \right]$	χ^2_1
Score Test	$S = \frac{U(\beta_0)^2}{I(\beta_0)}$	χ^2_1

Chapter 3

Markov Models

1. Markov Models
   - Introduction
   - 2-state Markov Models
   - Multi-state Markov Models
   - Notable relationships
2. Data and Estimation
   - D_i and V_i
   - MLE of μ in 2-state Markov Models
   - MLE of μ's in Multi-state Markov Models

Markov Models: Introduction

- In examining T and μ_{x+t} previously, we were limited to the waiting time of one event of interest.

  What if we want more states/multiple decrements?

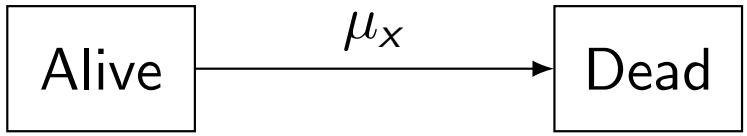
- In the Kaplan-Meier and Cox Regression models, we had the assumption of right-censored times.

  What if there are other censoring types involved?

- Markov models are a more general alternative that deals with both issues.

Markov Models: 2-state Models - Example

- We can re-frame what we've used thus far as a Markov model:



- This Markov model makes two fundamental assumptions:
  - Probabilities regarding the future only depend on current information: [Markov property] If current age is x, for any $t > 0$,

$$\begin{aligned} &\mathbb{P}(\text{Alive or Dead at age } x+t \mid \text{Alive or Dead before/at } x) \\ =\ &\mathbb{P}(\text{Alive or Dead at age } x+t \mid \text{Alive or Dead at age } x) \end{aligned}$$

  - Probability of death in a short time interval is approximately proportional to the force of mortality at that time: For $t > 0$,

$$_{dt}q_{x+t} = \mu_{x+t}dt + o(dt)$$

Markov Models: 2-state Models - Example

- Does the Alive-Dead model agree with the results in Chapter 1? Yes.
- Chapter 1 theory $\Rightarrow$ Markov assumption 2:

$$ {}_tp_x = \exp\left(-\int_0^t \mu_{x+s}\, ds \right) \Longrightarrow \; {}_{dt}q_{x+t} = \mu_{x+t}dt + o(dt) $$

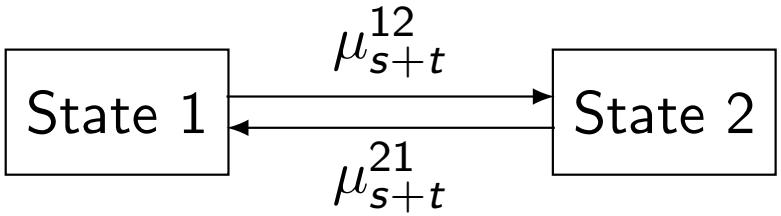
- Markov assumptions 1 & 2 $\Rightarrow$ Chapter 1 theory:

$$ {}_{dt}q_{x+t} = \mu_{x+t}dt + o(dt) \Longrightarrow \; {}_tp_x = \exp\left(-\int_0^t \mu_{x+s}\, ds \right) $$

[For proofs, see Lecture Notes.]

Markov Models: 2-state Models - General

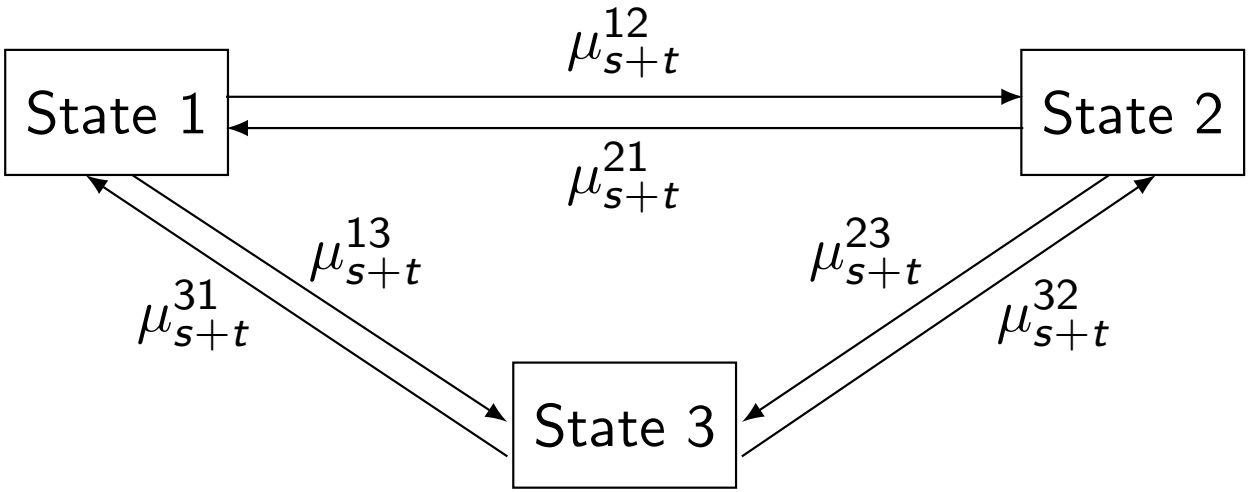
- Naturally, we can generalize past the Alive-Dead model.
  A general 2-state Markov Model:



- Notations:
  - μ_x^{gh} = Transition intensity from state g to state h at age x
  - ${}_tp_x^{gh} = \mathbb{P}(\text{In state } h \text{ at time } x+t \mid \text{In state } g \text{ at time } x)$
    (analogous to ${}_tq_x$ in Alive-Dead model).
- $\mu_x^{gh} = 0$ if transition from state g to h is impossible.

Markov Models: Multi-state Models - General

- A general 3-state Markov Model:



- Notations:
  - μ_x^{gh} = Transition intensity from state g to state h at time x
  - ${}_tp_x^{gh} = \mathbb{P}(\text{In state } h \text{ at time } x+t \mid \text{In state } g \text{ at time } x)$
- Some μ_x^{gh} can be 0; further generalizes to $n \in \mathbb{Z}^+$ states.

Markov Models: General Assumptions

- The assumptions under the general Markov model become:
  - Probabilities regarding the future only depend on current information: [Markov property] For $t > 0$,

$$\begin{aligned} &\mathbb{P}(\text{In state } h \text{ at time } s+t \mid \text{In state } g \text{ at time } s \text{ and all history}) \\ =\ &\mathbb{P}(\text{In state } h \text{ at time } s+t \mid \text{In state } g \text{ at time } s) \end{aligned}$$

  - Probability of transition between distinct states in a short time interval is approximately proportional to the transition intensity at that time: For $t > 0$ and $i \neq j$,

$$_{dt}p_{x+t}^{gh} = \mu_{x+t}^{gh} dt + o(dt)$$

  - Probability of two or more transitions in time dt is $o(dt)$.

Markov Models: Notable Relationships

- $\mu_x^{gh} = \dfrac{\partial}{\partial t}\, {}_tp_x^{gh}\Big|_{t=0}$; (Recall in Chp. 1: $\mu_x = \dfrac{\partial}{\partial t}\, {}_tq_x\Big|_{t=0}$).
- ${}_tp_x^{\overline{gg}} = \mathbb{P}(\text{In state } g \text{ from } x \text{ to } x+t \mid \text{In state } g \text{ at time } x)$:

  $${}_tp_x^{\overline{gg}} = \exp\left(-\int_0^t \sum_{k\neq g} \mu_{x+s}^{gk}\, ds \right) \quad ; \quad (\text{Chp. 1: } {}_tp_x = e^{-\int_0^t \mu_{x+s}\, ds}),$$

  ${}_tp_x^{\overline{gg}} = {}_tp_x^{gg}$ if return to state g is impossible.
- Kolmogorov forward equations: For $t > 0$,

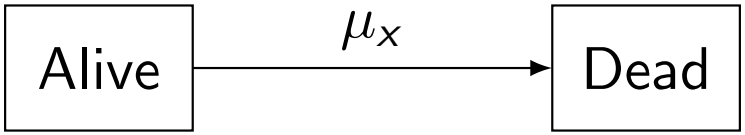
  $$\frac{\partial}{\partial t}\, {}_tp_x^{gh} = \sum_{j\neq h} \left({}_tp_x^{gj}\, \mu_{x+t}^{jh} - {}_tp_x^{gh}\, \mu_{x+t}^{hj} \right), \quad g \neq h.$$

  [For proof, see Theorem 8.1.1 in F79SP Notes.]

Data and Estimation: Introduction

- We now have Markov Models for the transition between states. But how do we estimate the transition intensities μ_x^{gh} ?
- What data structure allows us to estimate these transition intensities?
- What is the probability distribution of these estimators?

Data and Estimation: Introduction

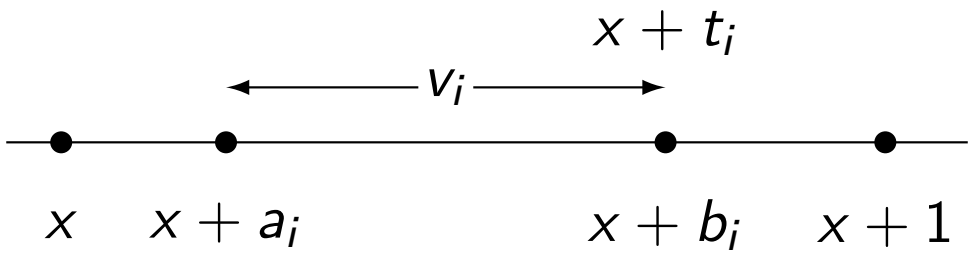


- Consider the following dataset:

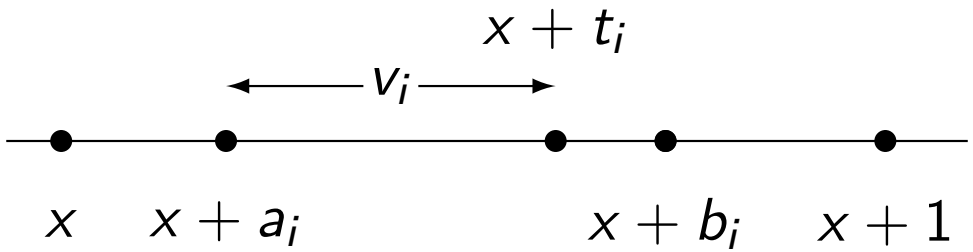
Observation began at age: $x + a_i$	Observation ended at age: $x + t_i$	Observation would have ended at age: $x + b_i$	Observation duration: v_i	Died? d_i
24.47	24.97	24.97	0.5	No
24.00	25.00	25.00	1.00	No
24.15	24.92	25.00	0.77	Yes
⋮	⋮	⋮	⋮	⋮
24.00	24.65	24.65	0.65	No

Data and Estimation: MLE formulation

- Case 1: Life does not die before observation ends:
  $d_i = 0, \quad t_i = b_i, \quad v_i = b_i - a_i$



- Case 2: Life dies before observation ends:
  $d_i = 1, \quad t_i \in (a_i, b_i), \quad v_i = t_i - a_i$



Data and Estimation: MLE formulation

- Estimating varying μ_{x+t} is too complicated; we require a simplifying assumption: $\mu_{x+t} = \mu$ over a certain period.
- Objective: To construct a likelihood for μ.
- Is there a way for us to encompass both cases with one probability function? What μ-dependent variables should we consider?
  - $D_i \in \{0, 1\}$: Indicator for death of life i;
  - V_i: Waiting time in alive state for life i

Data and Estimation: MLE formulation for 2 states

- Case 1: Life does not die $\Rightarrow d_i = 0,\ v_i = b_i - a_i$. Therefore:

$$\mathbb{P}(D_i = 0, V_i = b_i - a_i) = e^{-\mu v_i}$$

- Case 2: Life dies $\Rightarrow d_i = 1, v_i = t_i - a_i$. The probability of dying at any single point is 0, so we consider an interval:

$$\begin{aligned}
&\mathbb{P}(D_i = 1, v_i \leq V_i < v_i + dv_i) \\
&= \mathbb{P}(\text{survive for } v_i \mid \text{age } x + a_i)\ \mathbb{P}(\text{die within } dv_i \mid \text{age } x + a_i + v_i) \\
&= e^{-\mu v_i}(\mu\ dv_i + o(dv_i)) = e^{-\mu v_i}\mu\ dv_i + o(dv_i) \\
&\to e^{-\mu v_i}\mu\ dv_i \quad \text{as } dv_i \to 0.
\end{aligned}$$

- Therefore, we can encompass both cases using a joint probability function for D_i and V_i:

$$f_i(d_i, v_i) = e^{-\mu v_i}\mu^{d_i}$$

Data and Estimation: MLE formulation for 2 states

- Letting $D = \sum_{i=1}^{n} D_i,\ V = \sum_{i=1}^{n} V_i$, the total likelihood is:

$$L(\mu; D, V) = \prod_{i=1}^{n} e^{-\mu V_i} \mu^{D_i} = e^{-\mu V} \mu^{D}.$$

- Therefore the ML estimator $\tilde{\mu}$ is

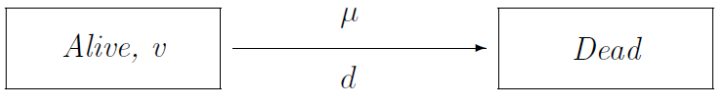
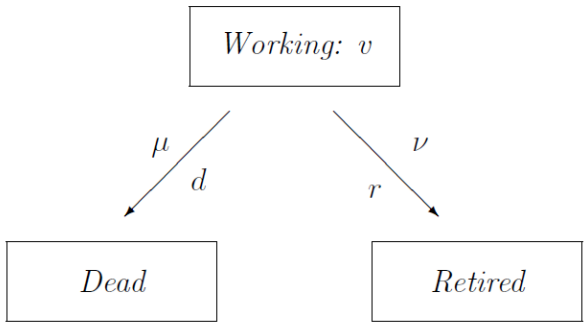
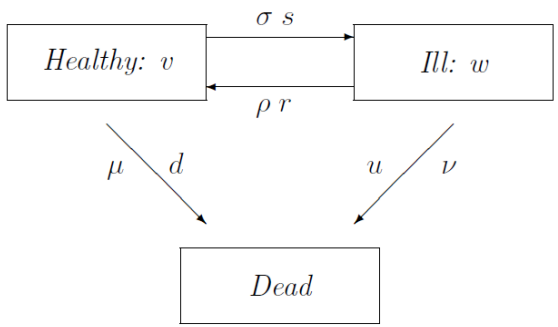
$$\tilde{\mu} = \frac{D}{V} = \frac{\text{Total number of deaths}}{\text{Total waiting time}}$$

- Applying the ML theorem, we have, asymptotically,

$$\tilde{\mu} \approx \mathcal{N}\left(\mu_0, \frac{\mu_0}{\mathbb{E}(V)}\right), \text{ with } \text{Var}(\tilde{\mu}) = \frac{\mu_0}{\mathbb{E}(V)} \approx \frac{\tilde{\mu}}{V} = \frac{D}{V^2},$$

where μ_0 is the true value of μ. Additionally, $V \equiv E_x^c$ is known as the "central exposed to risk".

Data and Estimation: MLE formulation for ≥ 2 states

Markov model	Estimations
	$L(\mu) = e^{-\mu W_a}\ \mu^{T_{ad}}$ $\tilde{\mu} = D/V = \frac{T_{ad}}{W_a}$
	$L(\mu,\nu) = e^{-(\mu+\nu)W_w}\ \mu^{T_{wd}}\ \nu^{T_{wr}}$ $\tilde{\mu} = \frac{T_{wd}}{W_w},\ \tilde{\nu} = \frac{T_{wr}}{W_w}$
	$L(\mu,\nu,\sigma,\rho) = e^{-(\mu+\sigma)W_h}\ \mu^{T_{hd}}\ \sigma^{T_{hi}}$ $e^{-(\rho+\nu)W_i}\ \rho^{T_{ih}}\ \nu^{T_{id}}$ $\tilde{\mu} = T_{hd}/W_h,\ \tilde{\sigma} = T_{hi}/W_h,\ \ldots$

Data and Estimation: MLE formulation for ≥ 2 states

- Each waiting time W_g in state g contributed a factor

$$e^{-\left(\sum_{h \neq g} \mu_{gh}\right) W_g} ,$$

- Each transfer number T_{gh} from state g to h contributes a factor

$$\mu_{gh}^{T_{gh}} ,$$

- The total likelihood is then the product of all factors:

$$L(\underline{\mu}) = \prod_g \left[e^{-\left(\sum_{h \neq g} \mu_{gh}\right) W_g} \prod_{h \neq g} \mu_{gh}^{T_{gh}} \right]$$

- The log-likelihood is thus a sum of separate terms of the parameters:

$$\ell(\underline{\mu}) = \sum_g \left[-\left(\sum_{h \neq g} \mu_{gh} \right) W_g + \sum_{h \neq g} \left(T_{gh} \log \mu_{gh} \right) \right]$$

Data and Estimation: MLE for multiple states

- Hence the MLE for the transition intensity from state g to h, $\tilde{\mu}_{gh}$ is:

$$\tilde{\mu}_{gh} = \frac{T_{gh}}{W_g} = \frac{\text{No. transfers from } g \text{ to } h}{\text{Total waiting time in } g} \ .$$

- The marginal asymptotic distribution of $\tilde{\mu}_{gh}$ is

$$\tilde{\mu}_{gh} \approx \mathcal{N}\left(\mu_{gh_0}, I(\mu_{gh_0})^{-1}\right), \ I(\mu_{gh_0}) \approx I(\tilde{\mu}_{gh}) = \frac{\mathbb{E}(W_g)}{\tilde{\mu}_{gh}} \ .$$

- The joint asymptotic distribution for the vector of MLEs $\underline{\tilde{\mu}}$ is

$$\underline{\tilde{\mu}} \approx \mathcal{N}\left(\underline{\mu}_0, \underbrace{I(\underline{\mu}_0)^{-1}}_{\substack{\text{covariance} \\ \text{matrix}}}\right), \ I(\underline{\mu}_0) \approx I(\underline{\tilde{\mu}}) = \text{diag}\left(\frac{\mathbb{E}(W_g)}{\tilde{\mu}_{gh}}\right) \ .$$

Since $I(\mu_0)$ is a diagonal matrix, $\underline{\tilde{\mu}}$ are asymptotically independent (see miscellaneous comments for details).

Data and Estimation: MLE for multiple states

Miscellaneous Comments: Covariances of MLEs

- The separate sum nature of the log-likelihood leads to other transition intensities vanishing when differentiating with respect to one particular transition intensity, i.e.

$$U(\mu_{gh}) = \frac{\partial \ell(\mu_{gh})}{\partial \mu_{gh}}$$

is only a function of μ_{gh} and not other μ's. Thus for $(g,h) \neq (i,j)$,

$$\frac{\partial^2 \ell}{\partial \mu_{gh} \partial \mu_{ij}} = \frac{\partial U}{\partial \mu_{gh} \partial \mu_{ij}} = 0,$$

$$\Rightarrow \text{Cov}(\tilde{\mu}_{gh}, \tilde{\mu}_{ij}) = \mathbb{E}\left(- \frac{\partial^2 \ell}{\partial \mu_{gh} \partial \mu_{ij}} \right) = \mathbb{E}(0) = 0.$$

Thus the diagonal entries of the covariance matrix (i.e. covariances between different MLEs), and hence the information matrix I, are 0.

Data and Estimation: MLE for multiple states

Miscellaneous Comments: Multivariate Normal & Independence

- In general, zero covariance does not imply independence.
- However, in the specific case of multivariate Normal distributions, zero covariance does imply independence.
- That is, if $\underline{\tilde{\mu}} = (\tilde{\mu}_1, \tilde{\mu}_2, ..., \tilde{\mu}_p)' \sim \mathcal{N}\left(\underline{\mu}, I(\underline{\mu})^{-1}\right)$, and all off-diagonal entries of $I(\mu)$ are zero (hence all off-diagonal entries of $I(\mu)^{-1}$ — which represent the covariances between different μ;s — are zero), then $\mu_1, \mu_2, ..., \mu_p$ are independent.
- This means that, in Markov models with constant transition intensities, the MLEs of the transition intensities are asymptotically independent.

Data and Estimation: MLE for multiple states

Miscellaneous Comments: Multivariate Normal $\rightarrow$ Marginal Normal

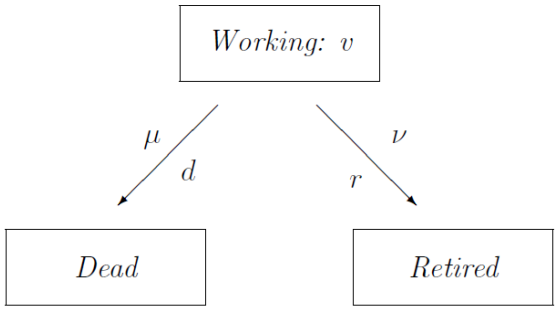
- The marginal distributions of a multivariate normal distribution are also normal, with mean and variance taken from the corresponding components. That is,

$$\underline{\tilde{\mu}} = \begin{pmatrix} \tilde{\mu}_1 \\ \vdots \\ \tilde{\mu}_p \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_1 \\ \vdots \\ \mu_p \end{pmatrix}, \begin{pmatrix} \mathrm{Cov}(\tilde{\mu}_1, \tilde{\mu}_1) & \cdots & \mathrm{Cov}(\tilde{\mu}_1, \tilde{\mu}_p) \\ \vdots & \ddots & \vdots \\ \mathrm{Cov}(\tilde{\mu}_p, \tilde{\mu}_1) & \cdots & \mathrm{Cov}(\tilde{\mu}_p, \tilde{\mu}_p) \end{pmatrix} \right)$$

$$\Rightarrow \begin{cases} \tilde{\mu}_1 \sim \mathcal{N}\Big(\mu_1, \mathrm{Cov}(\tilde{\mu}_1, \tilde{\mu}_1) = \mathrm{Var}(\tilde{\mu}_1)\Big), \\ \vdots \\ \tilde{\mu}_p \sim \mathcal{N}\Big(\mu_p, \mathrm{Cov}(\tilde{\mu}_p, \tilde{\mu}_p) = \mathrm{Var}(\tilde{\mu}_p)\Big) \end{cases}$$

Data and Estimation: MLE for multiple states

Miscellaneous Comments: Example



- Take the working-retired-dead model as example. The score vector, information matrix & covariance matrix are then:

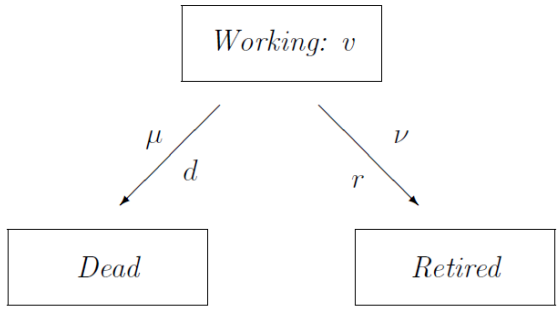
$$\underline{U}(\mu,\nu) = \begin{pmatrix} \frac{\partial \ell}{\partial \mu} \\ \frac{\partial \ell}{\partial \nu} \end{pmatrix} = \begin{pmatrix} -V + \frac{D}{\mu} \\ -V + \frac{R}{\nu} \end{pmatrix}$$

$$I(\mu,\nu) = -\mathbb{E}\begin{pmatrix} \frac{\partial \ell}{\partial \mu^2} & \frac{\partial \ell}{\partial \mu \partial \nu} \\ \frac{\partial \ell}{\partial \nu \partial \mu} & \frac{\partial \ell}{\partial \nu^2} \end{pmatrix} = \begin{pmatrix} \frac{\mathbb{E}(D)}{\mu^2} & 0 \\ 0 & \frac{\mathbb{E}(R)}{\nu^2} \end{pmatrix} = \begin{pmatrix} \frac{\mathbb{E}(V)}{\mu} & 0 \\ 0 & \frac{\mathbb{E}(V)}{\nu} \end{pmatrix}$$

$$\text{Cov}\begin{pmatrix} \tilde{\mu} \\ \tilde{\nu} \end{pmatrix} \equiv \begin{pmatrix} \text{Cov}(\tilde{\mu},\tilde{\mu}) & \text{Cov}(\tilde{\mu},\tilde{\nu}) \\ \text{Cov}(\tilde{\nu},\tilde{\mu}) & \text{Cov}(\tilde{\nu},\tilde{\nu}) \end{pmatrix} \equiv \begin{pmatrix} \text{Var}(\tilde{\mu}) & \text{Cov}(\tilde{\mu},\tilde{\nu}) \\ \text{Cov}(\tilde{\nu},\tilde{\mu}) & \text{Var}(\tilde{\nu}) \end{pmatrix}$$

$$= I(\mu_0,\nu_0)^{-1} = \begin{pmatrix} \frac{\mu_0}{\mathbb{E}(V)} & 0 \\ 0 & \frac{\nu_0}{\mathbb{E}(V)} \end{pmatrix}$$

Data and Estimation: MLE for multiple states



Miscellaneous Comments: Example

- Take the working-retired-dead model as example. We can therefore extract marginal variances of $\tilde{\mu}$ and $\tilde{\nu}$:

$$I(\mu_0, \nu_0)^{-1} = \begin{pmatrix} \frac{\mu_0}{\mathbb{E}(V)} & 0 \\ 0 & \frac{\nu_0}{\mathbb{E}(V)} \end{pmatrix} \Longrightarrow \begin{cases} \text{Var}(\tilde{\mu}) = \frac{\mu_0}{\mathbb{E}(V)}, \text{ and} \\ \text{Var}(\tilde{\nu}) = \frac{\nu_0}{\mathbb{E}(V)} \end{cases}$$

Chapter 4

Binomial and Poisson Models of Mortality

1. The Binomial Model
   - Introduction
   - Assumption 1: Uniform distribution of deaths
   - Assumption 2: Constant force of mortality
   - Likelihood & Estimation
2. The actuarial estimate of q_x: Heuristics & Derivation
3. The Poisson Model: Results & Comments

The Binomial Model: Introduction

- If we observe n lives aged x exactly for one year exactly, then the number of deaths, D follows a simple binomial distribution:

$$D \sim \mathcal{B}(n, q_x).$$

- Hence the likelihood is:

$$L(q_x; D, n) \propto q_x^D (1 - q_x)^{n-D} \ ,$$

which gives the ML estimator as

$$\tilde{q}_x = \frac{D}{n} \ , \text{ with } \mathbb{E}(\tilde{q}_x) = q_x, \ \text{Var}(\tilde{q}_x) = \frac{q_x(1 - q_x)}{n} \ .$$

- But the assumptions of exact ages & exact observation times may be too simplistic/restrictive.

The Binomial Model: Introduction

- Recall the following dataset from Chapter 3:

Observation began at age: $x + a_i$	Observation ended at age: $x + t_i$	Observation would have ended at age: $x + b_i$	Observation duration: v_i	Died? d_i
24.47	24.97	24.97	0.5	No
24.00	25.00	25.00	1.00	No
24.15	24.92	25.00	0.77	Yes
⋮	⋮	⋮	⋮	⋮
24.00	24.65	24.65	0.65	No

The Binomial Model: Introduction

- What if we didn't know the exact death/end times?:

Observation began at age: $x + a_i$		Observation would have ended at age: $x + b_i$		Died? d_i
24.47		24.97		No
24.00		25.00		No
24.15		25.00		Yes
⋮		⋮		⋮
24.00		24.65		No

The Binomial Model: Introduction

- We can extend the simple binomial model to allow for cases where we do not know the exact waiting time.
- For an observation period of $b_i - a_i$ starting from age $x + a_i$ on life i, the probability of death is ${}_{b_i-a_i}q_{x+a_i} =: q_i$. Thus, for each life i,

$$D_i \sim \mathcal{B}(1, q_i)$$

where q_i are potentially different $\forall i = 1, ..., n$. Then:

$$\mathbb{P}(D_i = d_i) = q_i^{d_i}(1 - q_i)^{1-d_i} \;\Rightarrow\; L(\underline{q}; \underline{D}) \propto \prod_{i=1}^{n} q_i^{D_i}(1 - q_i)^{1-D_i} \;.$$

- $\underline{q} = (q_1, ..., q_n)'$ is a vector of n potentially different components; estimations will be poor if we do not make some additional assumptions to smooth the estimates.

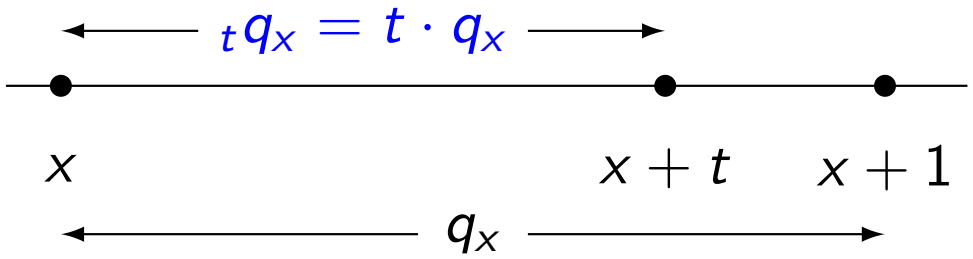
The Binomial Model: Introduction

2 main assumptions

- Uniform Distribution of Deaths (UDD)
- Constant Force of Mortality (CFM)

Additionally there is the Balducci Assumption (BA), but we only mention it briefly and do not explore further.

The Binomial Model: Assumption 1 - UDD



- Formally, uniform distribution of deaths (UDD) means we assume:

$$\mathbb{P}(\text{Die in } (x+t, x+t+dt) \mid \text{Die in } (x, x+1)) = dt + o(dt).$$

- Property 1:

$$_tq_x = t \cdot q_x \ , \ 0 \leq t \leq 1$$

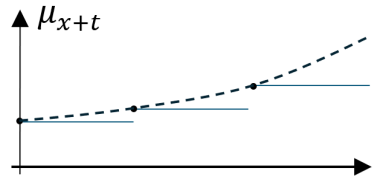
- Property 2:

$$q_i = {}_{b_i - a_i}q_{x+a_i} = \frac{(b_i - a_i)q_x}{1 - a_i \cdot q_x}$$

- Property 3:

$$\mu_{x+t} = \frac{q_x}{1 - t \cdot q_x}, \ 0 \leq t \leq 1$$

The Binomial Model: Assumption 2 - CFM



- Formally, constant force of mortality (CFM) means we assume:

$$\mu_{x+t} = \mu_x \ , \ 0 \leq t < 1,$$

by default; sometimes, we may also assume $\mu_{x+t} = \mu_{x+s}, 0 \leq t < 1$, for any $0 \leq s < 1$ (e.g. $s = 1/2$).

- Property 1:

$$_tq_x = 1 - \exp(-\mu_x t) \ , \ 0 \leq t < 1$$

- Property 2:

$$q_i = {}_{b_i - a_i}q_{x+a_i} = 1 - \exp\{-(b_i - a_i)\mu_x\}$$

- Property 3:

$$\mu_{x+t} = \mu_x = -\log(1 - q_x), \ 0 \leq t < 1$$

The Binomial Model: Likelihood & Estimation

- Hence, under UDD we reduce the likelihood from $L(\underline{q}; \underline{D})$ to

$$L(q_x; \underline{D}) \propto \prod_{i=1}^{n} \left[\frac{(b_i - a_i) q_x}{1 - a_i \cdot q_x} \right]^{D_i} \left[1 - \frac{(b_i - a_i) q_x}{1 - a_i \cdot q_x} \right]^{1 - D_i}$$

- And, under CFM we reduce the likelihood from $L(\underline{q}; \underline{D})$ to

$$L(\mu_x; \underline{D}) \propto \prod_{i=1}^{n} \left[1 - e^{-(b_i - a_i)\mu_x} \right]^{D_i} \left[e^{-(b_i - a_i)\mu_x} \right]^{1 - D_i}$$

- MLEs and their asymptotic distributions thus follow.
- One issue: CFM (& Markov) gives $\hat{\mu}_x$; how do we get $\hat{q}_x$ & $\text{Var}(\hat{q}_x)$? The Delta method: $\text{Var}\left[f(X) \right] \approx \left[f'\left(\mathbb{E}(X) \right) \right]^2 \text{Var}(X)$.

The actuarial estimate of q_x

- When observation ages x are not exact, and the observation periods are known but not exactly 1 year (similar to the case of Markov models), can we estimate q_x?
- Markov models offer estimates of μ, which gives $\hat{q}_x = 1 - e^{-\mu}$.
- But it is useful to have a more intuitive and direct estimate. Hence, actuaries often use the formula:

$$\hat{q}_x = \frac{d}{v + \frac{1}{2}d}$$

to estimate q_x. But why?

Initial & Central Exposed to Risk

- First, assume we are observing n lives over perfect one-year age intervals, from age x to $x+1$.
- If observation ages are exact at x and $x+1$, then each life is initially exposed to 1 year of risk of death.

  Hence the total number of years initially exposed to risk of death is $n \equiv E_x$, the "initial exposed to risk".
- At the end of the observation period, the lives that *have* died ended up being exposed to <1 year of risk of death.

  Hence the total number of years exposed to risk of death is the total waiting time $v \equiv E_x^c < E_x$, the "central exposed to risk".

Initial & Central Exposed to Risk

- However, in real-life datasets, the initial exposed to risk E_x is unknown, because of:
  - Conflicting definitions of age x,
  - Mismatch between exact age and age labels,
  - Other sources of inaccuracies in data.

  Sometimes even the central exposed to risk E_x^c needs to be adjusted/estimated! (See Chapter 8.)

The actuarial estimate of q_x - Heuristics

- To estimate death rates of lives between ages x and $x+1$, can we also have something like $\hat{q}_x = d/n$ even when datasets are not perfect?
- A natural idea would be to set $\hat{q}_x = \dfrac{d}{E_x}$ where E_x "replaces" n from the binomial model, to handle imperfections.
- But, as established, E_x are often unknown and need to be estimated.
- The actuarial estimate of q_x assumes deaths occur halfway at $x + \frac{1}{2}$, hence each death causes E_x^c to be $\frac{1}{2}$ year less than E_x. Thus, d deaths shave $\frac{1}{2}d$ off of E_x:

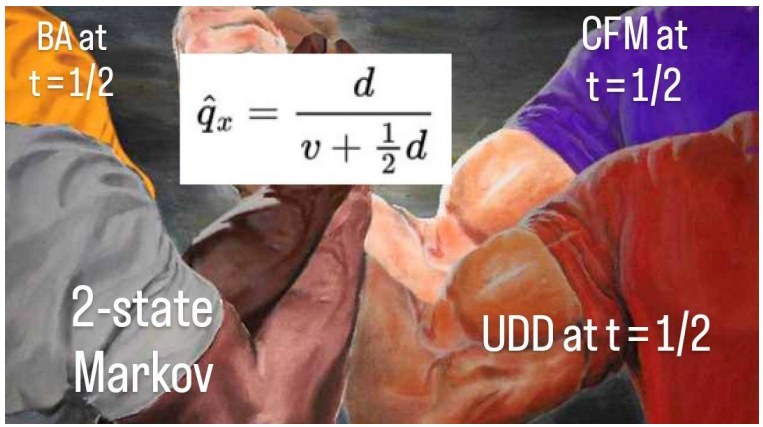
$$E_x = E_x^c + d \equiv v + \frac{1}{2}d \quad \Rightarrow \quad \hat{q}_x = \frac{d}{E_x} = \frac{d}{v + \frac{1}{2}d} \, .$$

The actuarial estimate of q_x - Mathematical Derivation

- We can, in fact, derive the actuarial estimate $\hat{q}_x = d/(v + \frac{1}{2}d)$ mathematically, using assumptions we've seen before.

1. CFM at $t = \frac{1}{2}$: Set $\mu_{x+t} = \mu_{x+\frac{1}{2}}$ for $0 \leq t < 1$.
2. 2-state Markov: $\hat{\mu}_{x+\frac{1}{2}} = \frac{d}{v}$.
3. UDD at $t = \frac{1}{2}$: $\mu_{x+\frac{1}{2}} = \frac{q_x}{1 - \frac{1}{2}q_x}$

   (BA at $t = \frac{1}{2}$ as well).



Combining all $\Rightarrow \frac{d}{v} = \frac{\hat{q}_x}{1 - \frac{1}{2}\hat{q}_x} \Rightarrow \hat{q}_x = \frac{d}{v + \frac{1}{2}d} \equiv \frac{d}{E_x}$.

The actuarial estimate of q_x - Comments

- $\{\text{Markov } \hat{\mu}_{x+\frac{1}{2}}\}$ & $\{\text{UDD } \mu_{x+\frac{1}{2}}\} \implies \{\text{actuarial } \hat{q}_x\}$,

  but note $\{\text{actuarial } \hat{q}_x\}$ & $\{\text{UDD } \mu_{x+\frac{1}{2}}\} \not\implies \{\text{Markov } \hat{\mu}_{x+\frac{1}{2}}\}$.

  The difference: $-\log(1-\hat{q}_x) - \dfrac{\hat{q}_x}{1-\frac{1}{2}\hat{q}_x} \approx \dfrac{1}{12}\hat{q}_x^3$ (very small).

- We can show using the Delta method (proof not required) that $\text{Var}(\hat{q}_x) \approx \dfrac{\hat{q}_x(1-\hat{q}_x)}{E_x}$, approximately matching results from the simple binomial model.

- If we have exact ages of entry and exit, then computing E_x is pointless and Markov model is simpler and better.

- If we have > 1 decrements then Markov model is simpler and better.

The Poisson Model

- Motivated by the Markov model: recall the likelihood

$$L(\mu; D, V) \propto e^{-\mu V} \mu^{D} = e^{-\mu \sum_{i=1}^{n} V_i} \; \mu^{\sum_{i=1}^{n} D_i}$$

- Compare this with a Poisson likelihood for λ in $X_i \sim \text{Poi}(\lambda)$:

$$L(\lambda; \underline{X}) \propto \prod_{i=1}^{n} e^{-\lambda} \lambda^{X_i} = e^{-\lambda n} \; \lambda^{\sum_{i=1}^{n} X_i}$$

- This inspires a Poisson model for mortality where μ is constant:

$$D|V \sim \mathcal{P}(\mu V) \equiv \mathcal{P}(\mu E_x^c)$$

$$\Rightarrow \tilde{\mu} = \frac{D}{V} \equiv \frac{D}{E_x^c}, \quad \mathbb{E}(\tilde{\mu}) = \frac{\mu E_x^c}{E_x^c} = \mu, \quad \text{Var}(\tilde{\mu}) = \frac{\mu E_x^c}{(E_x^c)^2} = \frac{\mu}{E_x^c}.$$

where $V \equiv E_x^c$ is the central exposed to risk.

The Poisson Model: Comments

- Note the similarity in results to Markov model!
- One issue: Poisson model results in $\mathbb{P}(D = d) > 0$ for $d > n$, which does not make sense! But in practice, if n is large and q_x is small, this is not an issue.
- An alternative view of the Poisson model:
  Constant transition intensity μ
  $\Rightarrow$ Waiting times are Exp(μ)
  $\Rightarrow$ Its counting process over 1 unit time is $\mathcal{P}(\mu)$
  $\Rightarrow$ Its counting process over E_x^c time is $\mathcal{P}(\mu E_x^c)$.

[See F79SP Section 5.9]

Chapter 4 Summary/Wrap-Up

- Binomial Model: $D_i \sim \mathcal{B}(1, {}_{b_i-a_i}q_{x+a_i})$.

  Simplifying assumptions to estimate q_x:

  - UDD: ${}_tq_x = tq_x$, ${}_{b_i-a_i}q_{x+a_i} = \dfrac{(b-a)q_x}{1-aq_x}$

  - CFM: $\mu_{x+t} = \mu$, ${}_{b_i-a_i}q_{x+a_i} = 1 - e^{-(b_i-a_i)\mu}$

- Actuarial Estimate: $\hat{q}_x = \dfrac{d}{v + \frac{1}{2}d} = \dfrac{d}{E_x}$

- Poisson Model: $D \sim \mathcal{P}(\mu E_x^c)$.

  To estimate μ, $\tilde{\mu} = \dfrac{D}{E_x^c}$ and properties follow.

Chapter 4 Summary/Wrap-Up

- If mortality rates/force of mortality/transition rates/transition intensities are small and the exposure is large, then there is little difference between Markov, Binomial and Poisson models.
- Markov approach allows for multiple decrements, Binomial and Poisson do not.

Introduction to Weighted Least Squares (WLS) Regression and Generalized Linear Models (GLMs)

(a pre-cursor to Chapter 5)

Introduction to WLS Regression & GLMs

- First, recall the linear regression model:

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2).$$

  or more succinctly

$$Y = \underline{\beta}^T \underline{X} + \epsilon, \quad \epsilon \sim \mathcal{N}(\underline{0}, \sigma^2).$$

- This is still a rather restrictive model for several reasons:
  - Homoscedasticity: Variance of Y, σ^2, is constant.
  - Y can only have linear relationships with $X_1, ..., X_p$.
  - Y can only be normally distributed.

Introduction to WLS Regression

- To combat heteroscedasticity (non-constant variance), we can use Weighted Least Squares (WLS) regression, instead of Ordinary Least Squares (OLS).
- Recall, in OLS, we found the coefficients $\beta_1, ..., \beta_p$ by minimizing the sum of squared errors/residuals:

$$\min_{\beta_1,...,\beta_p} \sum_{i=1}^{n} (y_i - \dot{y}_i)^2$$

where y_i are realization of y from the data, and $\dot{y}_i$ are fitted values from the linear regression model, i.e. $\dot{y}_i = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_p x_{pi}$.

Introduction to WLS Regression

- In WLS, to find $\beta_1, ..., \beta_p$, we minimize a weighted sum of squared errors/residuals instead:

$$\min_{\beta_1,...,\beta_p} \sum_{i=1}^{n} w_i (y_i - \dot{y}_i)^2$$

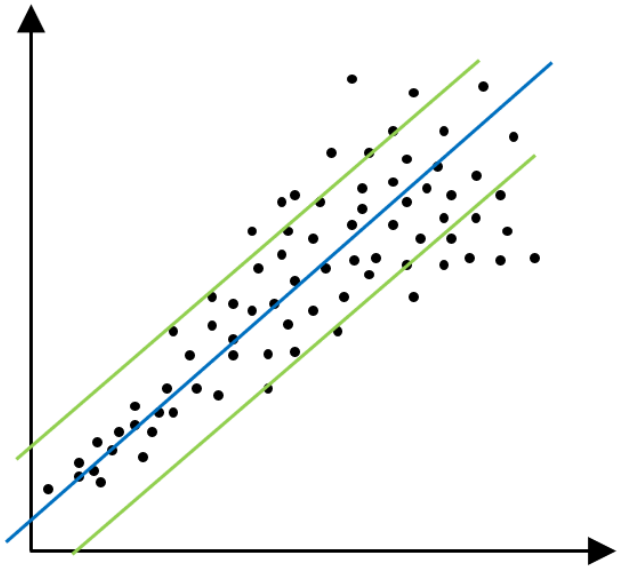
  where y_i and $\dot{y}_i$ are similar as before; w_i are weights given to these errors (informally, a measure of how "important" they are).

- Often, we set $w_i = \dfrac{1}{\text{Var}(y_i)}$, so that we minimize $\displaystyle\sum_{i=1}^{n} \frac{(y_i - \dot{y}_i)^2}{\text{Var}(y_i)}$.

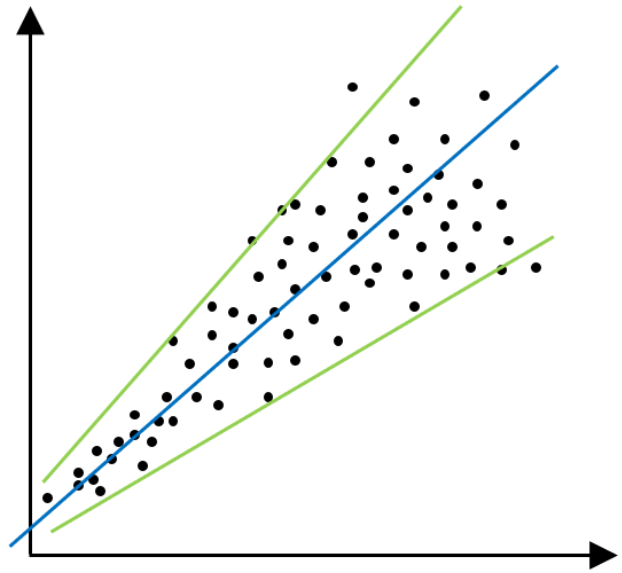
- By doing so, we assign more weight to observations with small variances, and less weight to observations with large variances.

Introduction to WLS Regression

- In heteroscedastic data, WLS is better than OLS since we don't "force" y_i with large variances to have a smaller overall variance.
- Similarly, we don't "force" y_i with small variances to have a larger overall variance.



OLS



WLS

Introduction to GLMs - Motivations

- We can rewrite the linear regression model as

$$\mathbb{E}(Y) = \beta_0 + \beta_1 X_1 + \cdots \beta_p X_p$$

noting that Y is normally distributed with fixed variance σ^2.

- Can we somehow modify this to allow for non-normal distributions for Y, and still maintain a relationship between $\mathbb{E}(Y)$ and $X_1, ..., X_p$?

Introduction to GLMs - Motivations

- Let's broaden the class of distributions that Y can take, to the exponential family of distributions, which have densities of the form

$$f_Y(y;\theta,\phi) = \exp\left\{\frac{y\theta - b(\theta)}{a(\phi)} + c(y,\phi)\right\}$$

for some parameterization θ, ϕ.

- Note that $\mathbb{E}(Y) = b'(\theta)$; $\mathrm{Var}(Y) = a(\phi)b''(\theta)$.
- Log-likelihood maximization with respect to θ is convenient.
- Therefore, it is natural to consider modelling

$$\theta = b'^{-1}(\mathbb{E}(Y)) := \beta_0 + \beta_1 X_1 + \cdots \beta_p X_p$$

$g(\cdot) := b'^{-1}(\cdot)$ is called the canonical link function.

Introduction to GLMs - Gaussian

- Let's consider a Normal model. What happens if we set $Y \sim \mathcal{N}(\mu, \sigma^2)$?

$$f_Y(y;\lambda) = \exp\left\{\frac{y\mu - \frac{1}{2}\mu^2}{\sigma^2} - \frac{1}{2}\log(2\pi\sigma^2) - \frac{y^2}{2\sigma^2}\right\}$$

  which suggests that $\theta = \mu$ and $b(\theta) = \frac{1}{2}\theta^2$.

- Therefore

$$b'(\theta) = \theta \Longrightarrow g(s) := b'^{-1}(s) = s$$

- This is the identity link function. We thus model

$$\mathbb{E}(Y) = \beta_0 + \beta_1 X_1 + \cdots \beta_p X_p$$

Introduction to GLMs - Poisson

- Let's consider a Poisson model. What happens if we set $Y \sim \mathcal{P}(\lambda)$?

$$f_Y(y;\lambda) = \exp\left\{\frac{y\log\lambda - e^{\log\lambda}}{1} - \log(y!)\right\}$$

  which suggests that $\theta = \log\lambda$ and $b(\theta) = e^{\theta}$.

- Therefore

$$b'(\theta) = e^{\theta} \Longrightarrow g(s) := b'^{-1}(s) = \log(s)$$

- This is the log link function. We thus model

$$\log(\mathbb{E}(Y)) = \beta_0 + \beta_1 X_1 + \cdots \beta_p X_p$$

Introduction to GLMs - Binomial

- Let's consider a Binomial model. What happens if we set $Y \sim \mathcal{B}(n, p)$?

$$f_Y(y;\lambda) = \exp\left\{ \frac{y \log\left(\frac{p}{1-p}\right) - (-n)\log(1-p)}{1} + \log\binom{n}{y} \right\}$$

  which suggests that $\theta = \log\left(\frac{p}{1-p}\right)$ and $b(\theta) = n\log(1+e^{\theta})$.

- Therefore

$$b'(\theta) = \frac{ne^{\theta}}{1+e^{\theta}} \Longrightarrow g(s) := b'^{-1}(s) = \log\left(\frac{\frac{s}{n}}{1-\frac{s}{n}}\right)$$

- This is the logit link function on s/n. We thus model

$$\log\left(\frac{\frac{\mathbb{E}(Y)}{n}}{1-\frac{\mathbb{E}(Y)}{n}}\right) = \beta_0 + \beta_1 X_1 + \cdots \beta_p X_p$$

Introduction to GLMs - Summary

Distribution of Y	Canonical link function
Normal, $\mathcal{N}(\mu, \sigma^2)$	Identity, $g(\mu) = \mu$
Poisson, $\mathcal{P}(\lambda)$	Log, $g(\lambda) = \log(\lambda)$
Binomial, $\mathcal{B}(n, q)$	Logit, $g(q) = \log\left(\frac{q}{1-q}\right)$

Introduction to GLMs - Summary

- Essentially, GLMs extend simpler linear regression models by:
  - Extending the types of distributions Y can take to the exponential family of distributions;
  - Allowing for link functions that elicit non-linear relationships between Y and $X_1, ...X_p$.
- In Chapter 5, we will apply GLMs to graduate/smooth estimates of μ_x ($\mathcal{N}$), d_x ($\mathcal{P}$) and q_x ($\mathcal{B}$).

Chapter 5

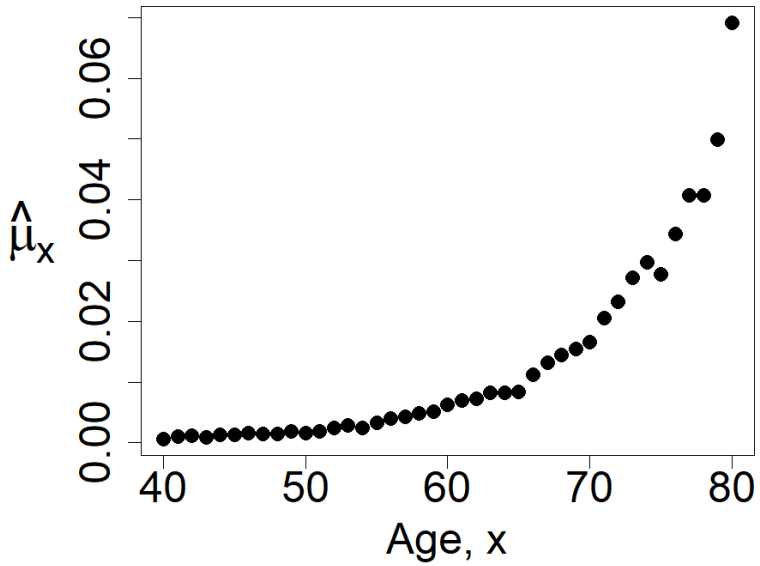
Methods of Graduation

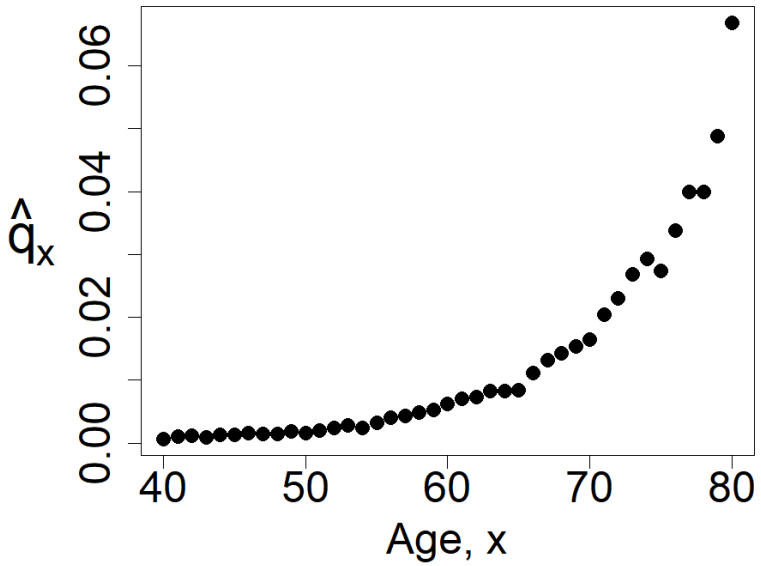
1. Gompertz Law
   - Gompertz law with normal, Poisson and Binomial errors
2. Residuals
   - Residuals with normal, Poisson and Binomial errors
3. Graduation by reference to a standard table
4. Graduation by mathematical formula
5. The problem of duplicates

Introduction

- Consider the CMI dataset, which provides deaths and exposures by age and by year.
- Fixing a year (say, 2000), we can make estimations using previous models for each age $x = 40, 41, ..., 80$:

$$\hat{\mu}_x = \frac{d_x}{E_x^c} \qquad \hat{q}_x = \frac{d_x}{E_x} = \frac{d_x}{E_x^c + \frac{1}{2} d_x}$$

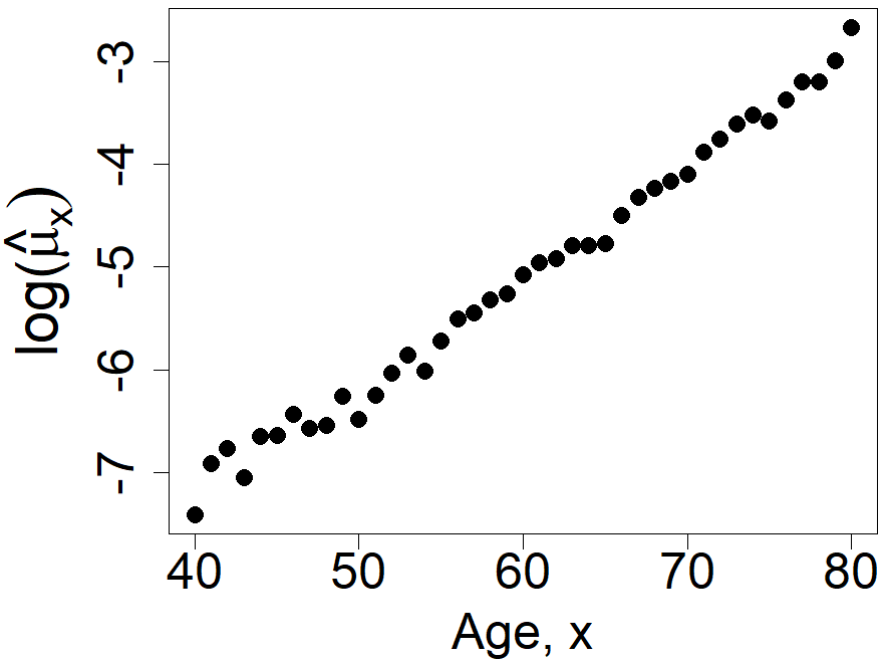




Gompertz Law

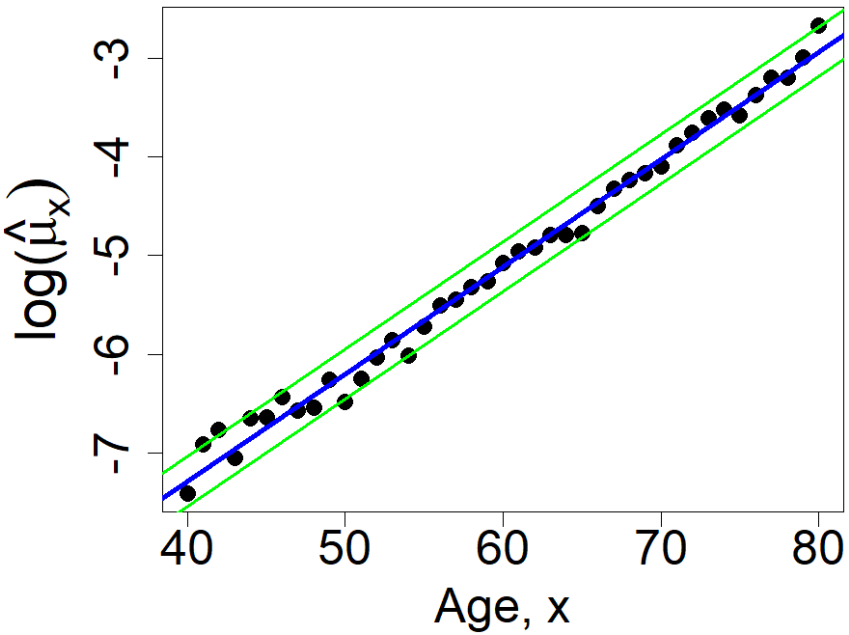
- Can we smooth or graduate the estimates above?
- The Gompertz Law says that $\log \mu_x$ is linear in age x:

$$\log \mu_x = a + bx$$



Gompertz Law with normal errors (OLS)

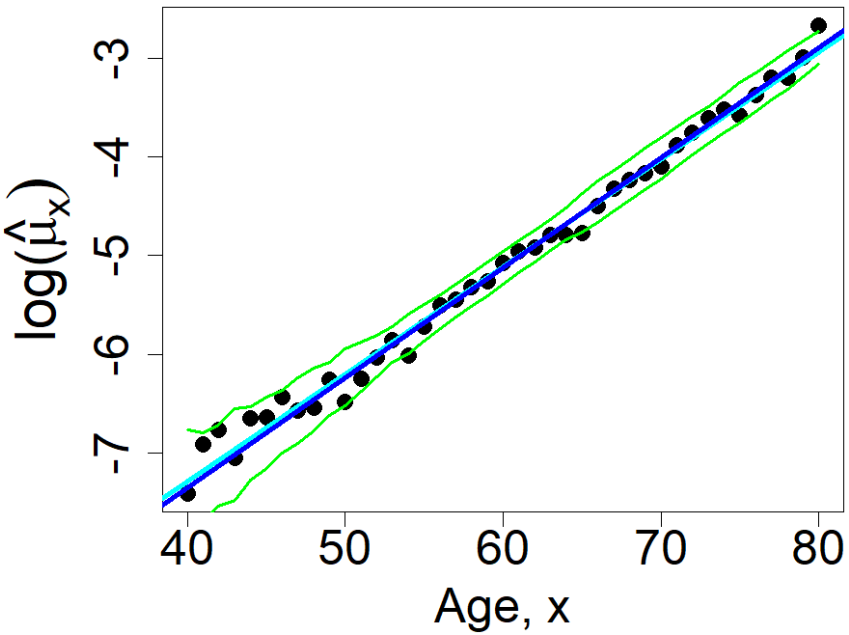
- Model: $\mathbb{E}\Big[\log(\hat{\mu}_x)\Big] = a + bx$, with $\log(\hat{\mu}_x) \sim \mathcal{N}(a + bx, \sigma^2)$
- Code: `Gomp.fit <- lm(Obs ~ AGE)`



$$\log(\hat{\mu}_x) = -11.629 + 0.109x$$

Gompertz Law with normal errors (WLS)

- Model: $\mathbb{E}\left[\overbrace{\log(\hat{\mu}_x)}^{\text{response}}\right] = a + bx$, with $\log(\hat{\mu}_x) \sim \mathcal{N}\left(a + bx, \frac{\sigma^2}{d_x}\right)$
- Code: `Gomp.fit.w <- lm(Obs ~ AGE, weights = DTH)`



$$\log(\hat{\mu}_x) = -11.806 + 0.111x$$

Gompertz Law with normal errors: What now?

- Using the graduated rates, we can now calculate death and survival probabilities, e.g.:

$$
\begin{aligned}
{}_tp_x &= \exp\left\{ -\int_0^t \exp\left\{\dot{\log}(\hat{\mu}_{x+s})\right\}\, ds \right\} \\
&= \exp\left\{ -\int_0^t e^{-11.806+0.111(x+s)}\, ds \right\}
\end{aligned}
$$

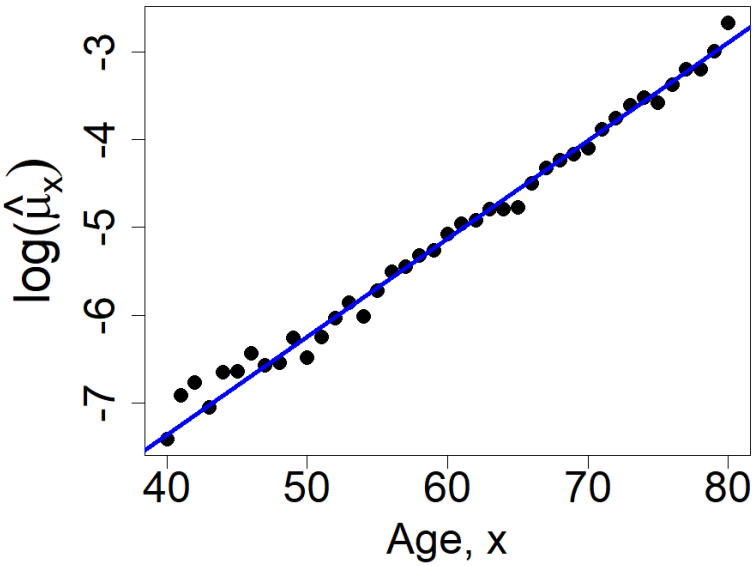
- We can also estimate the number of deaths at various ages, $\dot{d}_x$ with

$$
\dot{d}_x = \exp\left\{\dot{\log}(\hat{\mu}_x) + \log E_x^c\right\}
$$

and many other things.

Gompertz Law with Poisson errors

- Model: $\log\left[\mathbb{E}(\overbrace{D_x}^{\text{response}})\right] = \log(E_x^c) + a + bx$, with $D_x \sim \mathcal{P}(E_x^c \mu_x)$
- Code: `Gomp.glm <- glm(DTH ~ offset(log(EXP)) + AGE, family = poisson)`



$$\log(\hat{\mu}_x) = -11.824 + 0.112x$$

Gompertz Law with Poisson errors: What now?

- Using the graduated rates, we can now calculate death and survival probabilities, e.g.:

$$
\begin{aligned}
{}_tp_x &= \exp\left\{ -\int_0^t \frac{d_{x+s}}{E^c_{x+s}}\, ds \right\} \\
&= \exp\left\{ -\int_0^t e^{-11.824+0.112(x+s)}\, ds \right\}
\end{aligned}
$$

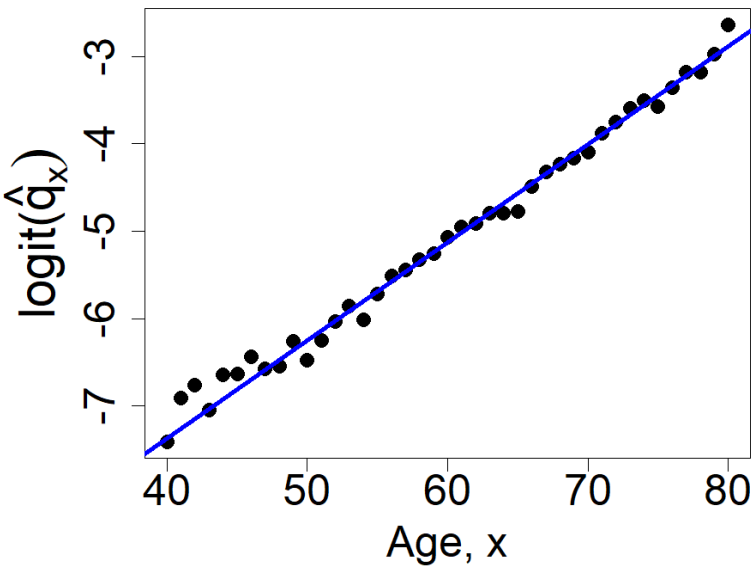
- We can also estimate the number of deaths d_x at various ages directly with the fitted value for the response variable, $\dot{d}_x$.

Gompertz Law with Binomial errors

- Model:

$$\text{logit}(\overbrace{q_x}^{\text{response}}) \equiv \text{logit}\left[\mathbb{E}\left(\frac{D_x}{E_x}\right)\right] = a + bx, \text{ with } D_x \sim \mathcal{B}(E_x, q_x)$$

- Code: `Gomp.bin <- glm(DTH/E.init ~ AGE, weights = E.init, family = binomial)`



$$\text{logit}(\hat{q}_x) = -11.858 + 0.112x$$

Gompertz Law with Binomial errors: What now?

- Using the graduated rates, we can now calculate death and survival probabilities, e.g.:

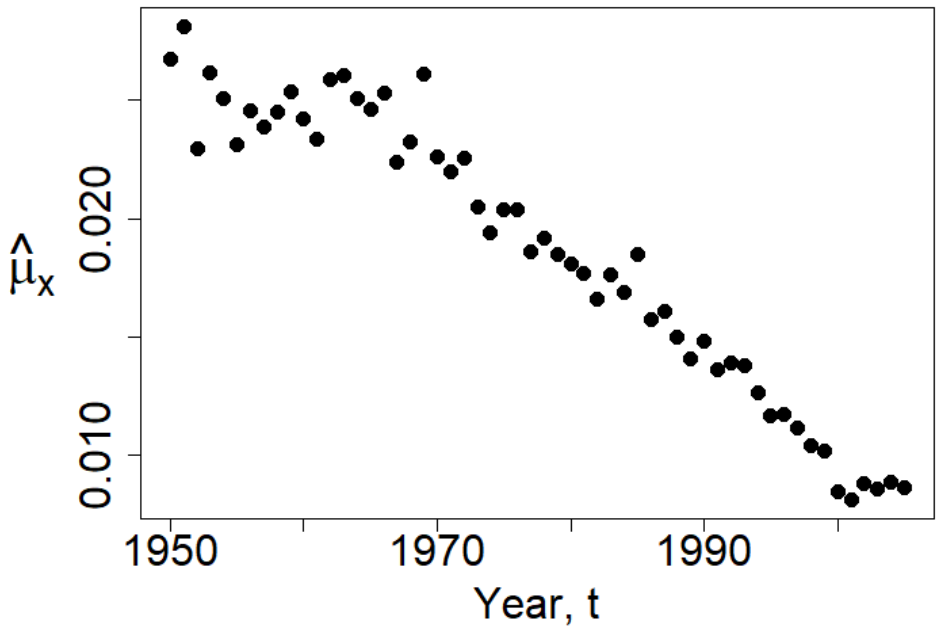
$$ {}_np_x = (1 - \dot{q}_x)(1 - \dot{q}_{x+1})(1 - \dot{q}_{x+2}) \cdots (1 - \dot{q}_{x+n-1}) $$

  or even obtain an estimate of μ_x and calculate probabilities with those.

- We can also estimate the number of deaths d_x at various ages with $\dot{d}_x = E_x \dot{q}_x$.

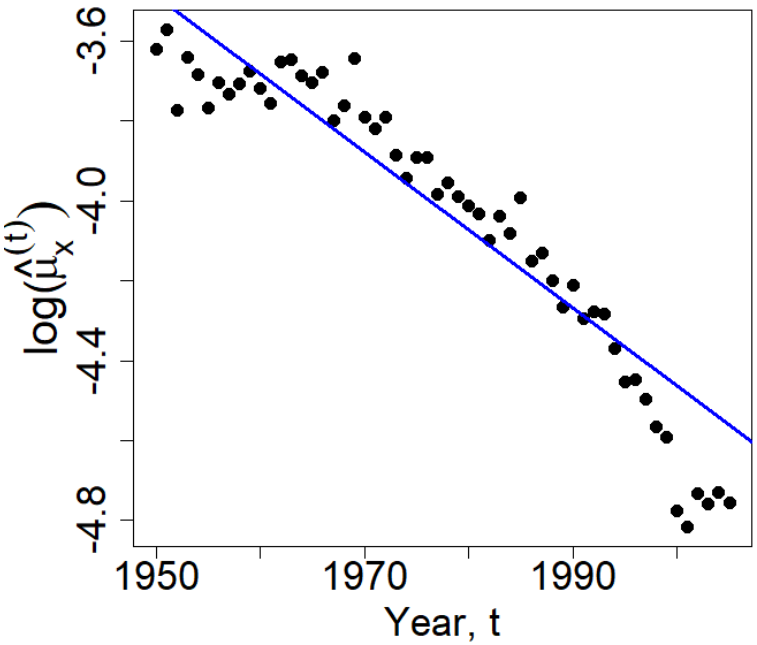
Gompertz models in *time*

- We can also fix an age, say 65, and model mortality over the years $t = 1950, 1951, ..., 2005$.

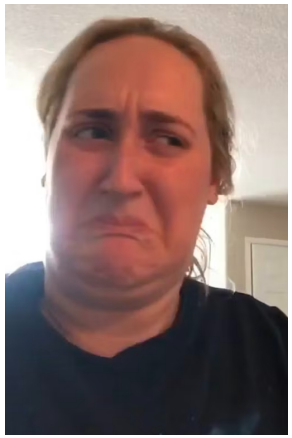


Gompertz-style models in time with Poisson errors

- Model: $\log\left[\mathbb{E}(D_x^{(t)})\right] = \log\left(E_x^{c(t)}\right) + a + bt$,
  with $D_x^{(t)} \sim \mathcal{P}(E_x^{c(t)}\mu_x^{(t)})$
- Code: `Gomp.bin <- glm(DTH ~ offset(log(EXP)) + Year, family = poisson)`

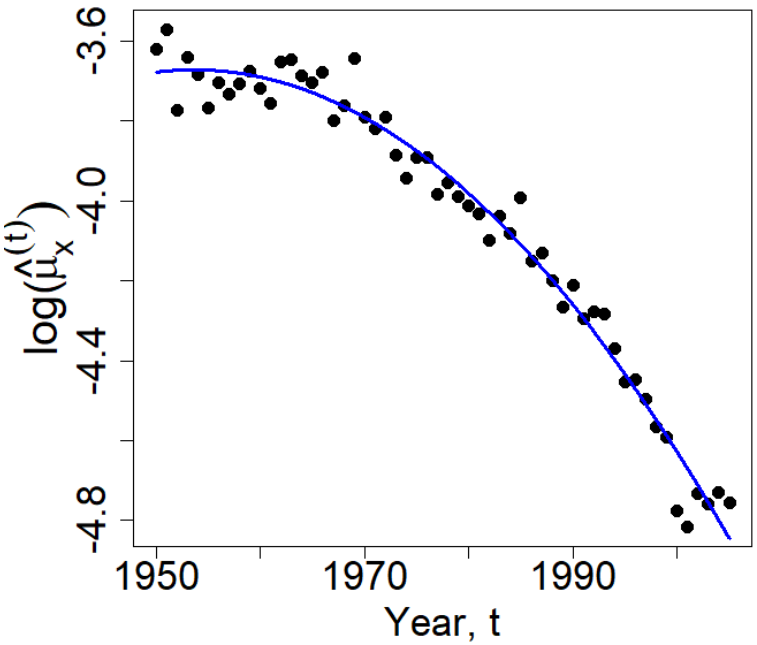


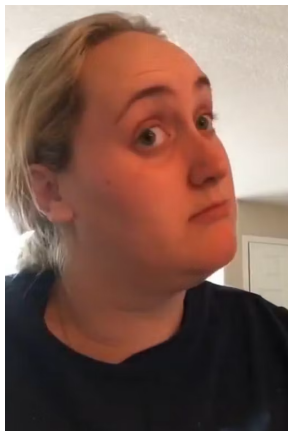
$$\log(\hat{\mu}_x^{(t)}) = 34.573 - 0.0195t$$



Gompertz-style models in time with Poisson errors

- Model: $\log\left[\mathbb{E}(D_x^{(t)})\right] = \log\left(E_x^{c(t)}\right) + a + bt + c(t-\bar{t})^2$,
  with $D_x^{(t)} \sim \mathcal{P}(E_x^{c(t)}\mu_x^{(t)})$
- Code: `Gomp.bin <- glm(DTH ~ offset(log(EXP)) + Year + Year2, family = poisson)`





$$\log(\hat{\mu}_x^{(t)}) = 38.025 - 0.0212t - 0.000445(t-\bar{t})^2$$

Residuals

- Residuals measure differences between the observed data and fitted values.
- Naturally, residuals are a good tool to investigate the goodness-of-fit of our models.
- The distribution of residuals should match our expectations.

Residuals with normal errors (OLS)

- Recall $\log(\hat{\mu}_x) \sim \mathcal{N}(a + bx, \sigma^2)$, rewrite as $y_i \sim \mathcal{N}(\dot{y}_i, \sigma^2)$.
- Then we would expect

$$\frac{y_i - \dot{y}_i}{\sigma} \sim \mathcal{N}(0, 1).$$

But σ^2 is unknown; we need to estimate it:

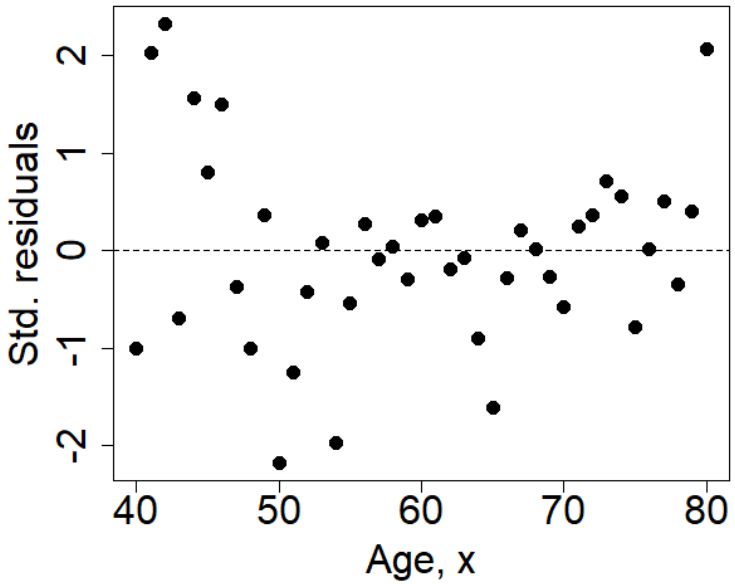
$$s^2 = \frac{\sum_{i=1}^{n} (y_i - \dot{y}_i)^2}{n - \mathrm{df}}.$$

- Therefore we define the standardized residual for the ith point:

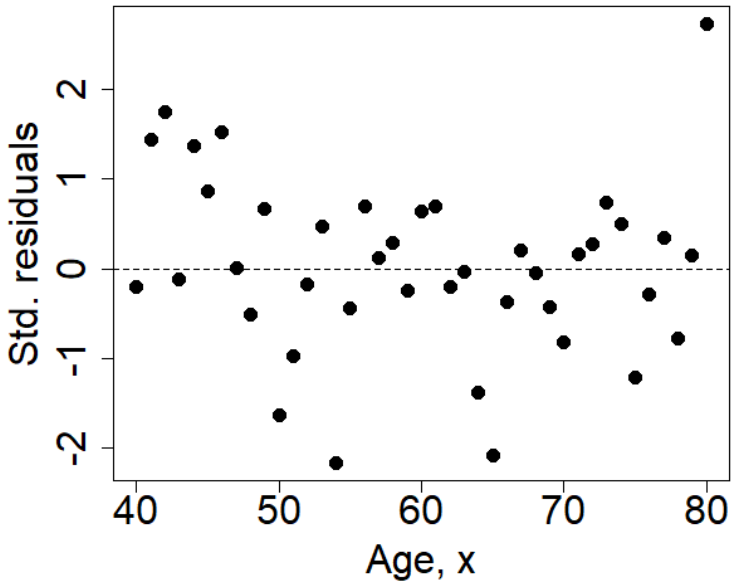
$$z_i := \frac{y_i - \dot{y}_i}{s}$$

and expect that $\{z_i\}_{i=1}^{n}$ to be approximately $\mathcal{N}(0, 1)$.

Residuals with normal errors



OLS



WLS

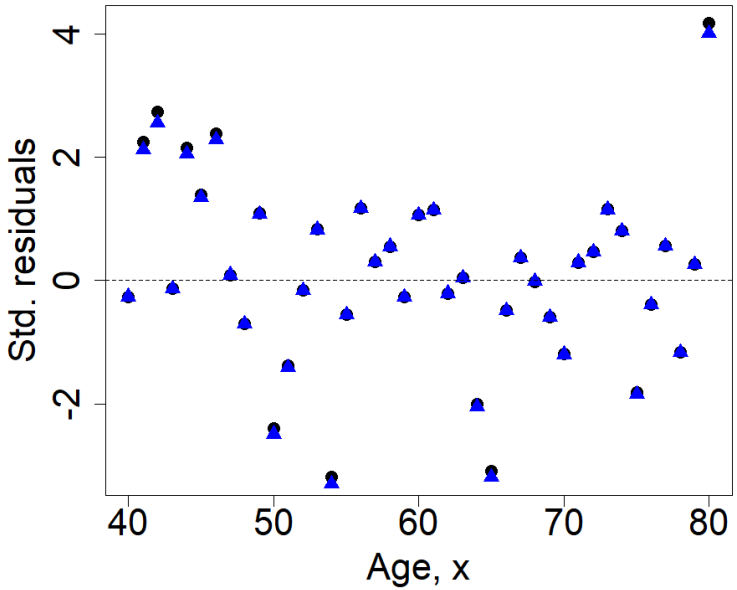
Residuals with Poisson errors

- Recall $\log\left[\mathbb{E}(D_x)\right] = \log(E_x^c) + a + bx$ with $D_x \sim \mathcal{P}(E_x^c \mu_x)$
  $\Rightarrow$ expect raw residuals $(d_i - \dot{d}_i)$ to be skewed/non-symmetric.
- Pearson residuals: `resid(Gomp.glm, type = "pear")`

$$z_i = \frac{d_i - \dot{d}_i}{\sqrt{\text{Var}(d_i)}} \approx \frac{d_i - \dot{d}_i}{\sqrt{\dot{d}_i}}$$

  since $\text{Var}(d_i) = E_i^c \mu_i \approx \dot{d}_i$.
- Deviance residuals: `resid(Gomp.glm)`
- Deviance residuals are preferred because they take account of the lack of symmetry;
- We expect no obvious patterns.

Residuals with Poisson errors



- Notice the difference between Pearson (black circle) and deviance (blue triangle) residuals: positive deviance residuals are closer to 0 to account for the lack of symmetry/skewness in the distribution of D_x.
- There are a few large residuals, possibly due to the problem of duplicates.

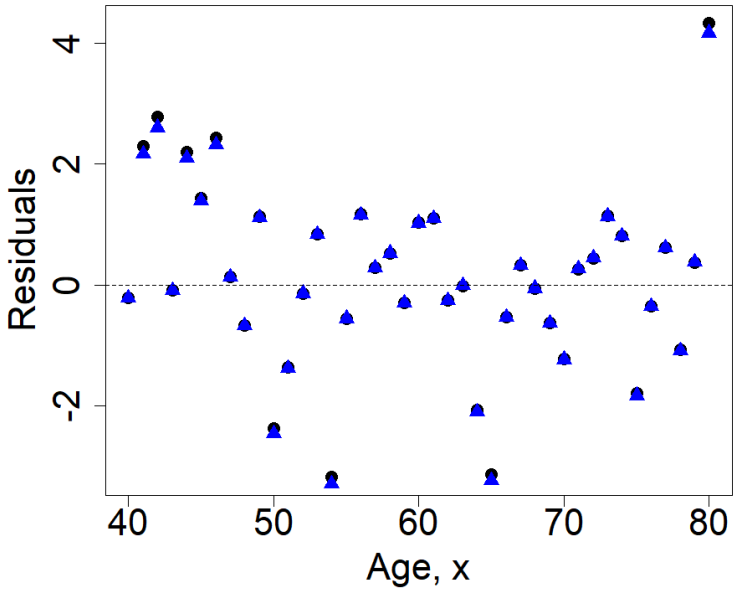
Residuals with Binomial errors

- Recall $\text{logit}(\hat{q}_x) = a + bx$ with $D_x \sim \mathcal{B}(E_x, q_x)$
  $\Rightarrow$ expect raw residuals $(q_i - \dot{q}_i)$ to be skewed/non-symmetric too.
- Pearson residuals: `resid(Gomp.bin, type = "pear")`

$$z_i = \frac{q_i - \dot{q}_i}{\sqrt{\text{Var}(q_i)}} \approx \frac{d_i - \dot{d}_i}{\sqrt{\dot{d}_i}}$$

  since $q_i = \frac{d_i}{E_i}$, $\dot{q}_i = \frac{\dot{d}_i}{E_i}$ and $\text{Var}(q_i) \approx \frac{\dot{q}_i(1-\dot{q}_i)}{E_i}$.
- Deviance residuals: `resid(Gomp.glm)`
- Deviance residuals are preferred because they take account of the lack of symmetry;
- We expect no obvious patterns.

Residuals with Binomial errors



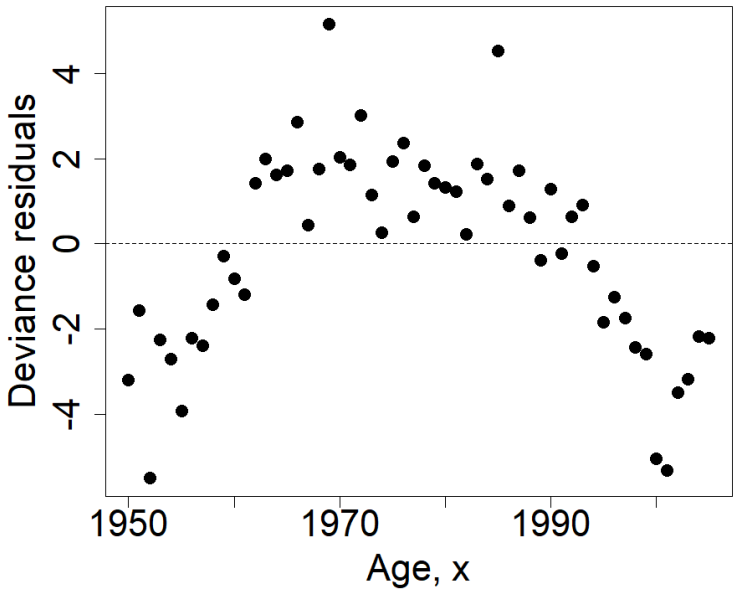
- Similar comments: No obvious patterns so Gompertz model for the mean seems okay; deviance residuals account for lack of symmetry, and a few large residuals possibly due to problem of duplicates.

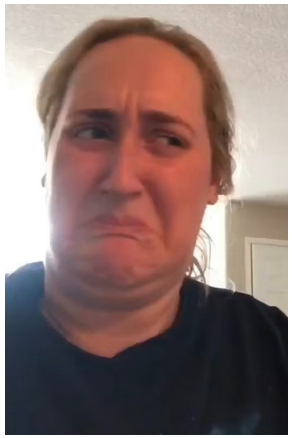
Example of non-ideal residuals

- Recall the Gompertz model in time with Poisson errors:

$$\log\left[\mathbb{E}(D_x^{(t)})\right] = \log\left(E_x^{c(t)}\right) + a + bt, \quad \text{with } D_x^{(t)} \sim \mathcal{P}(E_x^{c(t)}\mu_x^{(t)})$$

- Residuals exhibit an obvious pattern (quadratic-like?) that the model did not capture!





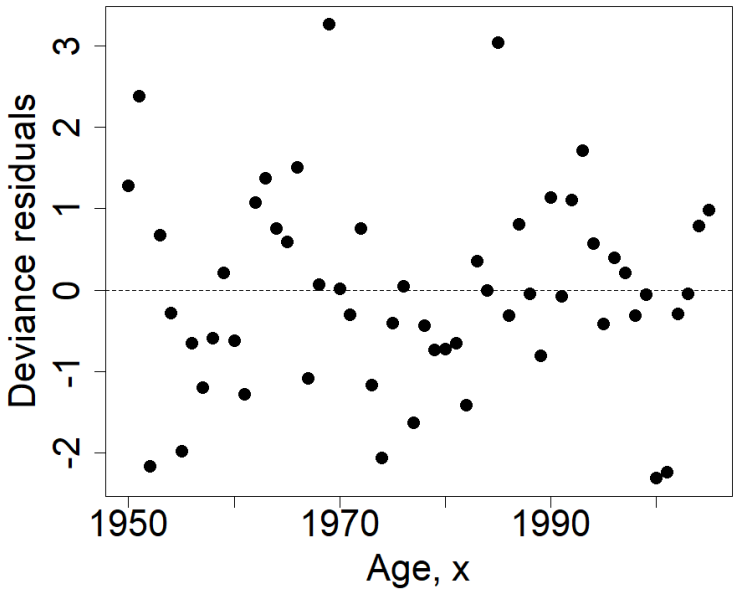
Example of non-ideal residuals: Fixing the model

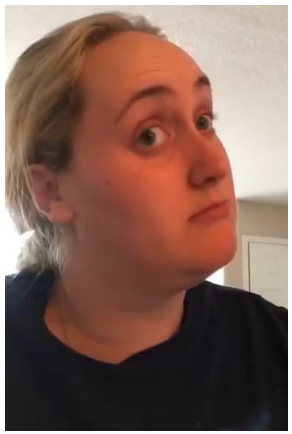
- Residuals seem to exhibit quadratic pattern. Try:

  $$\log\left[\mathbb{E}(D_x^{(t)})\right] = \log\left(E_x^{c(t)}\right) + a + bt + c(t-\bar{t})^2,$$

  with $D_x^{(t)} \sim \mathcal{P}(E_x^{c(t)}\mu_x^{(t)})$

- Now there are no more obvious patterns - seems good.





A summary of model outputs

- From the output directly (e.g. `Gomp.fit$___` or `Gomp.glm$___`):
  - `coef`: returns coefficients of the model, i.e. a and b (and others).
  - `fit` or `fitted.values`: returns fitted values for the response variable.

    In Gompertz law with normal errors, this is $\log(\hat{\mu}_i)$.
    In Gompertz law with Poisson errors, this is d_i.
    In Gompertz law with Binomial errors, this is $\dot{q}_i$.
  - `resid` or `residuals`: returns raw residuals on the response variable.

    In Gompertz law with normal errors, this is $\log(\hat{\mu}_i) - \log(\hat{\mu}_i)$.
    In Gompertz law with Poisson errors, this is $d_i - d_i$.
    In Gompertz law with Binomial errors, this is $\hat{q}_i - \dot{q}_i$.
  - `lin` or `linear.predictors`: returns the linear predictor.

    In Gompertz law with Poisson errors, this is $\log(E_x^c) + a + bx$.
    In Gompertz law with Binomial errors, this is $a + bx$.

A summary of model ...summaries

- From the summary of a linear regression model (i.e. `summary(Gomp.fit)$___`):
  - `resid` or `residuals`: returns raw residuals on the response variable.

    In Gompertz law with normal errors only; this is $\log(\hat{\mu}_i) - \log(\hat{\mu}_i)$.
  - `sig` or `sigma`: returns an estimate for σ^2.

    In Gompertz law with normal errors (OLS), this is $s^2 = \sum(y_i - \dot{y}_i)^2/(n - \text{df})$.
    In Gompertz law with normal errors (WLS), this is $s^2 = \sum d_i(y_i - \dot{y}_i)/(n - \text{df})$.
- From the summary of a generalized linear model/GLM (e.g. `summary(Gomp.glm)$___`):
  - `deviance.resid`: returns deviance residuals on the response variable.

    This can also be obtained with `resid(Gomp.glm)`;
    to obtain Pearson residuals, use `resid(Gomp.glm, type = "pear")`.

Graduation by standard table

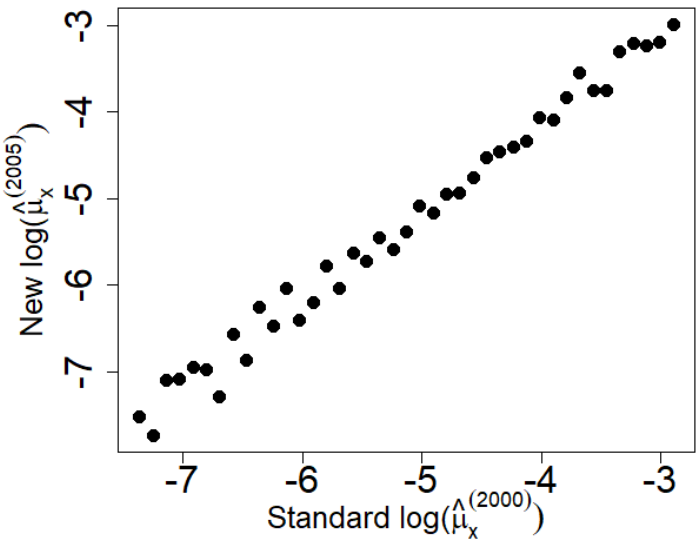
- We can use an established graduation, or a standard table, as a reference to graduate a new dataset/second experience.
- This is appropriate if:
  - The new dataset has relatively little data
  - The standard table uses a simple, smooth graduation (so that our new graduation is also smooth).
- Define:
  - $\{\mu_x^s\}$ = graduated mortality rates from the standard table,
  - $\{\hat{\mu}_x\}$ = ungraduated rates from the new dataset.

Graduation by standard table: Steps

1. Choose an appropriate standard table based on the population of interest.
2. Consider the features and selection characteristics of the standard table.
3. Determine a suitable relationship between the ungraduated rates $\hat{\mu}_x$ and the graduated standard table rates μ^s_x via plots.
   - $\hat{\mu}_x$ against μ^s_x; if this is linear then fit $\hat{\mu}_x = a\mu^s_x + b$.
   - $\hat{\mu}_x/\mu^s_x$ against x; if this is linear then fit $\hat{\mu}_x = (ax + b)\mu^s_x$.
   - $\log(\hat{\mu}_x)$ against $\log(\mu^s_x)$; if this is linear then fit $\log(\hat{\mu}_x) = a + b\log(\mu^s_x)$.
   - $\hat{\mu}_x$ against μ^s_{x+k}; if this seems equal then apply age-rating.
   - Any other transformations (of μ_x or q_x) and their corresponding relationships.

Graduation by standard table: Example

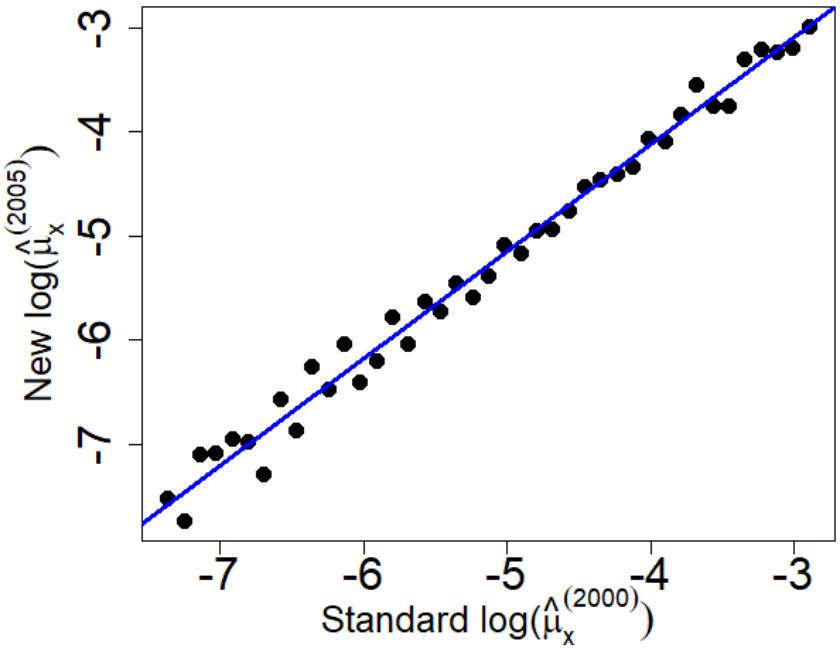
1. Choose standard table: Take Poisson-GLM graduated rates $\log(\mu_x^s)$ from 2000 as standard table, and use this to graduate raw rates $\log(\hat{\mu}_x)$ from 2005.
2. Similar populations and selection characteristics; no extra considerations needed.
3. Rates from both years should approximately obey Gompertz law. Hence we check the plot of $\log \hat{\mu}_x$ against $\log(\mu_x^s)$:



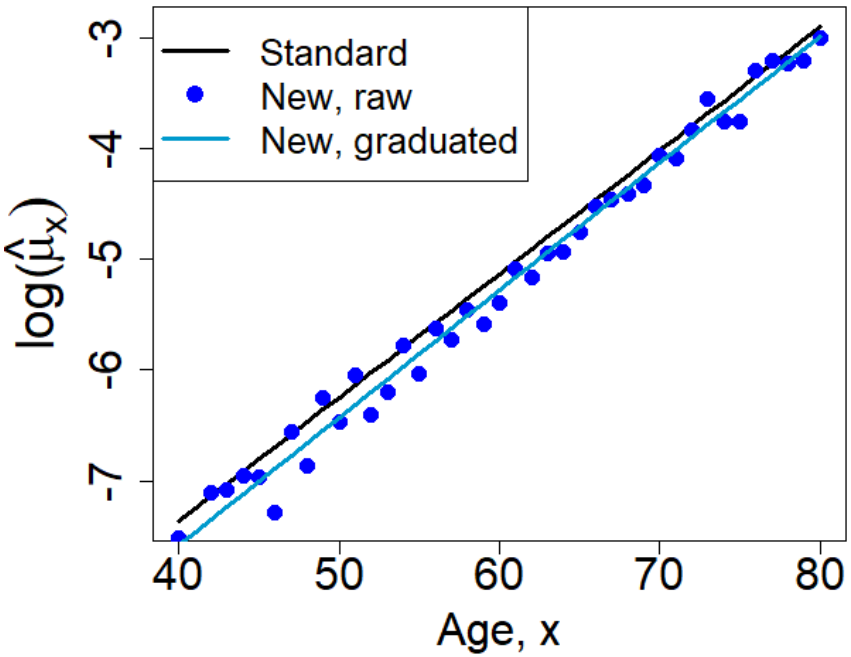
$$\text{Linear} \implies \log(\hat{\mu}_x) = a + b \log(\mu_x^s).$$

Graduation by standard table: WLS Example

- Model: $\mathbb{E}(\overbrace{\log(\hat{\mu}_x)}^{\text{response}}) = a + b\,\overbrace{\log(\mu_x^s)}^{\text{indep. var.}}$, with normal errors.



$\dot{\log}(\hat{\mu}_x) = 0.00889 + 1.031\log(\mu_x^s)$

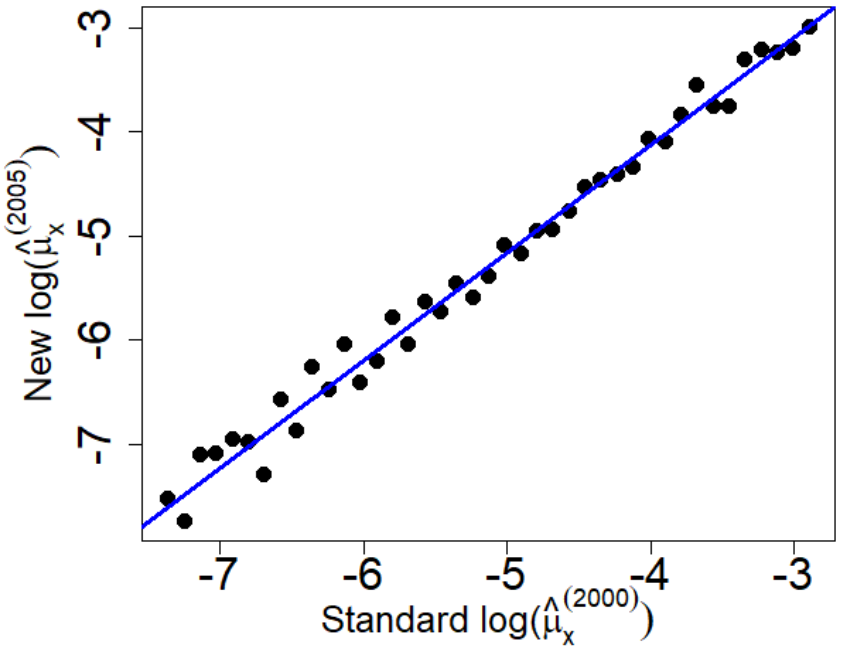


$\dot{\log}(\hat{\mu}_x) = -11.824 + 0.112x$

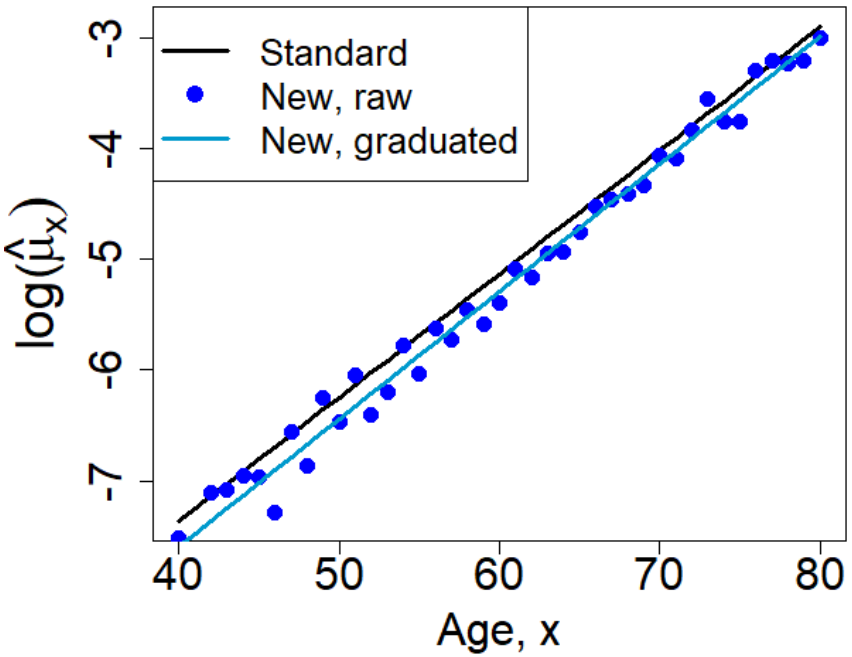
$\dot{\log}(\hat{\mu}_x) = -12.180 + 0.115x$

Graduation by standard table: Poisson GLM Example

- Model: $\log(\mathbb{E}(\overbrace{D_x}^{\text{response}})) = \log(E_x^c) + a + b\,\overbrace{\log(\mu_x^s)}^{\text{indep. var.}}$, with $D_x \sim \mathcal{P}(E_x^c \mu_x)$.



$$\log(\hat{\mu}_x) = 0.0149 + 1.034 \log(\mu_x^s)$$



$$\log(\hat{\mu}_x) = -11.824 + 0.112x$$
$$\log(\hat{\mu}_x) = -12.211 + 0.115x$$

Graduation by standard table: Pros and Cons

Advantages	Disadvantages
Uses information from graduation of more extensive, similar data	Not always obvious which standard table to use
Can be used with little data especially at extreme ages	May not adhere to data very well
Graduated rates will be smooth since standard rates are smoothed	Features in the standard table show in new graduated rates
Can incorporate prior expectations of graduation curve	Not appropriate for producing new standard tables

Graduation by Mathematical Formula

- Mortality rates (μ_x and q_x) may also be graduated using parametric formulae.
- This is the preferred method for large data sets.
- Finding a single formula that adheres well to the data can be difficult.
  - Gompertz: $\mu_x = Bc^x$, or $\log \mu_x = a + bx$
  - Makeham: $\mu_x = A + Bc^x$
  - Perks: $\mu_x = \dfrac{A + Bc^x}{Ec^{-x} + 1 + Dc^x}$, $\mu_x = \dfrac{A + Bc^x}{1 + Dc^x}$
  - Aus. National 1960/62: $\dfrac{q_x}{p_x} = \underbrace{A^{(x+b)^c}}_{\text{Infant mort.}} + \underbrace{De^{-E(\log(x)-\log(F))^2}}_{\text{Accident Hump}} + \underbrace{GH^x}_{\text{Aging}}$
- Splines (see Chapter 7) are used more widely by actuaries recently.

Graduation by Mathematical Formula

- We might want to consider a family of formulae we can use. Let's consider a function

$$\alpha_1 + \alpha_2 x + \alpha_3 x^2 + \cdots + \alpha_r x^{r-1} + \exp\left\{\alpha_{r+1} + \alpha_{r+2} x + \alpha_{r+3} x^2 + \cdots + \alpha_{r+s} x^{s-1}\right\}$$

- We call this the Gompertz-Makeham family, packaged as

$$GM_{\underline{\alpha}}^{r,s}(x) = \sum_{i=1}^{r} \alpha_i x^{i-1} + \exp\left\{\sum_{i=1}^{s} \alpha_{r+1} x^{i-1}\right\}$$

  - Gompertz's formula is $GM_{\underline{\alpha}}^{0,2}(x)$.
  - Makeham's formula is $GM_{\underline{\alpha}}^{1,2}(x)$.
  - Note $\text{logit}(q_x) = \beta_0 + \beta_1 x + \cdots \beta_n x^{n-1} \iff q_x = \dfrac{GM_{\underline{\alpha}}^{0,n}(x)}{1 + GM_{\underline{\alpha}}^{0,n}(x)}$.

Graduation by Mathematical Formula: Model Selection

- A larger model with more parameters will always adhere better to the data; where do we stop? How do we tell whether a smaller model is more "efficient"?
- Consider a model M_0 with n parameters (e.g. $GM_{\underline{\alpha}}^{n,0}(x)$), and a model M_1 with an additional k parameters (e.g. $GM_{\underline{\alpha}}^{n,k}(x)$), i.e. M_0 is nested in M_1.
- If M_0 has a likelihood of L_0 and M_1 has a likelihood of L_1 at their respective MLEs, then

$$-2\log\Lambda = -2\log\frac{L_0}{L_1} \sim \chi^2_k.$$

With this, we can carry out hypothesis tests to determine whether the k additional parameters are statistically significant.

The Problem of Duplicates

- Thus far, we've taken for granted that
  - E_x^c = number of person-years observed
  - d_x = number of deaths

  but in truth, many datasets are based on policies, not lives. Hence,
  - E_x^c = number of policy-years observed
  - d_x = number of policy claims due to death
- Essentially, claims are not independent due to 1 death possibly resulting in $\geq$1 claims.
- How does this affect the variability of the data?

The Problem of Duplicates: Toy Example

- Consider the following scenarios:
  - a) Scenario 1: 2 policies held by 2 identical, independent lives.
  - b) Scenario 2: 2 policies held by the same life.
- Define $C :=$ number of claims out of the 2 policies, and $q :=$death probability (resulting in claim(s)). Then
  - a) $C \sim \mathcal{B}(2, q) \implies \mathbb{E}(C) = 2q, \mathrm{Var}(C) = 2q(1-q)$.
  - b) $C \sim 2 \times \mathcal{B}(1, q) \implies \mathbb{E}(C) = 2q, \mathrm{Var}(C) = 4q(1-q)$.
- Therefore non-independence of claims inflates the variabilit y of claim numbers.

The Problem of Duplicates: General

- Now consider the following scenarios:

  a) Scenario 1: $N\sum_i i\pi_i$ policies held by $N\sum_i i\pi_i$ identical, independent lives.

  b) Scenario 2: $N\pi_1$ lives hold 1 policy, $N\pi_2$ lives hold 2 policies, $N\pi_3$ lives hold 3 policies, etc. (total $N\sum_i i\pi_i$ policies also).

- Define $C :=$ number of claims out of the $N\sum_i i\pi_i$ policies,
  $D_i :=$ number of deaths among the $N\pi_i$ individuals who hold i policies
  $\sim \mathcal{B}(N\pi_i, q)$, and $q :=$ death probability. Then

  a) $C \sim \mathcal{B}(N\sum_i i\pi_i, q) \Rightarrow \mathbb{E}(C) = Nq\sum_i i\pi_i, \mathrm{Var}(C) = Nq(1-q)\sum_i i\pi_i$

  b) $C = iD_i \implies \mathbb{E}(C) = Nq\sum_i i\pi_i, \mathrm{Var}(C) = Nq(1-q)\sum_i i^2\pi_i$

- Therefore to counter the problem of duplicates, we need to increase the variance of the number of claims in (a) by a variance ratio of $r = \left(\sum_i i^2\pi_i\right)/\left(\sum_i i\pi_i\right)$, and account for it during graduation.

Chapter 6

Graduation and Statistical Tests

1. Pearson Residuals
2. Statistical Tests
   - χ^2 test
   - Standardized deviations tests
   - Sign test
   - Change of sign test
   - Grouping of signs/Stevens test
   - Serial Correlation test

Recap: Hypothesis Testing & Test Distributions

Pearson Residuals

- In Chapter 5 we evaluated the residuals of our models visually. Is there a way to do this formally instead?
- We will work with Pearson residuals. Recall, for the Poisson GLM:

$$z_x = \frac{d_x - \dot{d}_x}{\sqrt{\text{Var}(d_x)}} \approx \frac{d_x - \dot{d}_x}{\sqrt{\dot{d}_x}},$$

and in the Binomial GLM:

$$z_x = \frac{\hat{q}_x - \dot{q}_x}{\sqrt{\text{Var}(\hat{q}_x)}} \approx \frac{d_x - \dot{d}_x}{\sqrt{\dot{d}_x}}$$

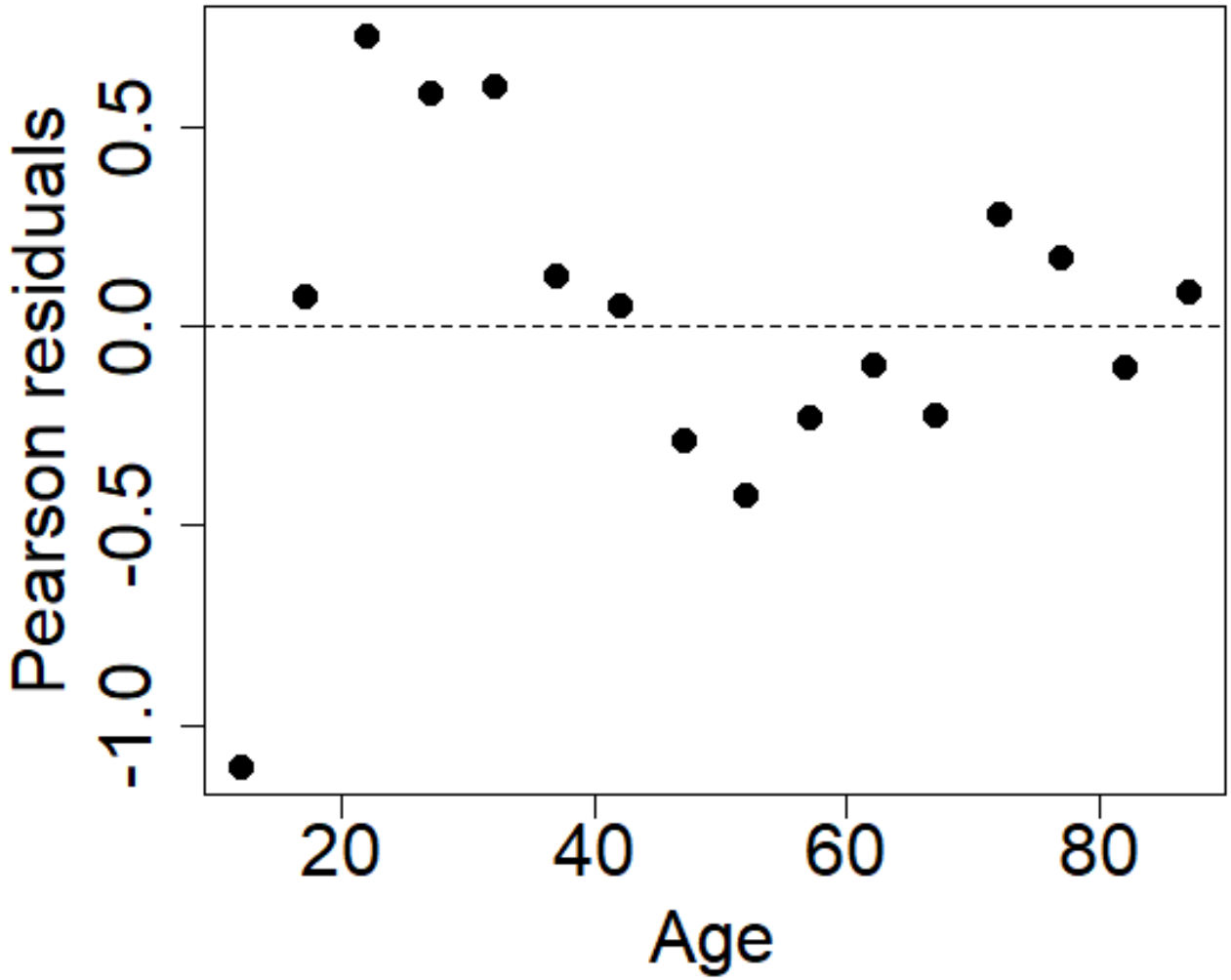
Pearson Residuals: Data & Model

$$\text{logit}(q_x) = a_0 + a_1 x + a_2 x^2, \text{ with } D_x \sim \mathcal{B}(E_x, q_x)$$

x	E_x	d_x	$\dot{d}_x$	z_x
12	8119	14	18.78	-1.104
17	7750	20	19.68	0.072
22	6525	22	18.83	0.730
27	5998	23	20.37	0.583
32	5586	26	23.10	0.603
37	5245	28	27.35	0.125
42	4659	32	31.70	0.054
47	4222	37	38.77	-0.285

x	E_x	d_x	$\dot{d}_x$	z_x
52	3660	44	46.92	-0.427
57	3012	54	55.70	-0.228
62	2500	68	68.81	-0.097
67	2113	87	89.10	-0.222
72	1469	100	97.26	0.278
77	883	95	93.36	0.169
82	418	70	70.87	-0.103
87	181	49	48.40	0.086

Pearson Residuals: Data & Model



1A. χ^2 test: Standard Table

- H_0: d_x are consistent with a standard table d_x^s.

  H_1: d_x are not consistent with a standard table d_x^s.

- Test statistic: under H_0,

$$\sum_x z_x^2 = \sum_x \frac{(d_x - d_x^s)^2}{d_x^s} \approx \chi_N^2$$

  where N is the number of age groups.

- One-sided test; we only reject if $\sum_x z_x^2$ is too large.

1B. χ^2 test: Graduated Rates

- Let there be p parameters involved in graduation $\dot{d}_x$ of deaths.

  H_0: d_x are consistent with a set of graduated deaths $\dot{d}_x$.

  H_1: d_x are not consistent with a set of graduated deaths $\dot{d}_x$.

- Test statistic: under H_0,

$$\sum_x z_x^2 = \sum_x \frac{(d_x - \dot{d}_x)^2}{\dot{d}_x} \approx \chi^2_{N-p}$$

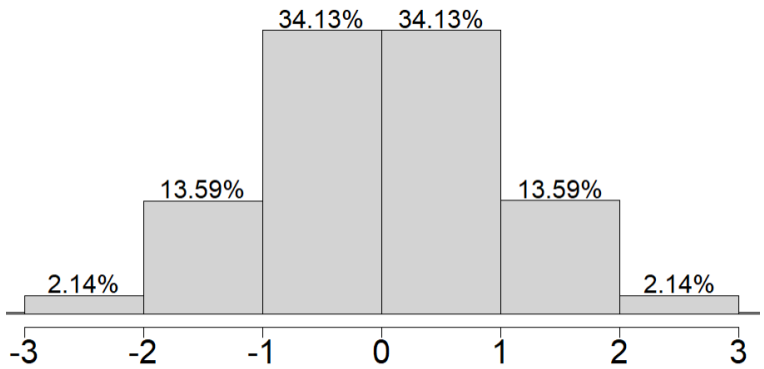
  where N is the number of age groups.

- One-sided test; we only reject if $\sum_x z_x^2$ is too large.

2A. Standardized Deviations Test (Equal-Width)

- If $\{z_i\} \sim \mathcal{N}(0,1)$, then its histogram should look like a histogram from $\mathcal{N}(0,1)$.
- Divide values into k equal-width intervals (with Expected > 5).
- H_0: $\{z_i\} \sim \mathcal{N}(0,1)$.
  H_1: $\{z_i\} \not\sim \mathcal{N}(0,1)$.
- Test statistic: under H_0,

$$\sum_{1}^{k} \frac{(\text{Observed - Expected})^2}{\text{Expected}} \approx \chi^2_{k-1}$$

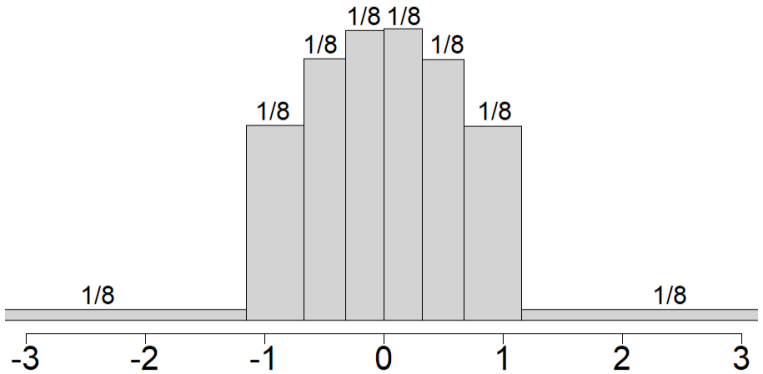


- One-sided test; we only reject if the test statistic is too large.

2B. Standardized Deviations Test (Equal-Area)

- If $\{z_i\} \sim \mathcal{N}(0,1)$, then its histogram should look like a histogram from $\mathcal{N}(0,1)$.
- Divide values into k equal-area/equal-probability intervals (with Expected > 5).
- H_0: $\{z_i\} \sim \mathcal{N}(0,1)$.
  H_1: $\{z_i\} \not\sim \mathcal{N}(0,1)$.
- Test statistic: under H_0,

$$\sum_{1}^{k} \frac{(\text{Observed - Expected})^2}{\text{Expected}} \approx \chi^2_{k-1}$$



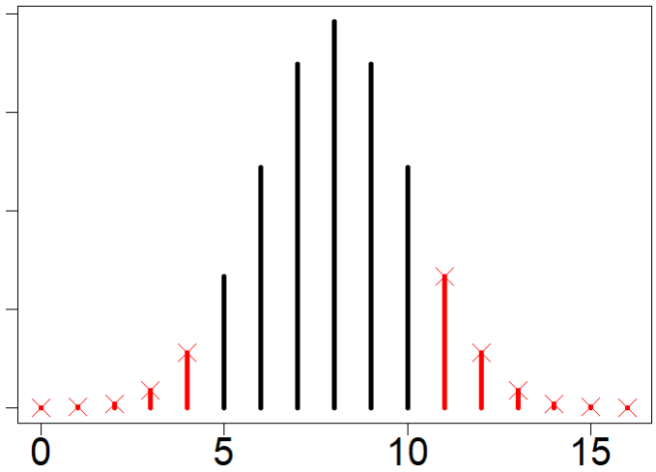
- One-sided test; we only reject if the test statistic is too large.

3. Sign Test

- H_0: There is no positive or negative bias in $\{z_i\}$.
  H_1: There is bias in $\{z_i\}$.
- Test statistic: under H_0,

$$S = \text{number of positive } z_i \sim \mathcal{B}\left(n, \frac{1}{2}\right)$$

$$p\text{-value} = \begin{cases} 2\,\mathbb{P}(S \leq s) & \text{if } s < n/2 \\ 2\,\mathbb{P}(S \geq s) & \text{if } s > n/2 \end{cases}$$



- If n large, can use Normal approx. $S \approx \mathcal{N}\left(\frac{n}{2}, \frac{n}{4}\right)$, with c.c.
- Two-sided test; we reject if the test statistic is too small or too large.

4. Change of Sign Test

- H_0: $\{z_i\} \sim \mathcal{N}(0,1)$.
  H_1: $\{z_i\} \not\sim \mathcal{N}(0,1)$.
- Test statistic: under H_0,

$$C = \text{number of sign changes} \sim \mathcal{B}\left(n-1, \frac{1}{2}\right)$$

$$p\text{-value} = \mathbb{P}(C \leq c)$$

- If n large, can use $\mathcal{N}$ approx. $C \approx \mathcal{N}\left(\frac{n-1}{2}, \frac{n-1}{4}\right)$, with c.c.
- One-sided test; we reject if the test statistic is too small.

5. Grouping of Signs Test (Stevens Test)

- Suppose we have m residuals, with n_1 positive & n_2 negative.
- H_0: The n_1 positive and n_2 negative residuals are in random order.
  H_1: The n_1 positive and n_2 negative residuals are not in random order.
- Test statistic: under H_0,

$$G = \text{number of groups of positive residuals},$$

$$\mathbb{P}(G = t) = \binom{n_1 - 1}{t - 1}\binom{n_2 + 1}{t} \Big/ \binom{n_1 + n_2}{n_1},$$

$$p\text{-value} = \mathbb{P}(G \leq g)$$

- If m large, can use $\mathcal{N}$ approx $G \approx \mathcal{N}\left(\frac{n_1(n_2+1)}{n_1+n_2}, \frac{(n_1 n_2)^2}{(n_1+n_2)^3}\right)$, with c.c.
- One-sided test; we reject if the test statistic is too small.

6. Serial Correlation Test

- H_0: The residuals are not serially correlated.

  H_1: The residuals are serially positively correlated.

- Test statistic: under H_0,

$$R_k = \text{Corr}\Big(\{z_1, \cdots, z_{n-k}\}, \{z_{k+1}, \cdots, z_n\}\Big) \approx \mathcal{N}\left(0, \frac{1}{n-k}\right)$$

- One-sided test; we reject if the test statistic is too large (too positive).

Summary

Test name	Test Statistic & Distribution	Side
χ^2-test (S.T.)	$\sum_x z_x^2 = \sum_x \frac{d_x - d_x^s}{d_x^s} \approx \chi^2_N$	Right
χ^2-test (Grad.)	$\sum_x z_x^2 = \sum_x \frac{d_x - \dot{d}_x}{\dot{d}_x} \approx \chi^2_{N-p}$	Right
S.D. Test (= Width)	$\sum_1^k \frac{(\text{Observed - Expected})^2}{\text{Expected}} \approx \chi^2_{k-1}$	Right
S.D. Test (= Area)	$\sum_1^k \frac{(\text{Observed - Expected})^2}{\text{Expected}} \approx \chi^2_{k-1}$	Right

Summary

Test name	Test Statistic & Distribution	Side
Sign Test	$S = \#$ positive residuals $\sim \mathcal{B}\left(n, \frac{1}{2}\right) \approx \mathcal{N}\left(\frac{n}{2}, \frac{n}{4}\right)$	Both
Change of Sign	$C = \#$ sign changes $\sim \mathcal{B}\left(n-1, \frac{1}{2}\right) \approx \mathcal{N}\left(\frac{n-1}{2}, \frac{n-1}{4}\right)$	Left
Grouping of Signs	$G = \#$ positive groups $\mathbb{P}(G = t) = \frac{\binom{n_1 - 1}{t-1}\binom{n_2+1}{t}}{\binom{n_1+n_2}{n_1}}$ $G \approx \mathcal{N}\left(\frac{n_1(n_2+1)}{n_1+n_2}, \frac{(n_1 n_2)^2}{(n_1+n_2)^3}\right)$	Left
Serial Correlation	$R_k = \text{Corr}\left(\{z_i\}_{i=1}^{n-k}, \{z_i\}_{i=k+1}^{n}\right) \approx \mathcal{N}\left(0, \frac{1}{n-k}\right)$	Right

Chapter 7

Mortality Projection

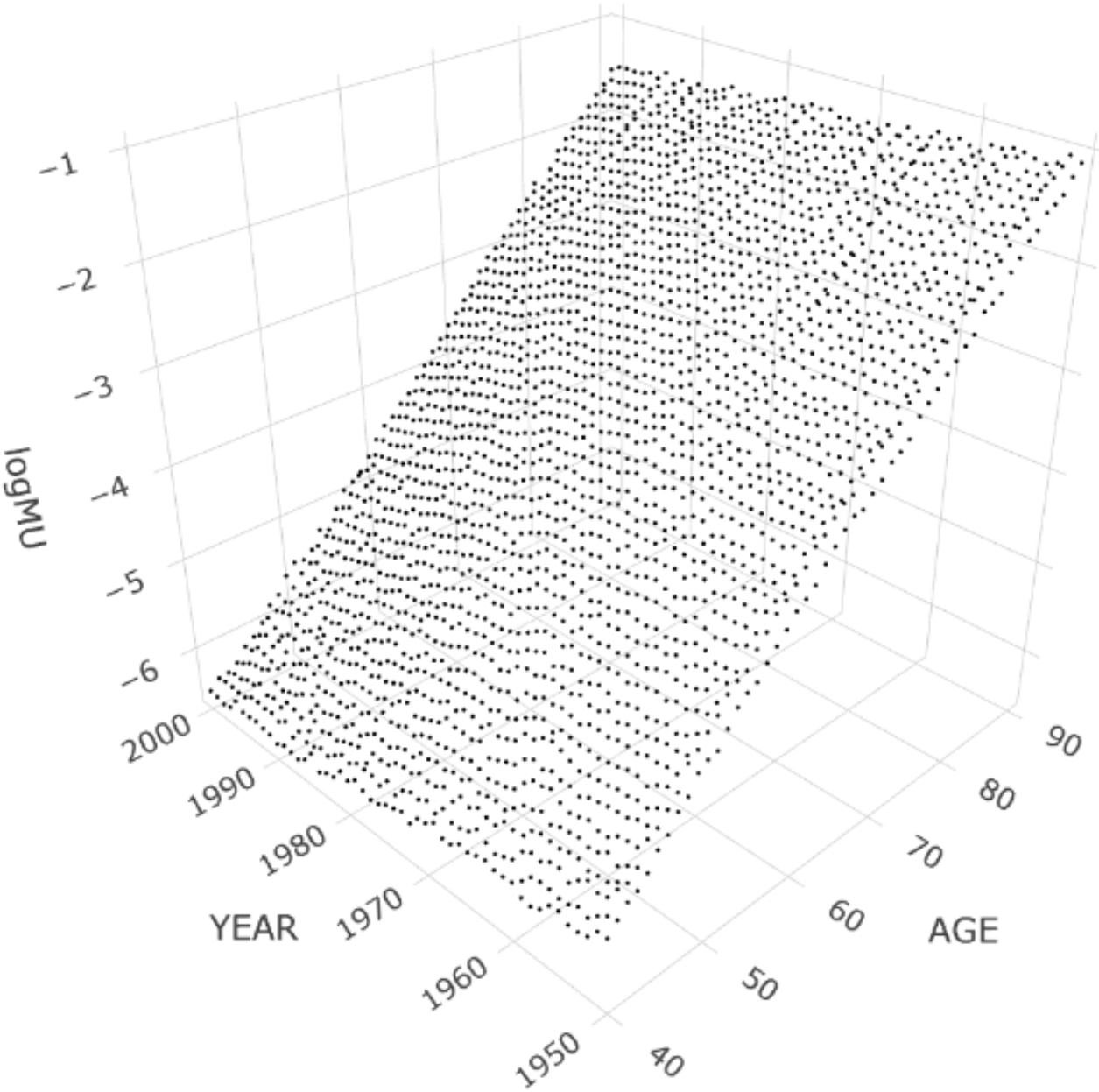
1. Introduction
2. Deterministic Projections
3. Stochastic Models
   - Lee-Carter Model
   - Cairns-Blake-Dowd Model
   - P-spline Model
   - Age-Period-Cohort Model
4. Cause-of-Death Models
5. Sources of Forecast Error

Introduction

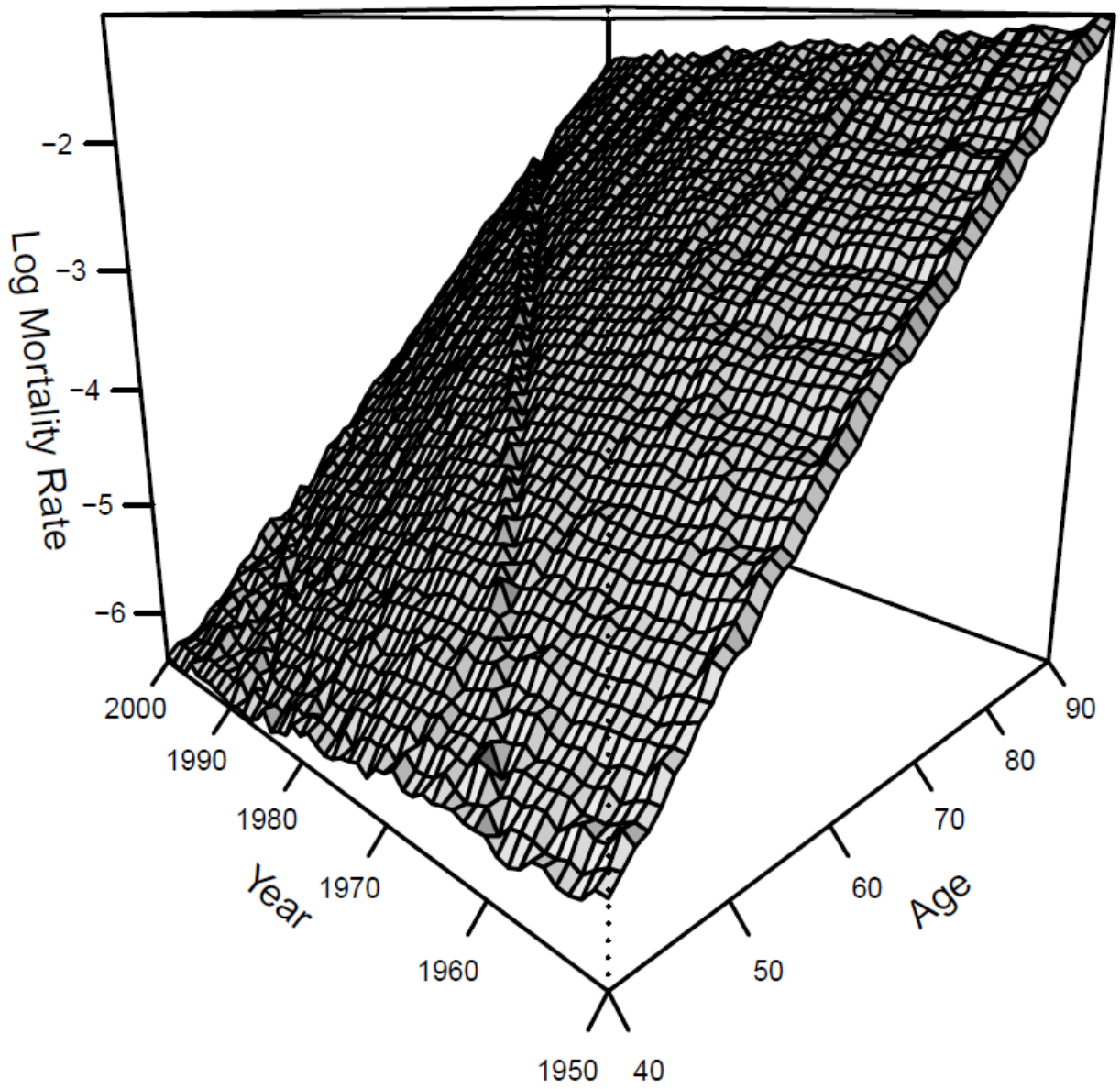
- In graduations, we only smooth existing data points of mortality and do not extrapolate or project past the limits of the data.
- However, mortality projection, i.e. estimating future mortality rates, is an important task for pension funds and writers of annuities.
- Human mortality will likely improve over time, and insurers need to take that into account in pricing their products, while also graduating mortality over ages.
- This motivates a view of mortality in two dimensions: age and time.

$$\mu_{x,t} \qquad \mu_{x,y} \qquad q_{x,t} \qquad q_{x,y}$$

Introduction



Introduction



Deterministic Projections

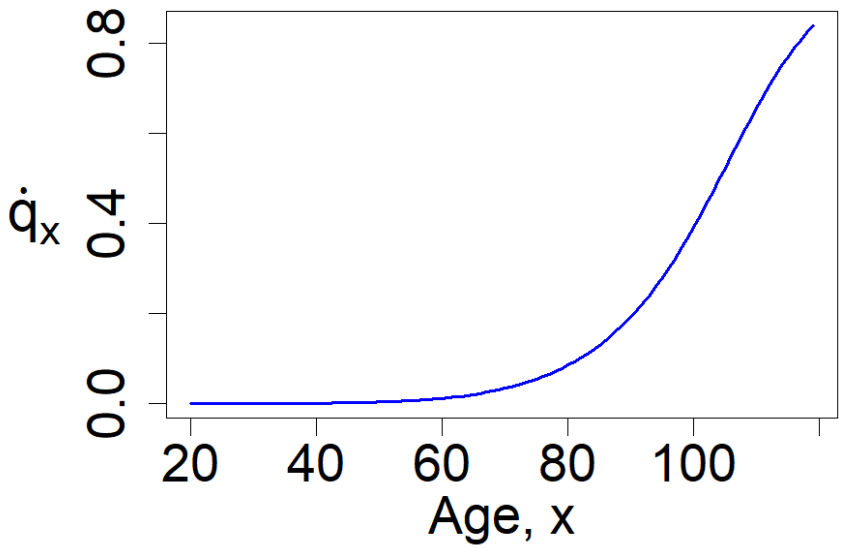
- Deterministic projections are simplistic, but do not offer uncertainty quantifications for their projections.
- Steps:
  1. Choose a base table of graduated rates.
  2. Study recent changes in $\mu_{x,t}$ or $q_{x,t}$, and extrapolate into the near future.
  3. Set, based on expert opinion or reference to low-mortality experiences, mortality rates $\mu_{x,\infty}$ or $q_{x,\infty}$ in the distant future.
  4. Choose a formula that blends the near-future and distant-future extrapolations.

Deterministic Projections: Example

- For example, consider a graduated model for the base year:

$$\text{logit}(\dot{q}_x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_5 x^5$$

$$\implies q_{x,0} = \left(1 + \exp\left\{-\left(a_0 + a_1 x + a_2 x^2 + \cdots + a_5 x^5\right)\right\}\right)^{-1}$$



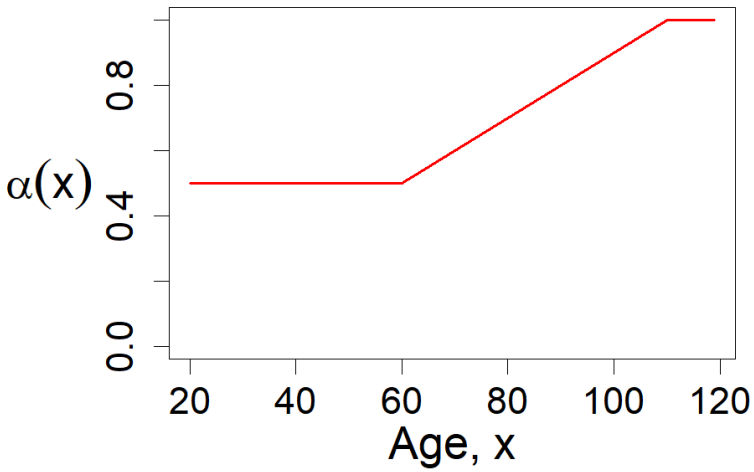
Deterministic Projections: Example

- Propose that $q_{x,\infty}$, the death rate in the distant future, is a proportion $\alpha(x)$ of current rate $q_{x,0}$.

$$q_{x,\infty} = \alpha(x)\ q_{x,0}$$

where

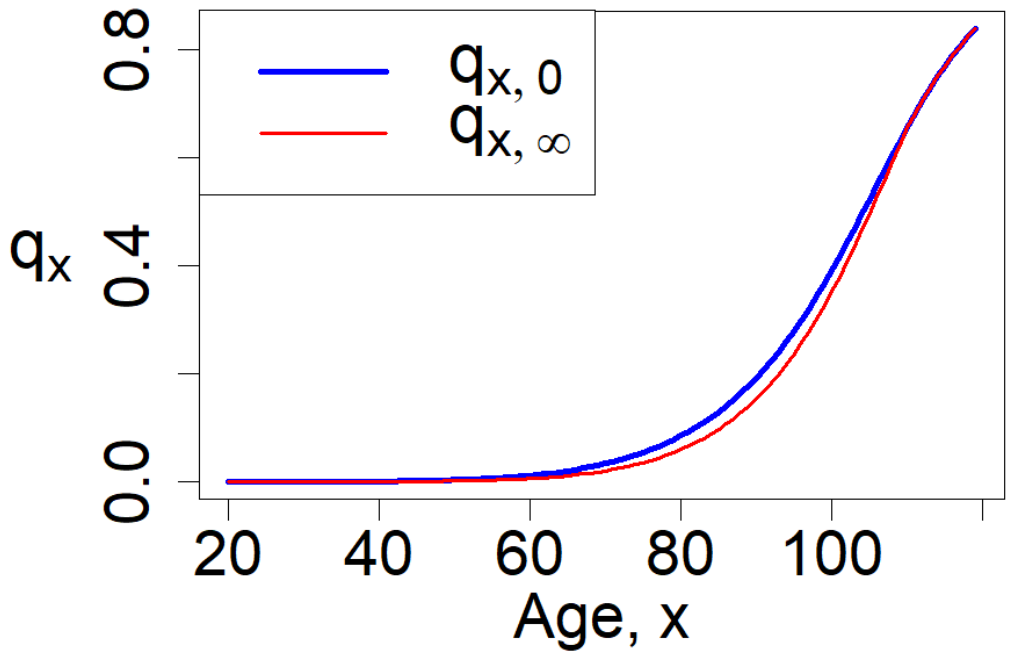
$$\alpha(x) = \begin{cases} 0.5 & \text{if } x \leq 60, \\ \frac{x-10}{100} & \text{if } 60 \leq x \leq 110, \\ 1 & \text{if } x \geq 110. \end{cases}$$



is the ultimate reduction factor.

Deterministic Projections: Example

- Combining $q_{x,0}$ and $\alpha(x)$ we get the eventual/ultimate death rates $q_{x,\infty}$:



- What's left is to model how $q_{x,0}$ evolves to $q_{x,\infty}$ over time t.

Deterministic Projections: Example

- Consider a reduction factor $RF(x,t)$ that indicates the fraction of the base mortality $q_{x,0}$ left at time t.
- $RF(x,t)$ must tend to $\alpha(x)$ as $t \to \infty$. For example:

$$RF(x,t) = \alpha(x) + (1-\alpha(x))(0.4)^{t/20}$$

Then

$$q_{x,t} := RF(x,t)\ q_{x,0}$$

gives us the modeled mortality of age x at time t.

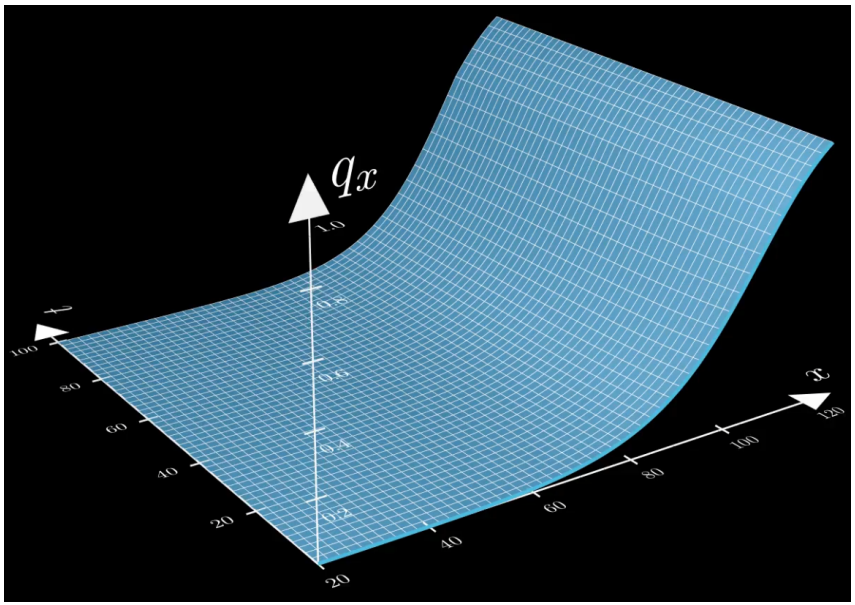
- Thus the full prediction surface is:

$$q_{x,t} = \left[\alpha(x) + (1-\alpha(x))(0.4)^{t/20}\right] \frac{1}{1+e^{-(a_0+a_1x+\cdots+a_5x^5)}}$$

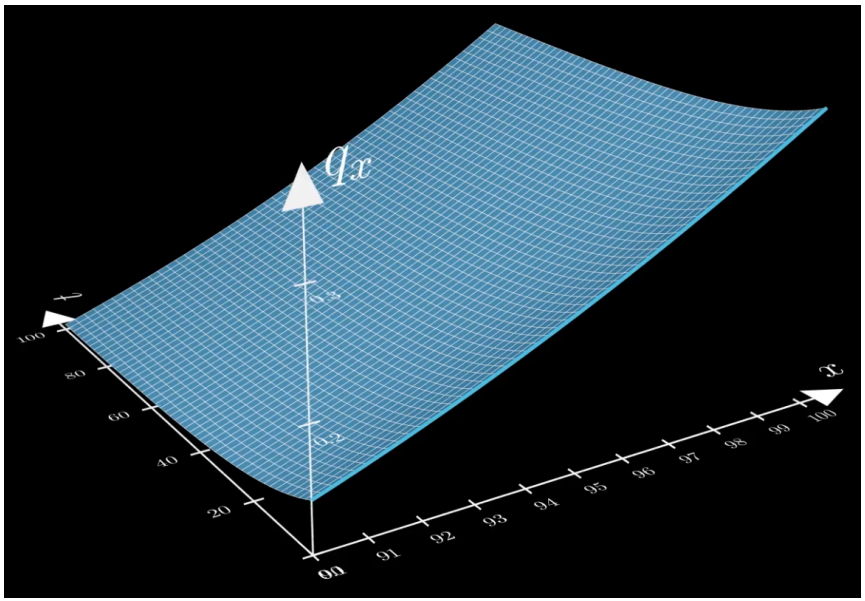
Deterministic Projections: Example

- The resulting prediction surface $q_{x,t}$ is plotted below:

Full surface



Zoomed into $90 \leq x \leq 100$



Deterministic Projections: Pros and Cons

- Pros #1: Simplicity.
- Cons #1: Relies on expert opinion (e.g. $q_{x,\infty}$).
- Cons #2: A single projection is almost certain to be wrong.
- Cons #3: No uncertainty quantifications; see stochastic models.

Stochastic 1: The Lee-Carter Model - Definition

- The Lee-Carter model is defined as:

$$\log(\mu_{x+1/2,y+1/2}) = \alpha_x + \beta_x \kappa_y$$

for $x = x_1, x_2, ..., x_{n_x}$ and $y = y_1, y_2, ..., y_{n_y}$.

- This models the log force of mortality at age x in calendar year y, applied at the mid-point of the year of age and the mid-point of the calendar year.

- The parameters to be estimated are $\underline{\alpha} = \left(\alpha_{x_1}, \alpha_{x_2}, ..., \alpha_{x_{n_x}}\right)$, $\underline{\beta} = \left(\beta_{x_1}, \beta_{x_2}, ..., \beta_{x_{n_x}}\right)$, $\underline{\kappa} = \left(\kappa_{y_1}, \kappa_{y_2}, ..., \kappa_{y_{n_y}}\right)$, totaling to $2n_x + n_y$ parameters.

Stochastic 1: The Lee-Carter Model - Definition

- In this initial model, the parameters have the following interpretations:
  - α_x represents the effect of age x alone on log mortality, without considering effects of time/calendar years (so this would be akin to...?)
  - κ_y reflects a general shape of how log mortality changes over time/calendar years
  - Since each age's mortality is affected by time differently, we use β_x to modulate the effects of time on each age.

    This means that if a certain β_x is large (resp. small), the effect of time is magnified (resp. diminished) for that age x.

Stochastic 1: The Lee-Carter Model - Identifiability

- Notice that, for all $c \in \mathbb{R} \setminus \{0\}$,

$$\alpha_x + \beta_x \kappa_y \equiv \alpha_x + (c\beta_x)\left(\frac{\kappa_y}{c}\right)$$

$\Longrightarrow$ the model is unidentifiable; its optimization does not have a unique solution.

- To force a unique optimum/unique representation, we will need to impose constraints on $\underline{\alpha}, \underline{\beta}, \underline{\kappa}$. A conventional choice:

$$\sum_{x=x_1}^{x_{n_x}} \beta_x = 1, \qquad \sum_{y=y_1}^{y_{n_y}} \kappa_y = 0$$

Stochastic 1: The Lee-Carter Model - Identifiability

- Transformations to the coefficients to satisfy constraints (to make the model identifiable):

$$\begin{aligned} \alpha'_x &:= \alpha_x + \beta_x \bar{\kappa} \\ \beta'_x &:= \beta_x / (n_x \bar{\beta}) \\ \kappa'_y &:= n_x \bar{\beta} (\kappa_y - \bar{\kappa}) \end{aligned}$$

- Notice that, after these transformations,

$$\sum_{x=x_1}^{x_{n_x}} \beta'_x = 1, \qquad \sum_{y=y_1}^{y_{n_y}} \kappa'_y = 0, \qquad \alpha'_x = \frac{1}{n_y} \sum_{y=y_1}^{y_{n_y}} \log \dot{\mu}_{x+\frac{1}{2},y+\frac{1}{2}} \quad ,$$

and yet we retain $\log \dot{\mu}_{x+\frac{1}{2},y+\frac{1}{2}} = \alpha'_x + \beta'_x \kappa'_y = \alpha_x + \beta_x \kappa_y$.

Stochastic 1: The Lee-Carter Model - Fitting

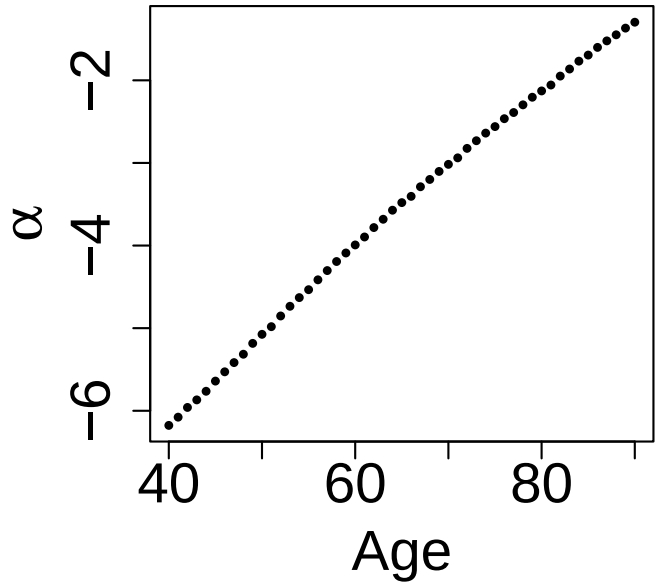
- This is an instance of generalized nonlinear models; use the `gnm` package in R.
- Since we are fitting a coefficient for every discrete-valued age and year, it's as if we are treating age and year as levels/factors, hence:

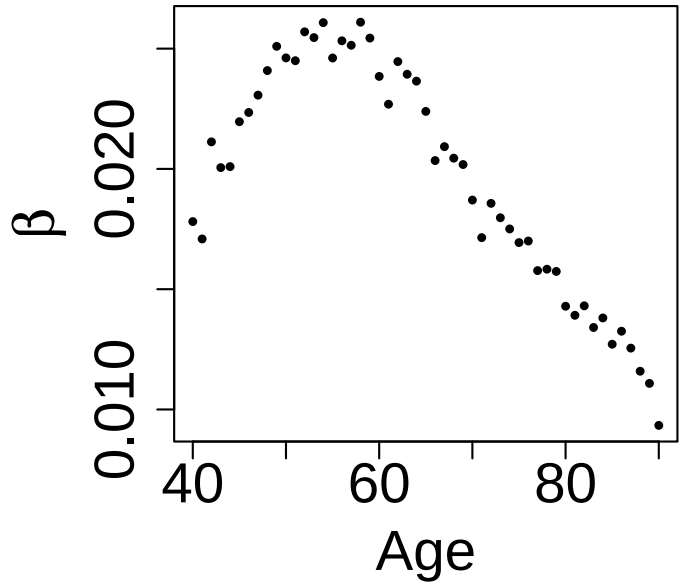
```
Age.F = factor(rep(AGE, ncol(Dth)))
Year.F = factor(rep(YEAR, each = nrow(Dth)))
LC = gnm(Dth.V ~ -1 + Age.F + Mult(Age.F, Year.F),
         offset=log(Exp.V), family = poisson)
```

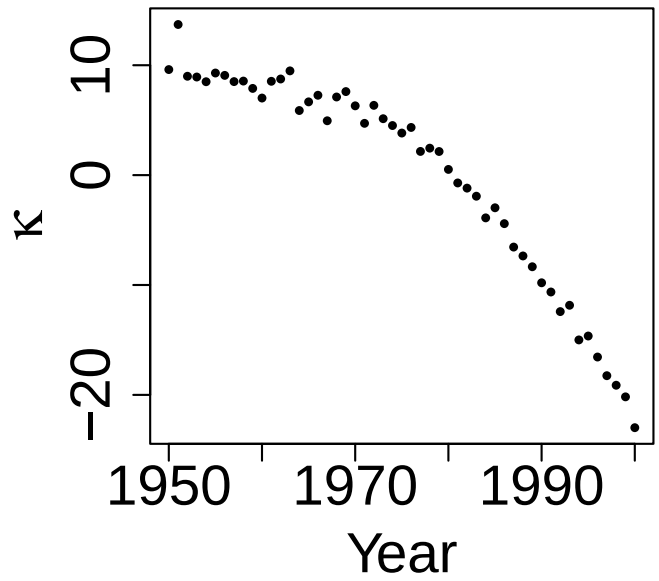
  This returns $2n_x + n_y$ estimated coefficients $\underline{\alpha}, \underline{\beta}, \underline{\kappa}$.

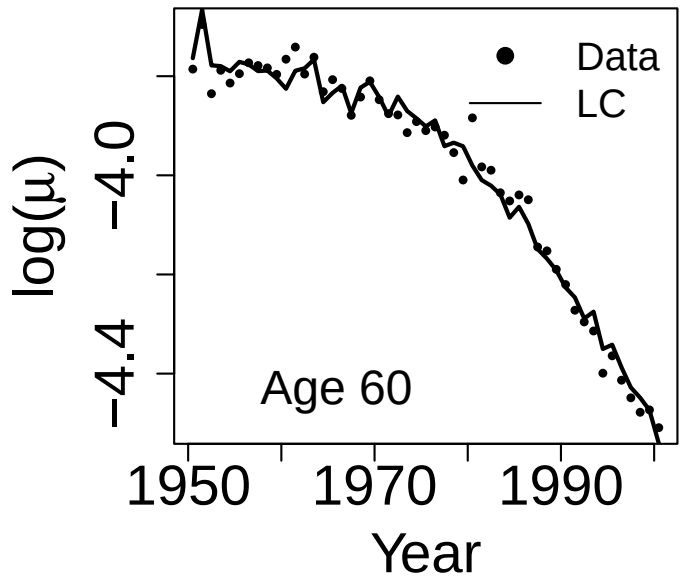
- Note, however, we still need to make transformations to the parameters to satisfy identifiability constraints! This then produces $\underline{\alpha}', \underline{\beta}', \underline{\kappa}'$, which we can examine.

Stochastic 1: The Lee-Carter Model - Results

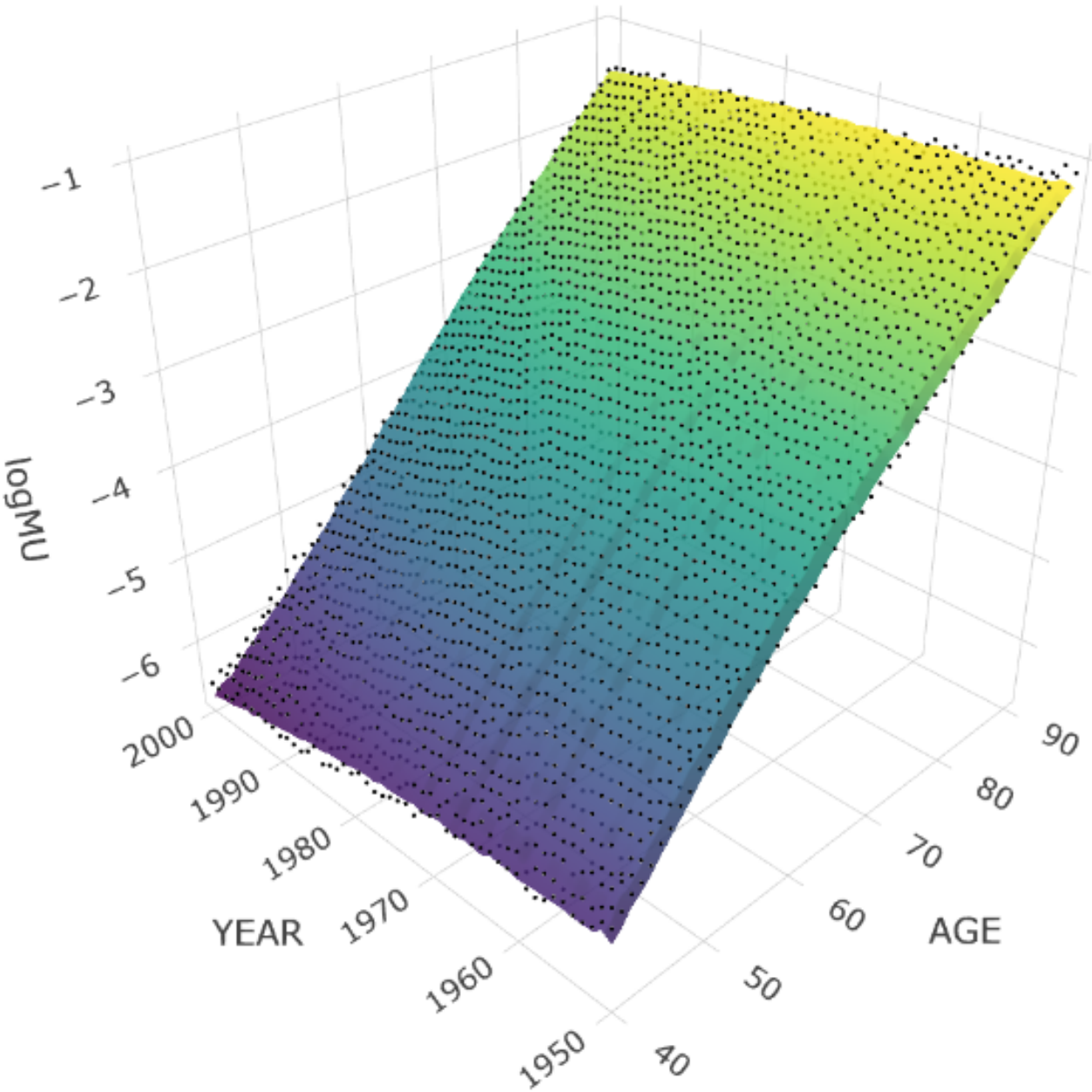








Stochastic 1: The Lee-Carter Model - Results



Stochastic 1: The Lee-Carter Model - Projection

- We first assume a time series model for κ_y (why not α_x or β_x?):

$$\kappa_{y+1} - \kappa_y = \delta + \epsilon_{y+1} \quad \Longrightarrow \quad \kappa_{y+1} = \kappa_y + \delta + \epsilon_{y+1}$$

$$\text{or more generally,} \quad \kappa_{y+m} = \kappa_y + m\delta + \sum_{i=1}^{m} \epsilon_{y+i}$$

  where δ is the trend and $\{\epsilon_y\} \overset{\text{i.i.d}}{\sim} \mathcal{N}(0, \sigma^2)$ are Gaussian error terms.

- Note that this is a rather simplistic model; other more time series models may be used, though this may also complicate forecasting.

Stochastic 1: The Lee-Carter Model - Projection

- To make predictions beyond y_{n_y}, we may estimate δ as the sample mean of all existing $(\kappa_y - \kappa_{y-1})$'s:

$$\hat{\delta} = \frac{1}{n_y - 1} \sum_{y=y_2}^{y_{n_y}} (\kappa_y - \kappa_{y-1}) = \frac{\kappa_{y_{n_y}} - \kappa_{y_1}}{n_y - 1}.$$

  with $\mathbb{E}(\hat{\delta}) = \delta$ and $\text{Var}(\hat{\delta}) = \dfrac{\sigma^2}{n_y - 1}$.

- To quantify estimation variances we also require an estimate for σ^2. We take the sample variance for all existing $(\kappa_y - \kappa_{y-1})$'s:

$$\hat{\sigma}^2 = \frac{1}{n_y - 2} \sum_{y=y_2}^{y_{n_y}} (\kappa_y - \kappa_{y-1} - \hat{\delta})^2$$

Stochastic 1: The Lee-Carter Model - Projection

- Hence, extending $\kappa_{y+1} = \kappa_y + \delta + \epsilon_{y+1}$ beyond y_{n_y}, we estimate

$$\hat{\kappa}_{y_{n_y}+1} = \kappa_{y_{n_y}} + \hat{\delta}, \quad \text{or}$$

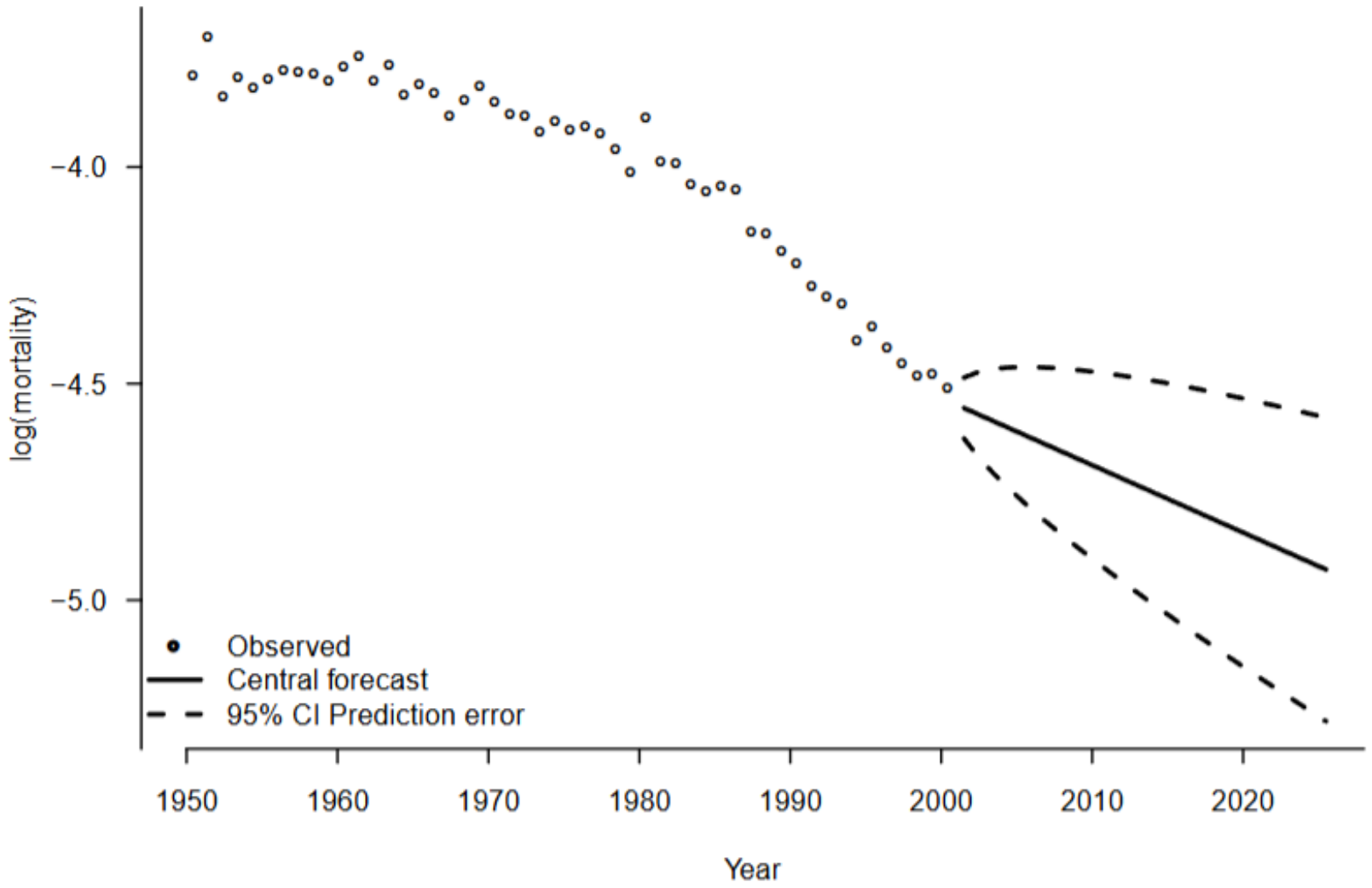
$$\hat{\kappa}_{y_{n_y}+m} = \kappa_{y_{n_y}} + m\hat{\delta} \quad \text{in general.}$$

- We also deduce from $\kappa_{y+m} = \kappa_y + m\delta + \sum_{i=1}^{m} \epsilon_{y+i}$ that

$$\text{Total Forecast SE}(\hat{\kappa}_{y_{n_y}+m}) = \sqrt{\text{Var}(m\delta) + \text{Var}\left(\sum_{i=1}^{m} \epsilon_{y_{n_y}+i}\right)}$$

$$= \sqrt{\underbrace{m^2 \frac{\sigma^2}{n_y - 1}}_{\substack{\text{parameter} \\ \text{variance}}} + \underbrace{m\sigma^2}_{\substack{\text{prediction} \\ \text{variance}}}}$$

Stochastic 1: The Lee-Carter Model - Projection



Stochastic 1: The Lee-Carter Model - Discussion

- Pros #1: Well-suited to forecasting:
  - The effect of calendar time is confined to one variable κ_y;
  - This effect can be modelled with various time series techniques;
  - This gives statistical properties of estimated mortality rates.
- Cons #1: Model restrictions:
  - Assumes α_x and β_x do not change over time - as a result, any calendar year effects not captured by κ_y could affect estimates of β_x and increase its variability;
  - As a result, estimated mortality rates have the same general "shape" for all ages - unrealistic.
- Cons #2: Large number of parameters, thus not smooth.
- Cons #3: No cohort term.

Stochastic 2: CBD Model - Definition

- The Cairns-Blake-Dowd model is defined as:

$$\log(\mu_{x+1/2,y+1/2}) = \kappa_y^{(1)} + \kappa_y^{(2)}(x - \bar{x})$$

for $x = x_1, x_2, ..., x_{n_x}$ and $y = y_1, y_2, ..., y_{n_y}$, where $\bar{x}$ is the mean age.

- We can also rewrite the model as:

$$\log(\mu_{x+1/2,y+1/2}) = \underbrace{(\kappa_y^{(1)} - \kappa_y^{(2)}\bar{x})}_{a} + \underbrace{\kappa_y^{(2)}}_{b} x$$

which means we are essentially fitting a Gompertz model for each calendar year y.

- The two time series $\kappa_y^{(1)}$ and $\kappa_y^{(2)}$ determine how the Gompertz "surface" develops over time.

Stochastic 2: CBD Model - Fitting

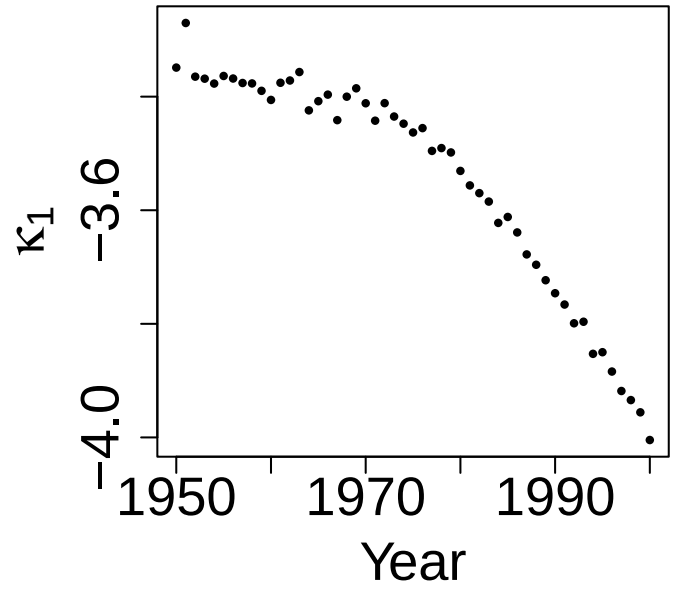
- The code below fits the CBD model:

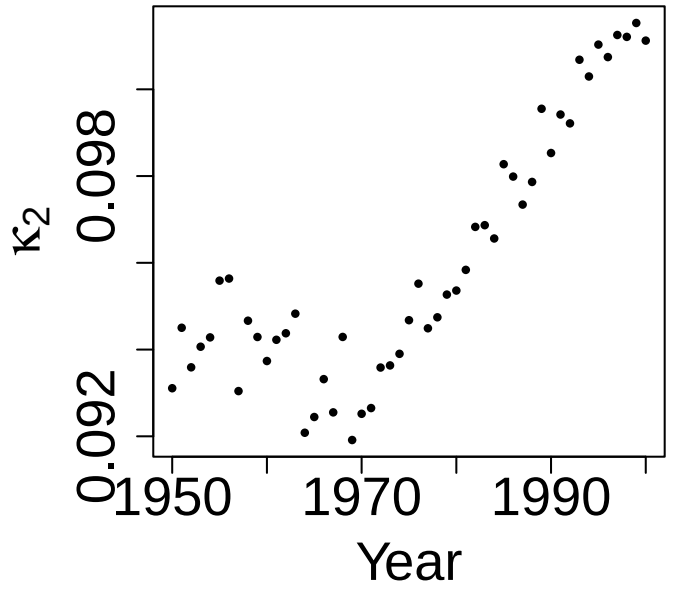
```
X1 = kronecker(diag(n.y), rep(1, n.x))
Age.mean = AGE - mean(AGE)
X2 = kronecker(diag(n.y), Age.mean)
X = cbind(X1, X2)
CBD.Male = glm(Dth.V ~ -1 + X, offset = log(Exp.V),
family = poisson)
```

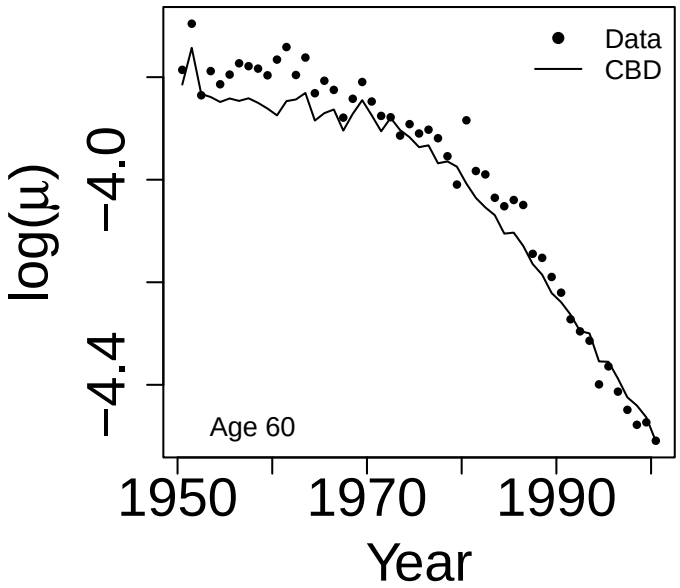
  Here, `Dth.V` and `Exp.V` are the death and exposure matrices flattened into vectors, `n.x` and `n.y` are n_x and n_y, and `AGE` is x.

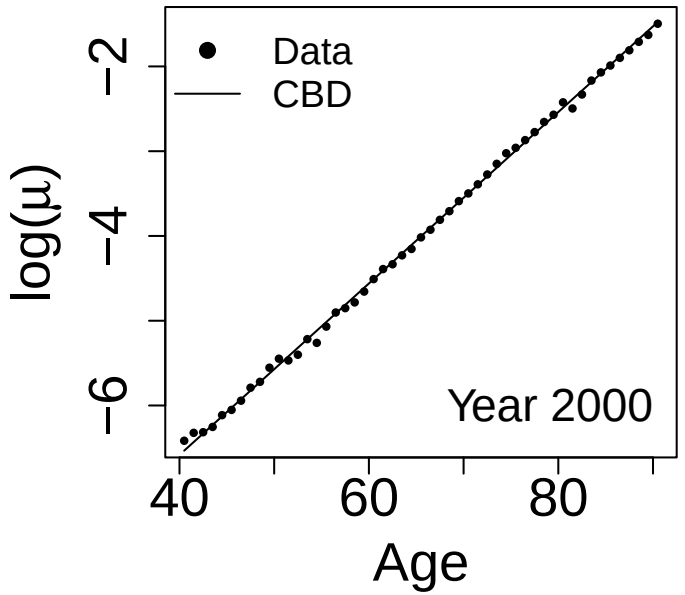
  We omit details of what `kronecker()` does to construct `X`; it simply gets the data and parameters lined up correctly for fitting.

Stochastic 2: CBD Model - Results

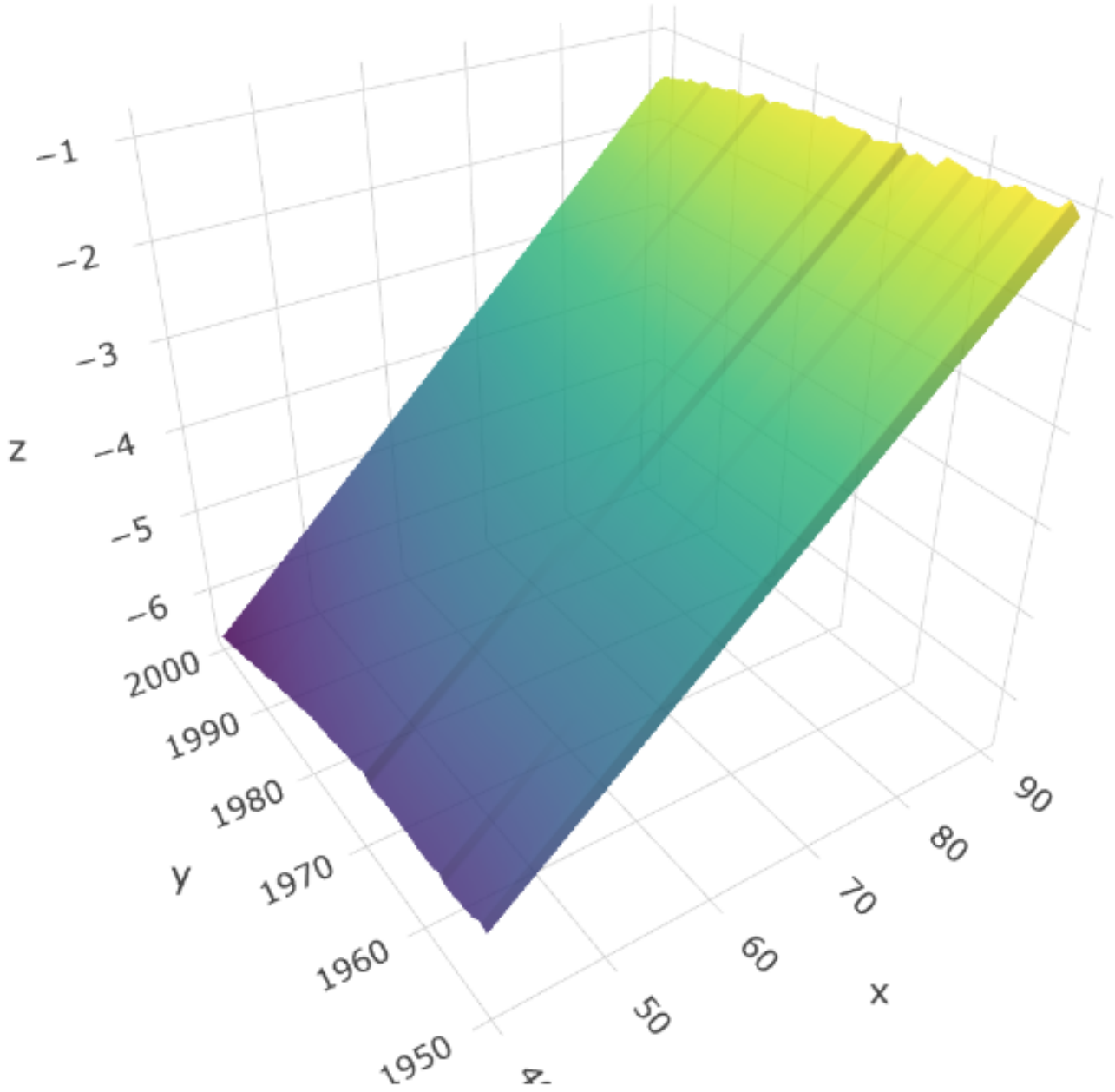








Stochastic 2: CBD Model - Results



Stochastic 2: CBD Model - Projection

- In the CBD model, we have a bivariate time series on $(\kappa_y^{(1)}, \kappa_y^{(2)})^T$.
- To project, we similarly assume:

$$\begin{pmatrix} \kappa_{y+1}^{(1)} \\ \\ \kappa_{y+1}^{(2)} \end{pmatrix} = \begin{pmatrix} \kappa_y^{(1)} \\ \\ \kappa_y^{(2)} \end{pmatrix} + \begin{pmatrix} \delta^{(1)} \\ \\ \delta^{(2)} \end{pmatrix} + \begin{pmatrix} \epsilon_{y+1}^{(1)} \\ \\ \epsilon_{y+1}^{(2)} \end{pmatrix}$$

  where $(\delta^{(1)}, \delta^{(2)})^T$ is the drift term and $(\epsilon_{y+1}^{(1)}, \epsilon_{y+1}^{(2)})^T$ is a bivariate Gaussian error term.

- The details are similar as before, but bivariate projections are beyond the scope of this course.

Stochastic 2: CBD Model - Discussion

- Pros #1: It fits in the GLM framework.
- Pros #2: Simplicity of forecasting.
- The CBD model can be extended to include quadratic terms, or even cohort terms, to improve the fit; however this makes forecasting much more difficult.

Stochastic 3: P-spline Model - Definition

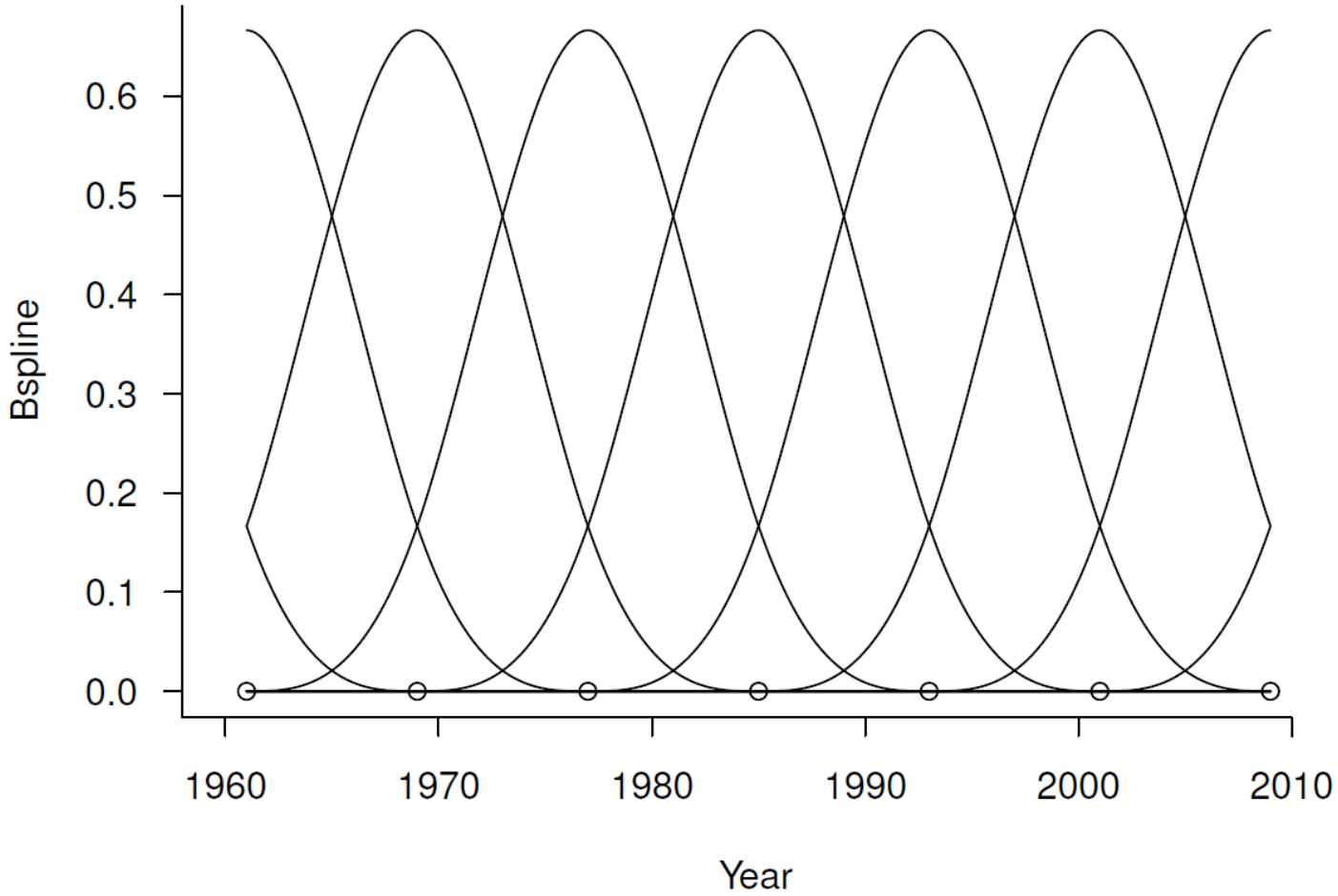
- We start by introducing basis splines ("B-splines"), made up of different functions joined at certain knots to get a continuous and differentiable function.
- An example of a quadratic B-spline:

$$B(t) = \begin{cases} \frac{t^2}{2}, & 0 \leq t \leq 1, \\ \frac{1}{2}(-2t^2 + 6t - 3), & 1 \leq t \leq 2, \\ \frac{1}{2}(3-t)^2, & 2 \leq t \leq 3, \\ 0, & \text{otherwise.} \end{cases}$$

- We can obtain a basis, e.g. $\{B_1(t), ..., B_n(t)\}$, of these splines.
- Then modeling $\log \mu_{x,t} = \sum_{j=1}^{n} \theta_j B_j(t)$ gives a smooth function!

Stochastic 3: P-spline Model - Definition

- An example of a cubic spline basis:



Stochastic 3: P-spline Model - Fitting

- To find $\boldsymbol{\theta} = (\theta_1, ..., \theta_n)$ that fits the data, we may assume $D_{x,t} \sim \mathcal{P}(E^c_{x,t}\mu_{x,t})$ and maximize a penalized likelihood $\ell_P(\boldsymbol{\theta})$ w.r.t. $\boldsymbol{\theta}$:

$$\ell_P(\boldsymbol{\theta}) = \ell(\boldsymbol{\theta}) - \frac{1}{2}\lambda P(\boldsymbol{\theta})$$

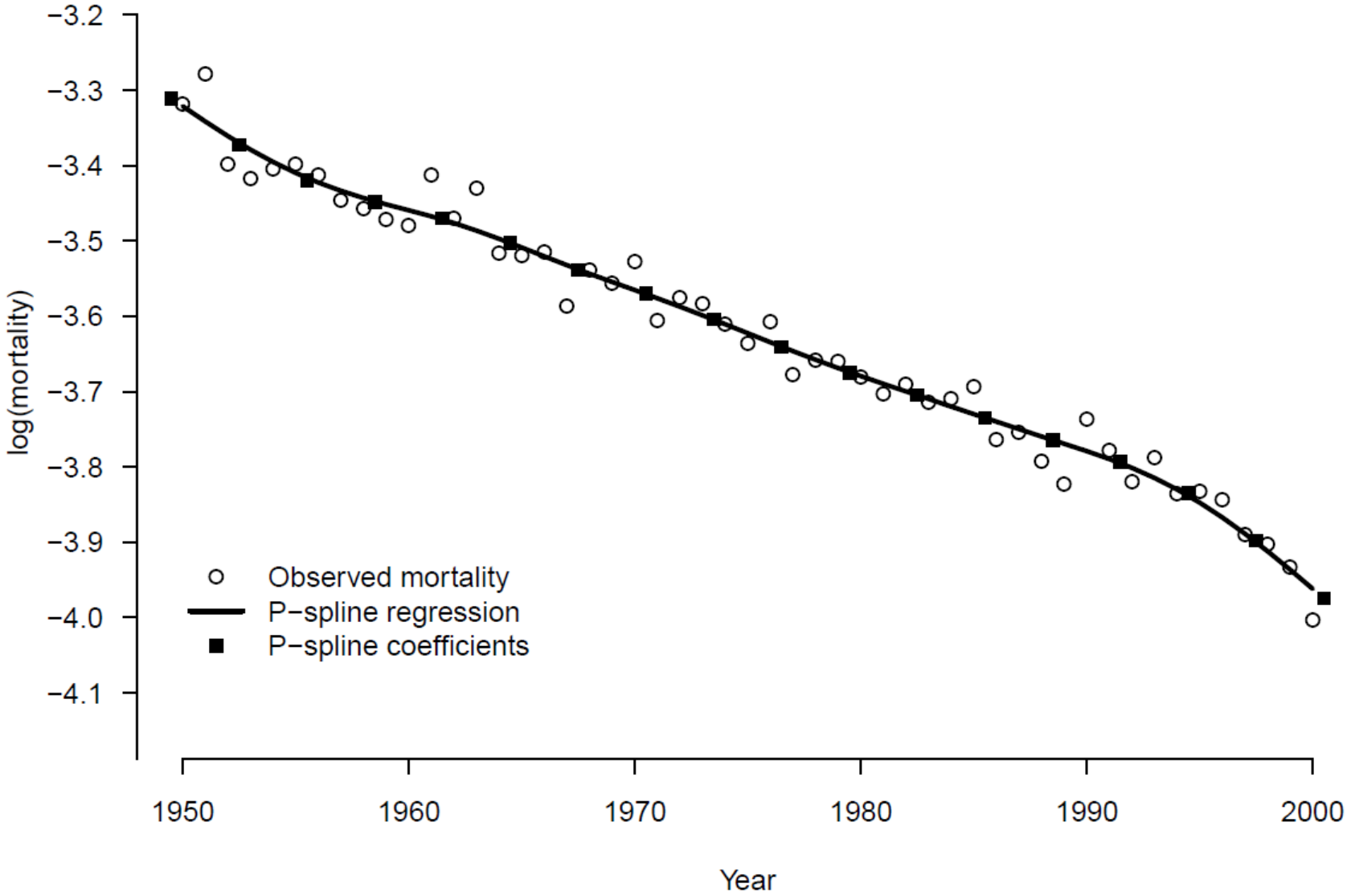
where

$$\ell(\boldsymbol{\theta}) = \sum_x \left(d_{x,t} \overbrace{\sum_{j=1}^{n} \theta_j B_j(t)}^{\text{fitted } \log \mu_{x,t}} - E^c_{x,t} \overbrace{e^{\sum_{j=1}^{n} \theta_j B_j(t)}}^{\text{fitted } \mu_{x,t}} \right),$$

$$P(\boldsymbol{\theta}) = (\theta_1 - 2\theta_2 + \theta_3)^2 + \cdots + (\theta_{n-2} - 2\theta_{n-1} + \theta_n)^2.$$

- Here $P(\boldsymbol{\theta})$ is the penalty (hence the name penalized splines or P-splines) which measures roughness.
- λ therefore controls the smoothness of the fitted curve.

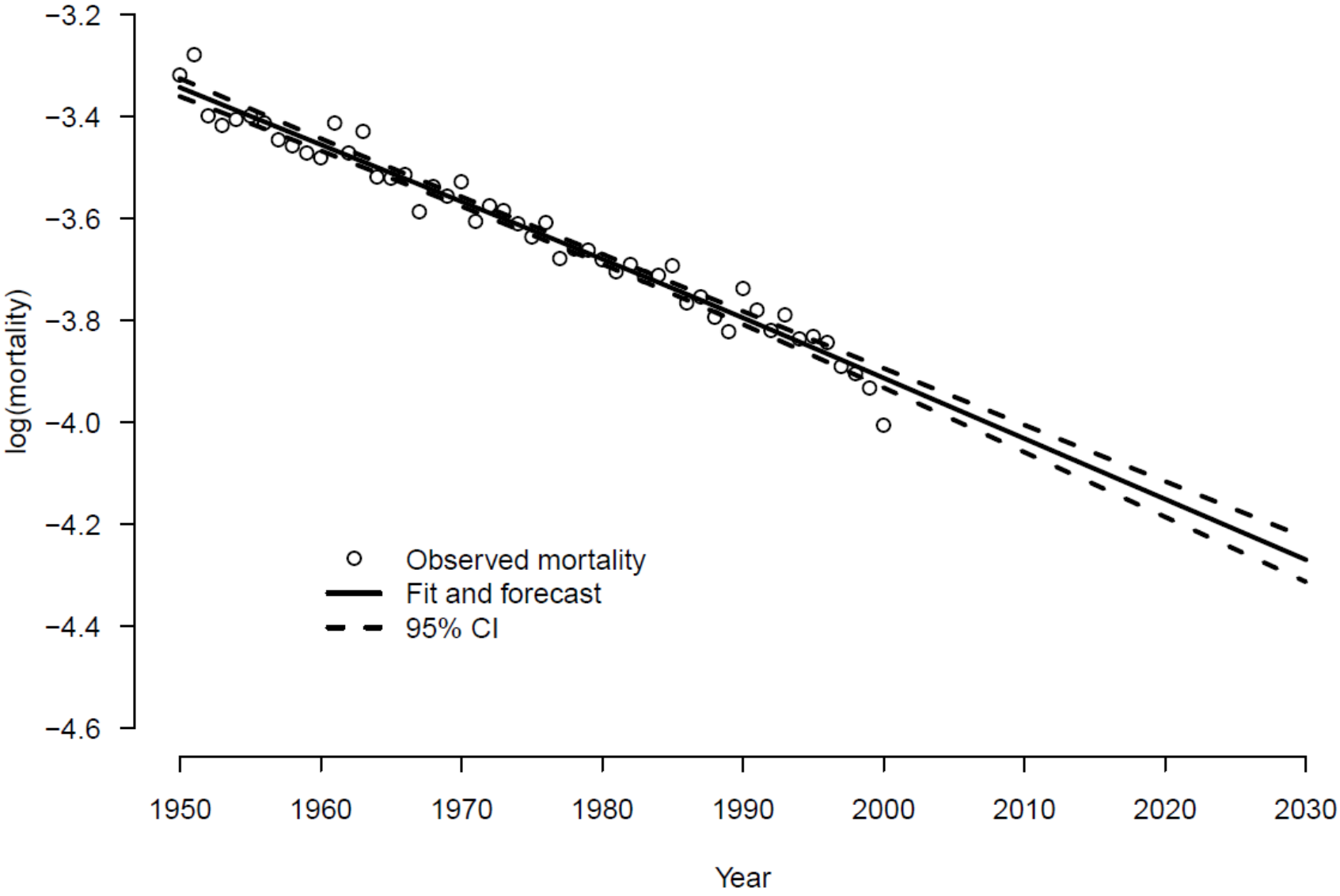
Stochastic 3: P-spline Model - Results



Stochastic 3: P-spline Model - Projections

- Define dummy death and exposure data for future years.
- Extend the spline bases to cover all future years.
- Set the weights of existing data to 1 and weights of dummy data to 0.
- Calculate the likelihood using only data with weight 1.
- Choose regression coefficients θ_j for the additional basis splines covering the future such that their contribution to the penalty term is 0.

Stochastic 3: P-spline Model - Projections



Stochastic 3: P-spline Model - Discussion

- Pros #1: Long-range effects of polynomials are minimized with the usage of splines.
- Pros #2: Spline models are regression models and so fitted coefficients are interpretable as regression coefficients.
- Pros #3: P-splines reduces the choice of knots and bases to the choice of the smoothing parameter λ.
- Pros #4: Fitting and projection are done simultaneously; projection is automatic from the penalty function.
- Cons #1: There is no interpretable basis for allowing the penalty to determine the forecast, only a mathematical convenience.
- Cons #2: Localization effect: Forecasts can be very sensitive to the last few years of fitted data.

Stochastic 4: APC Models - Definition

- Age-Period-Cohort (APC) models take account of age, calendar year and cohort effects.
- The simplest APC model can be written as

$$\log(\mu_{x+1/2,y+1/2}) = \beta_x + \kappa_y + \gamma_c$$

  - β_x is a parameter for each age x,
  - κ_y is a parameter for each calendar year y,
  - γ_c is a parameter for each cohort c.

- If c denotes birth year, then we may also write $\gamma_c = \gamma_{y-x}$.
- A more general form of the APC model can thus be written as:

$$\log(\mu_{x+1/2,y+1/2}) = \sum_{i=1}^{n} \beta_x^{(i)} \kappa_y^{(i)} \gamma_{y-x}^{(i)}$$

Stochastic 4: APC Models - Definition

- Some examples of APC Models:
  - Lee-Carter model. Here $n = 2$ and $\kappa_y^{(1)} = \gamma_{y-x}^{(1)} = \gamma_{y-x}^{(2)} = 1$:

$$\log(\mu_{x+1/2,y+1/2}) = \beta_x^{(1)} + \beta_x^{(2)}\kappa_y^{(2)} \qquad \text{(no cohort eff.)}$$

  - CBD model. Here $n = 2$, $\beta_x^{(1)} = \gamma_{y-x}^{(1)} = \gamma_{y-x}^{(2)} = 1$ and $\beta_x = x - \bar{x}$:

$$\log(\mu_{x+1/2,y+1/2}) = \kappa_y^{(1)} + \beta_x^{(2)}\kappa_y^{(2)} \qquad \text{(no cohort eff.)}$$

  - Renshaw-Haberman model. Here $n = 3$ and $\kappa_y^{(1)} = \kappa_y^{(3)} = \gamma_{y-x}^{(2)} = 1$:

$$\log(\mu_{x+1/2,y+1/2}) = \beta_x^{(1)} + \beta_x^{(2)}\kappa_y^{(2)} + \beta_x^{(3)}\gamma_{y-x}^{(3)}$$

  - Model M6. Here $n = 3$ and $\beta_x^{(1)} = \gamma_{y-x}^{(1)} = \gamma_{y-x}^{(2)} = \beta_x^{(3)} = \kappa_y^{(3)} = 1$:

$$\log(\mu_{x+1/2,y+1/2}) = \kappa_y^{(1)} + \beta_x^{(2)}\kappa_y^{(2)} + \gamma_{y-x}^{(3)}$$

Stochastic 4: APC Models - Identifiability

- Similar to the Lee-Carter model, APC models are generally unidentifiable, i.e. there are many sets of parameters that maximise the likelihood.
- Once again, we require identifiability constraints.
- E.g. constraints for Renshaw-Haberman model:

$$\sum_y \kappa_y^{(2)} = 0, \ \sum_x \beta_x^{(2)} = 1, \ \sum_{x,y} \gamma_{y-x}^{(3)} = 0, \ \sum_x \beta_x^{(3)} = 1.$$

- E.g. constraints for Model M6:

$$\sum_{x,y} \gamma_{y-x}^{(3)} = 0, \qquad \sum_{x,y} (y-x)\gamma_{y-x}^{(3)} = 0$$

Stochastic 4: APC Models - Fitting & Projection

- APC models are fitted using maximum likelihood, similar to other stochastic models discussed before.
- Time series approaches can be used to project all $\kappa_y^{(i)}$ and $\gamma_{y-x}^{(i)}$ simultaneously.

Stochastic 4: APC Models - Discussion

- Pros #1: Includes cohort effects that can be projected.
- Cons #1: Unidentifiability - requires constraints.
- Cons #2: Very demanding of the data.
- Cons #3: Some comment that cohort effects are not real.

Cause-of-Death Models

- Death arise from several major causes, e.g. cancer, cardiovascular disease, kidney failure etc.
- In cause-of-death models, we first model and project mortality by each cause separately.
- We can then combine these to form a model of aggregate mortality.
- In practice, however, cause-of-death modelling is every difficult.

Cause-of-Death Models

- Some difficulties with cause-of-death modelling:
  - Cause of death is not always accurately recorded on death certificates, due to errors and social/cultural reasons.
  - More than one cause of death can be recorded in death certificates - not sure which one is primary.
  - Different causes of death may show strong correlations due to lifestyle or genetics, which must be considered in modelling.
  - Causes of death evolve and are replaced in ways that are hard to predict.
  - Data may not be available for the long periods of time needed to project mortality using causes of death.

Sources of Forecast Error

1. Model risk
2. Parameter error
3. Stochastic or prediction risk
4. Basis risk
5. Volatility
6. Idiosyncratic risk
7. Mis-estimation risk

Chapter 8

Exposed to Risk

1. Introduction
2. Exact Age x vs. Age Labels for d_x, E_x^c
3. Rate Intervals on μ_x and q_x
4. Census Data
   - Calculation of E_x^c from census data
   - Adjustment of census data

Introduction

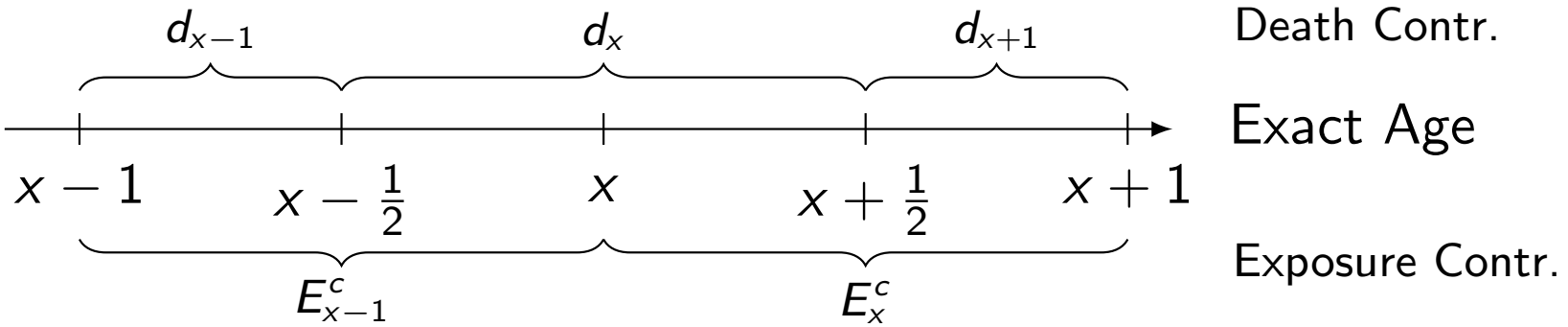
- We assume all lives are subdivided into more or less homogeneous groups, e.g. by age, sex, policy type, smoking status, occupation, etc.
- The central exposed to risk, E_x^c, is the observed waiting time in the alive state, after accounting for deaths, for "age x".
- The initial exposed to risk, E_x, is the waiting time that would have been observed for "age x" if no deaths occur.
- These quantities are often defined in units of "person-years" or "policy-years".

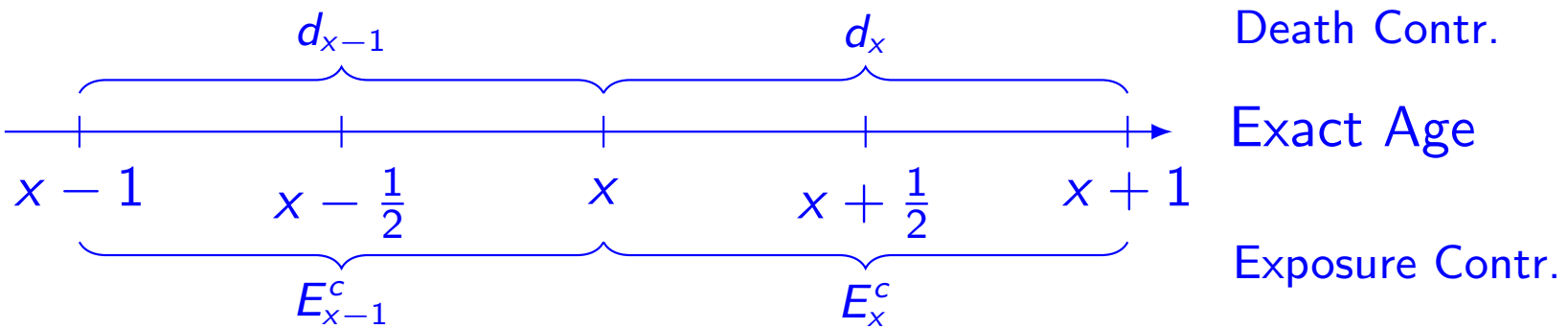
Exact Age vs. Age Labels

- "x" in d_x and E_x^c often refers to the age label, not the exact age.
- Age labels define what is considered to be "age x" in death and exposure data.
- Some examples of age labels:
  - Age last birthday
  - Age nearest birthday
  - Age next birthday
  - Age last birthday on preceding 1 Jan
  - Age nearest birthday on preceding 30 Jun
  - Age attained on birthday in current calendar year
- Sometimes, d_x and E_x^c might use different age labels - evidently, this throws off our estimates for μ and q!

Exact Age vs. Age Labels

- Principle of correspondence: Death data and exposure data must use the same age definition. That is, if a life is counted as being exposed to risk of death at age label x then the life would be counted in the deaths at the same age label.





Exact Age vs. Age Labels

- Golden Rule: We will always adjust the exposure data to fit the deaths data.
- This is because d_x is subject to more random variation than E_x^c.

  $\Longrightarrow$ E_x^c is a smoother variable.

  $\Longrightarrow$ More chance of getting the "correct" adjustment.
- But first, we need to examine methods to calculate E_x^c.

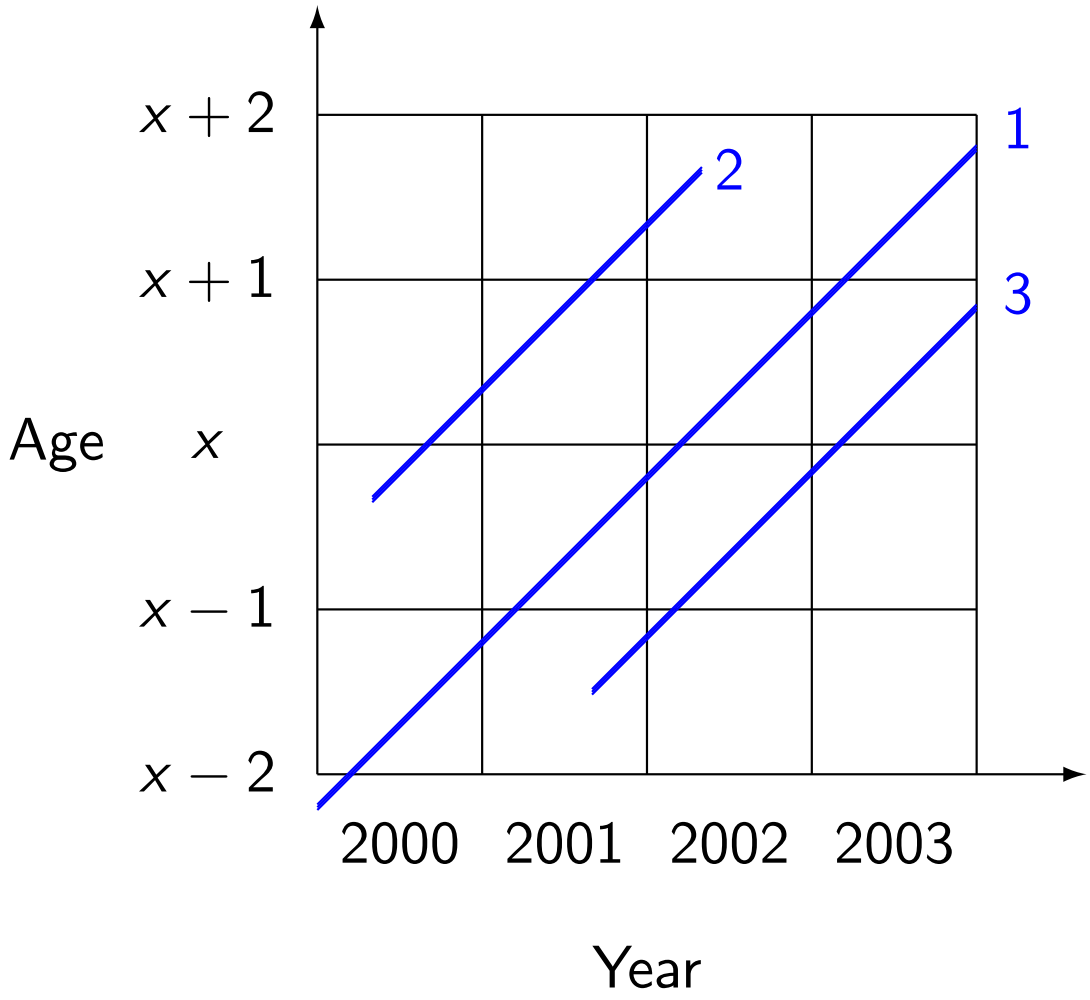
Exact Calculation of E_x^c: Lexis diagram

- Consider an example:

  Life 1:
  Observation starts at exact age $x - 2.2$, on $1/1/2000$
  Observation ends at exact age $x + 1.8$, on $31/12/2003$

  Life 2:
  Observation starts at exact age $x - 0.4$, on $2000+0.3$
  Observation ends at exact age $x + 1.7$, on $2000+0.4$

  Life 3:
  Observation starts at exact age $x - 1.5$, on $2001+0.6$
  Observation ends at exact age $x + 0.9$, on $31/12/2003$



Exact Calculation of E_x^c: Lexis diagram

- Horizontal cross-sections:

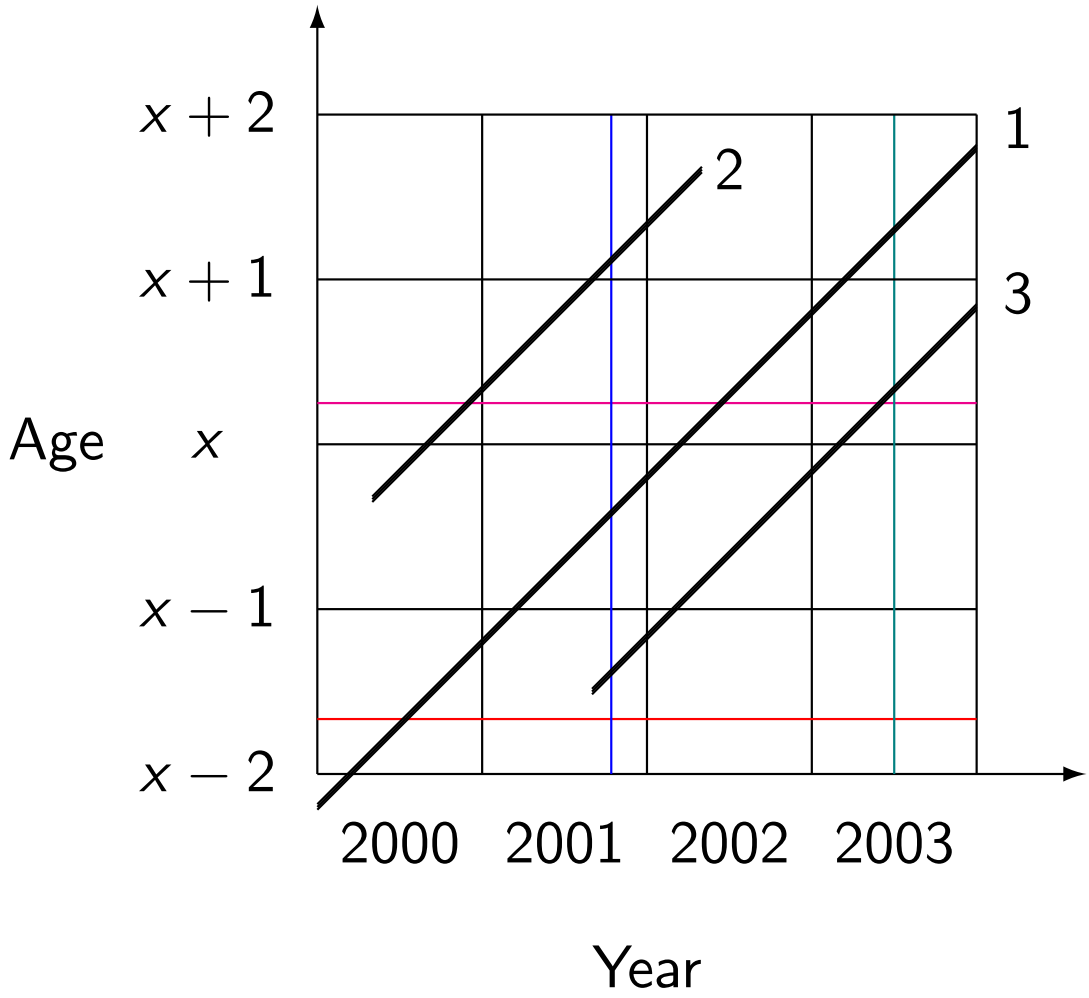
  Exact age $= x - 1.67$:
  We observed 1 life that attained exact age $x - 1.67$.

  Exact age $= x + 0.25$:
  We observed 3 lives that attained exact age $x + 0.25$.

- Vertical cross-sections:

  Exact time $= 2001.8$:
  We observe 3 lives alive at exact calendar time 2001.8.

  Exact time $= 2003.5$:
  We observe 2 lives alive at exact calendar time 2003.5.



Exact Exposure Method

- We often want to calculate each life's (or several lives') contributions to the exposures in each age,

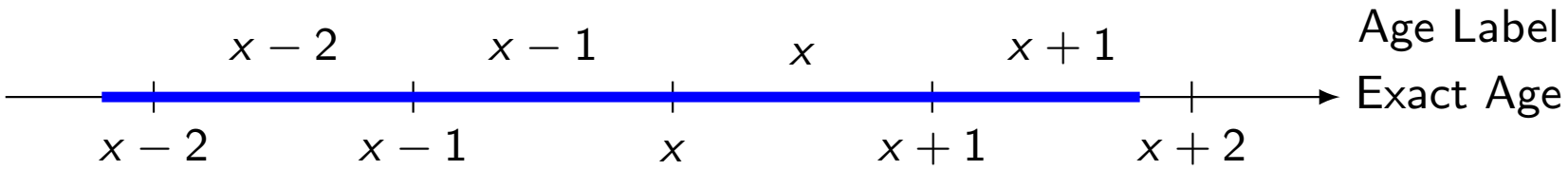
  i.e. contributions to ..., E^c_{x-1}, E^c_x, E^c_{x+1}, ... etc.

- To calculate these exactly, we use the exact exposure method.
  We will need the following information for this:
  - date of birth
  - date of entry
  - date of exit/leaving

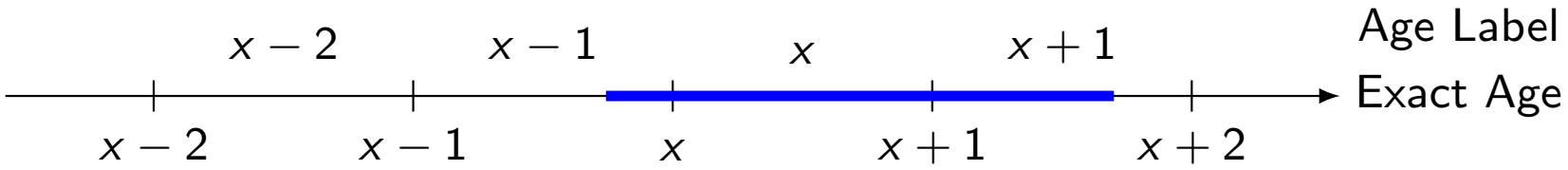
Exact Exposure Method: Example 1

- Previous example, with age label defined as "age last birthday".

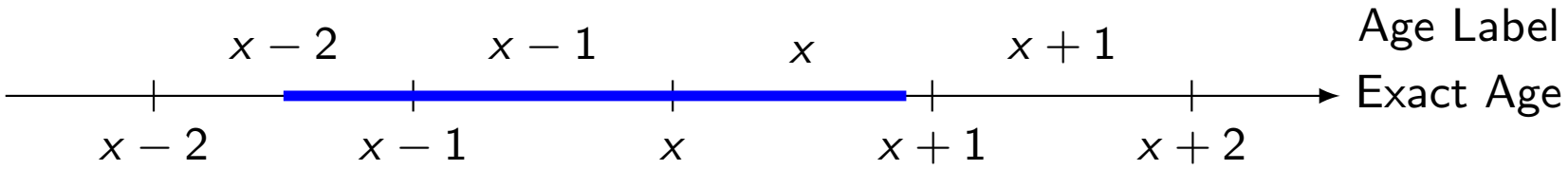
Life 1: contributes 0.2 to E^c_{x-3}, 1 to E^c_{x-2}, 1 to E^c_{x-1}, 1 to E^c_x, 0.8 to E^c_{x+1}.



Life 2: contributes 0.4 to E^c_{x-1}, 1 to E^c_x, 0.7 to E^c_{x+1}.



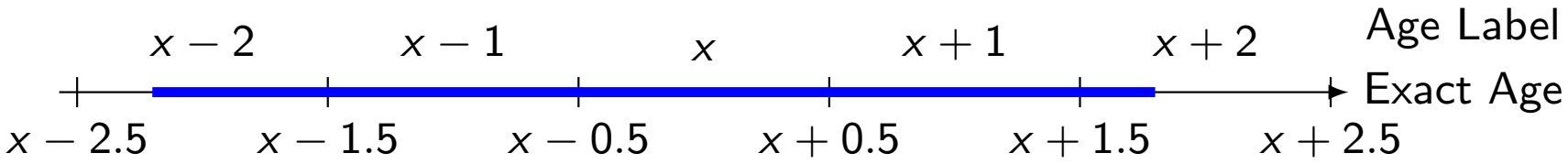
Life 3: contributes 0.5 to E^c_{x-2}, 1 to E^c_{x-1}, 0.9 to E^c_x.



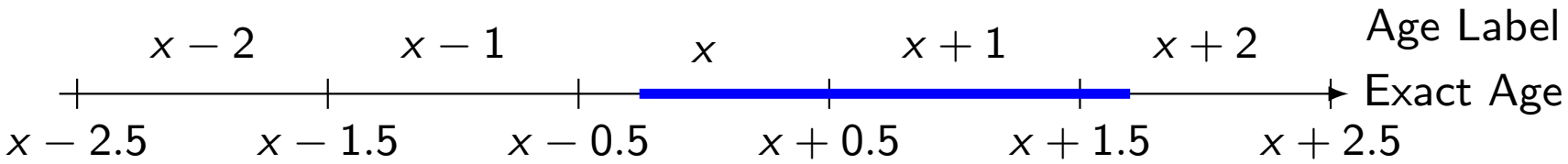
Exact Exposure Method: Example 2

- Same example, with age label defined as "age nearest birthday".

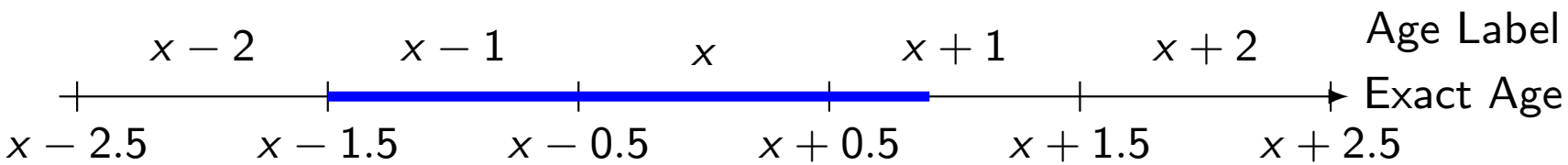
Life 1: contributes 0.7 to E^c_{x-2}, 1 to E^c_{x-1}, 1 to E^c_x, 1 to E^c_{x+1}, 0.3 to E^c_{x+2}.



Life 2: contributes 0.9 to E^c_x, 1 to E^c_{x+1}, 0.2 to E^c_{x+2}.



Life 3: contributes 1 to E^c_{x-1}, 1 to E^c_x, 0.4 to E^c_{x+1}.



Rate Intervals

- Once the age definitions for d_x and E_x^c are aligned, we can then make estimations for μ and q. But, at which point do the estimations apply?
- We assume μ applies at the center of age intervals, and also that q applies at the beginning of age intervals.
- The age interval should correspond with the age labels used.
- Note that the age labels can be defined in reference to:
  1. the birthday of the policyholder,
  2. a fixed point in the calendar year,
  3. the policy year.

Rate Intervals 1: Life year rate intervals

- When we define age with reference to birthdays, we obtain a life year rate interval for the estimates of μ and q.

Age Definition	Age Interval	$\hat{\mu}$ Estimates	$\hat{q}$ Estimates
x last birthday	$[x, x+1)$	$\mu_{x+\frac{1}{2}}$	q_x
x nearest birthday	$[x-\frac{1}{2}, x+\frac{1}{2})$	μ_x	$q_{x-\frac{1}{2}}$
x next birthday	$(x-1, x]$	$\mu_{x-\frac{1}{2}}$	q_{x-1}

- Life year rate intervals require exact age details (i.e. more data requirements).

Rate Intervals 2: Calendar year rate intervals

- When we define age with reference to a fixed point in the calendar year, we obtain a calendar year rate interval for the estimates of μ and q.
- Some examples of such age labels are:
  - Age last birthday on the preceding 1 January
  - Age nearest birthday on the preceding 30 June
  - Age attained on the birthday in the current calendar year
- This is common where the only information available is
  - calendar year of birth
  - calendar year of death

Rate Intervals 2: Calendar year rate intervals

- Example: What happens when the age label is defined as age attained on the birthday in the current calendar year?
- Note: With calendar year rate intervals, the number of lives exposed to the risk of death is not a continuous function – age labels of all individuals change at the same time in a calendar year!
  (What about life year rate intervals?)

Rate Intervals 3: Policy year rate intervals

- When we define age with reference to policy anniversaries, we obtain a policy year rate interval for the estimates of μ and q.
- Some examples of such age labels are:
  - Age on birthday in the previous policy year
  - Age next birthday at entry plus number of completed policy years
  - Age last birthday at the nearest policy anniversary
- Policy anniversaries are often spread uniformly over the year, so the number of lives exposed to the risk of death is relatively continuous, but there are exceptions.
- Rates estimated correspond easily to policy investigations and policy durations.

Rate Intervals 3: Policy year rate intervals

- Example: Consider the following scenario.

  Date of birth: 14/3/46
  Date of death: 19/9/92
  Policy start date: 12/5/79

  1. What is the age at death if age is defined as age last birthday on 1 April in the policy year of death?

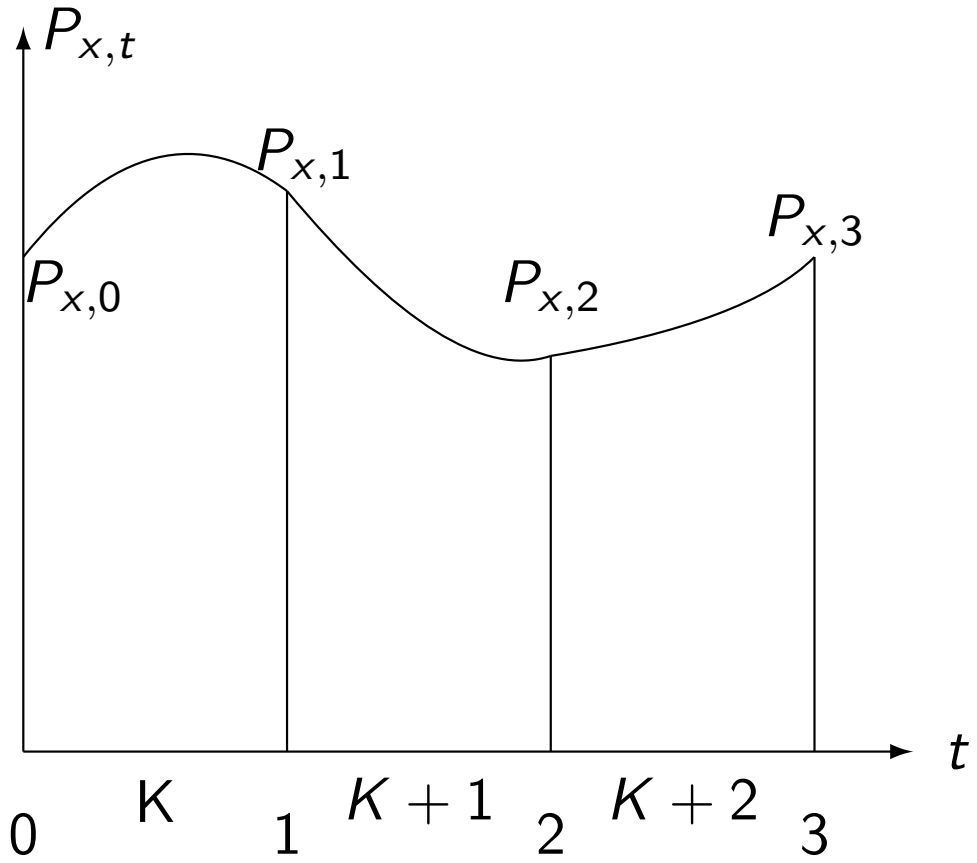
  2. Let the age label above be x'. At approximately what ages do the estimated rates $\hat{\mu}_{x'} = \dfrac{d_{x'}}{E^c_{x'}}$ and $\hat{q}_{x'} = \dfrac{d_{x'}}{E^c_{x'} + \frac{1}{2} d_{x'}}$ apply?

Census Data: Calculation of E_x^c

- Data on exposed to risk might not be complete. How do we calculate/approximate E_x^c in such cases then?
- We will attempt to use census data to calculate E_x^c.
  A census is simply a head count at a particular time instant.
- Define the following notations:
  - Observation window is $t = T_1$ to $t = T_2$,
  - $P_{x,t}$:= number of lives under observation with age label x at time t.

Census Data: Calculation of E_x^c

- Fixing an age x and varying t, we can visualize a curve for $P_{x,t}$:



Census Data: Calculation of E_x^c by a Census Formula

- If we knew $P_{x,t}$ for all $t \in [T_1, T_2]$ then we can calculate E_x^c with:

$$E_x^c = \int_{t=T_1}^{T_2} P_{x,t}\ dt$$

but this is rather unrealistic.

- Suppose now that we only have $P_{x,t}$ for $t = K,\ K+1,\ K+2,\ \ldots,\ K+N$, i.e. census counts on 1st Jan of each calendar year.

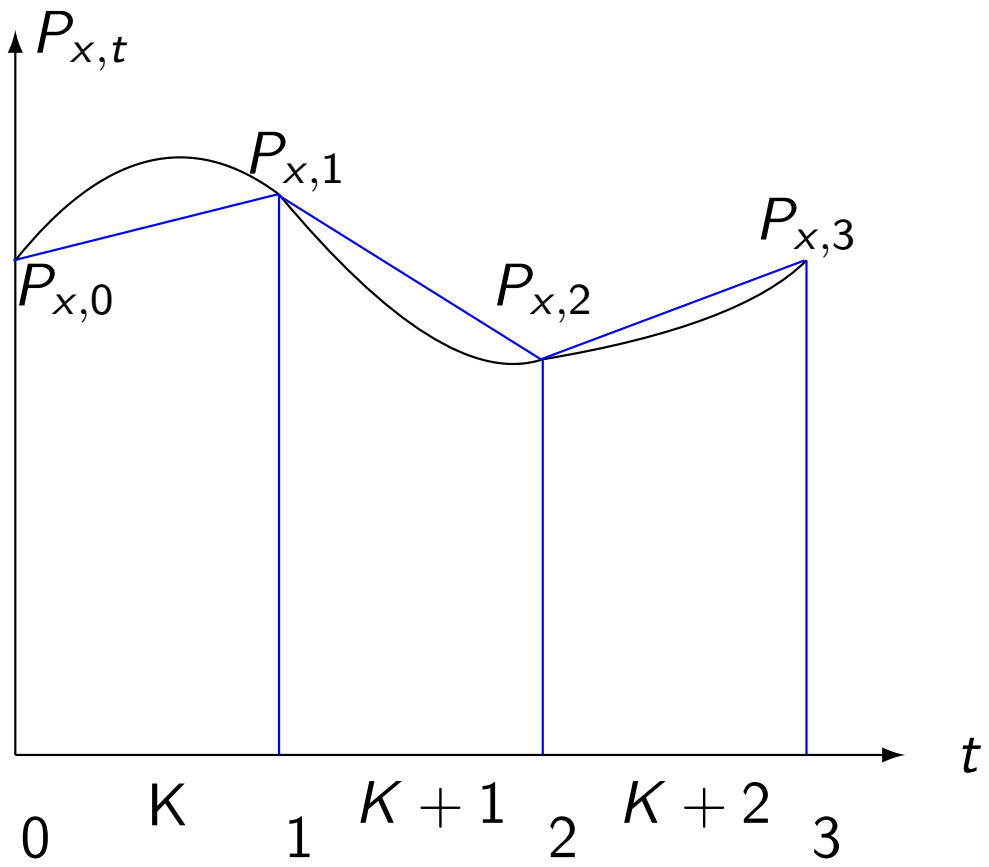
Then an approximation to the integral using the trapezium rule is

$$E_x^c \approx \frac{1}{2} \sum_{t=K}^{K+N-1} (P_{x,t} + P_{x,t+1}),$$

which implies the assumption that $P_{x,t}$ is linear between census dates.

Census Data: Calculation of E_x^c

- To illustrate the integral approximation for E_x^c:



Census Data: Adjustment of census data

- Recall the principle of correspondence & golden rule.
- If we calculate E_x^c using census data $P_{x,t}$, then adjusting the age label for E_x^c means adjusting the age label for $P_{x,t}$ as well.
- Let

  $P_{x,t}^C$ = no. of lives aged x at time t under census definition of age

  $P_{x,t}^D$ = no. of lives aged x at time t under definition of age for deaths

- Hence our objective is to estimate $P_{x,t}^D$ from $P_{x,t}^C$.

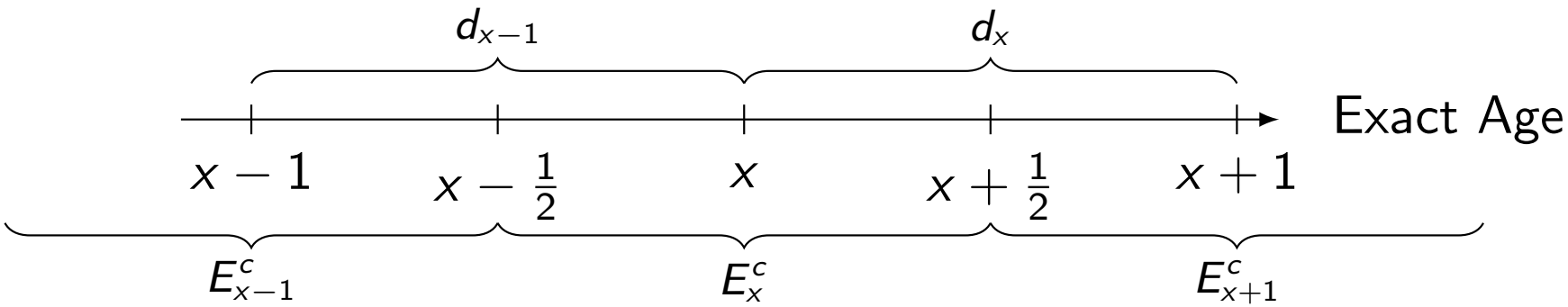
Census Data: Adjustment of census data - Example

- Example: Suppose that we define

$$d_x = \text{no. of deaths aged } x \text{ last birthday at date of death}$$

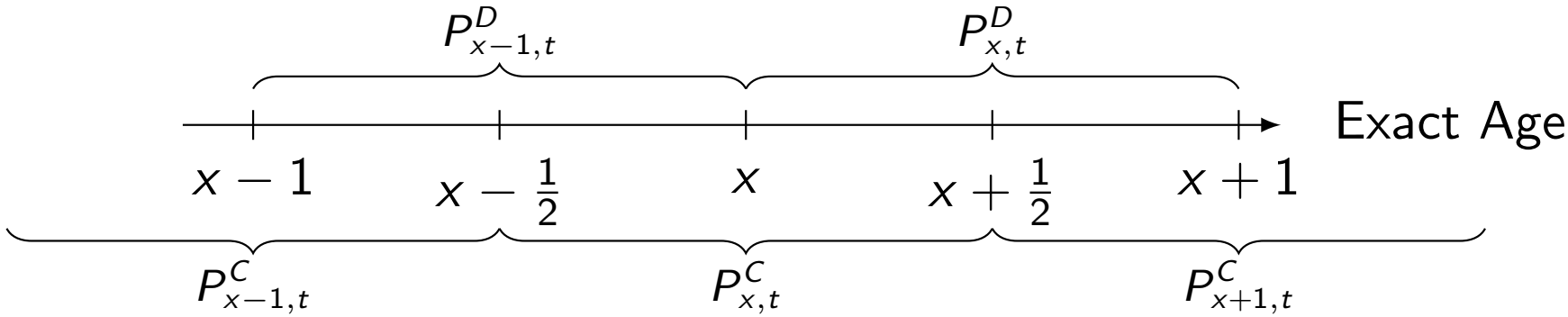
$$P^C_{x,t} = \text{no. of lives aged } x \text{ nearest birthday at time } t$$

i.e. we have different age definitions for death and census data:



Census Data: Adjustment of census data - Example

- $P^C_{x.t} \longleftrightarrow E^c_x$ and $P^D_{x,t} \longleftrightarrow d_x$, so at any time t,



If we assume uniform distribution of birthdays throughout the year, then by matching the intervals we get

$$P^D_{x-1,t} \approx \frac{1}{2}P^C_{x-1,t} + \frac{1}{2}P^C_{x,t}$$
$$P^D_{x,t} \approx \frac{1}{2}P^C_{x,t} + \frac{1}{2}P^C_{x+1,t}$$

with similar formulae for other ages.

Census Data: Adjustment of census data - Example

- We can then calculate E_x^c over the observation window $[T_1, T_2]$ as

$$E_x^c = \int_{T_1}^{T_2} P_{x,t}^D \, dt = \frac{1}{2} \int_{T_1}^{T_2} (P_{x,t}^C + P_{x+1,t}^C) \, dt$$

  using approximations previously derived.

- This adjustment needs to be adapted for age labels different from those in this example! In all cases, you should state the relevant assumptions applied.

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[not to be taken literally]